MAS2901: Statistical Inference

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Preface

Welcome to the online notes for **MAS2901 — Statistical Inference**.

This 10-credit module at Newcastle University introduces the two main approaches to *statistical inference*:

- **Frequentist inference**
- **Bayesian inference**

Teaching week	Week Beginning	Topic	Activities	Assessment
1	22nd September	Probability Theory, What is Statistical Inference? and Types of Inference	Two in-person lectures	
2	29th September	Frequentist Point Estimation	Two in-person lectures, Problems class (Friday 1pm)	Assignment 1 released (Wednesday)
3	6th October	Frequentist Point Estimation and Likelihood	Two in-person lectures	
4	13th October	Likelihood	Two in-person lectures, Problems class (Friday 1pm)	Assignment 1 due (Deadline Wednesday 4pm)
5	20th October	Bayesian Inference	Two in-person lectures	

Teaching week	Week Beginning	Topic	Activities	Assessment
6	27th October	Frequentist Interval Estimation	Two in-person lectures, Problems class (Friday 1pm)	Assignment 2 released (Wednesday)
Enrichment Week				
7	10th November	Frequentist Interval Estimation, Bayesian Intervals	Two in-person lectures	Assignment 2 Due (Deadline Wednesday 4pm)
8	17th November	Bayesian Intervals, Frequentist Hypothesis Testing	Two in-person lectures, Problems class (Friday 1pm)	Assignment 3 released (Wednesday)
9	24th November	Frequentist Hypothesis Testing	Two in-person lectures	
10	1st December	Bayesian Hypothesis Testing	Two in-person lectures, Problems class (Friday 1pm)	Assignment 3 Due (Deadline Wednesday 4pm)
11	8th December	Revision Week	Two in-person lectures	
Vacation				

What is Statistical Inference?

Statistical inference is the process of using data from a *sample* to learn about a *population*. Sometimes the population is concrete (e.g. "all people in the country today"). In other situations we want to generalise beyond those we sampled to a broader *target population* (sometimes called a *superpopulation*), such as "future patients with similar characteristics". Clear definitions and sensible assumptions are essential for valid conclusions.

There are two main types of statistical inference:

- **Frequentist inference.** Parameters are treated as **fixed but unknown**. Probability statements are about *data we might observe* under repeated sampling from the same model.
- **Bayesian inference.** Parameters are treated as **random** with a *prior distribution.* Probability statements are about the *parameter given the data* via the posterior distribution.

The examples below are only to build intuition about the core difference between frequentist and Bayesian inference. Don't worry about the details—we'll cover them carefully later in the course.

Example 0.1 (A comparison of Frequentist vs Bayesian inference)**.** A survey asks $n = 20$ students if they prefer study space A or B.

Suppose $x = 12$ prefer A. Compare a frequentist and a Bayesian estimate of the long-run proportion p who would choose A.

Frequentist. The natural estimator is the sample proportion

$$\hat{p} = \frac{x}{n} = \frac{12}{20} = 0.60.$$

Bayesian (with a uniform prior). Take a $\text{Beta}(1, 1)$ prior for p. The posterior is

$$p \mid x \sim \text{Beta}(1 + x,\, 1 + n - x) = \text{Beta}(13, 9),$$

whose mean is

$$\mathbb{E}[p \mid x] = \frac{13}{13 + 9} = \frac{13}{22} \approx 0.591.$$

Interpretation. The frequentist estimate $\hat{p}$ is a function of the data and treats p as fixed. The Bayesian estimate is an expectation *of p itself* under the posterior distribution.

Example 0.2 (Interpreting probability statements)**.** Decide whether each statement is *frequentist* or *Bayesian* in spirit.

1. "If we repeated the survey many times, the method we use would produce intervals that capture p about 95% of the time."

2. "Given the data from this survey and our prior, there is a 95% probability that p lies between 0.45 and 0.75."

1. **Frequentist.** This describes *long-run coverage* of a procedure over repeated samples with p fixed.

2. **Bayesian.** This is a probability statement *about p given the observed data* (a credible interval).

Useful Links

- Web based R: https://webr.sh/
- Web based Jupyter Lab (for Python and R): https://jupyter.org/try-jupyter/lab/

Further Reading

- Statistical Inference. *Casella, George.; Berger, Roger L.*
- A student's guide to Bayesian statistics. *Lambert, Ben.*
- Theory of statistics. *Schervish, Mark J.*

Part I

Part I — Foundations

1 Probability Theory

Learning objectives

- Define and work with events (union, intersection, complement) and apply inclusion–exclusion.
- Distinguish mutually exclusive, exhaustive, and independent events.
- Compute conditional probabilities and apply Bayes' theorem.
- Use partitions and the law of total probability.
- Define random variables and their distribution functions: cdf, pmf (discrete), and pdf (continuous).
- Compute probabilities from pmfs/pdfs/cdfs (using sums or areas) and check normalisation.

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Before we delve into statistical inference, we briefly review key ideas from probability theory.

Note on this chapter

- This chapter is provided **for review and reference only**.
- It will not be examined in detail, but the ideas are useful background for later topics.
- Much of the content should already be familiar.

1.1 Probability and Events

When working with probability we need three basic ingredients[1]

1. The **sample space** Ω. all possible outcomes of an experiment.
   Example: for a coin toss, $\Omega = \{\text{Heads}, \text{Tails}\}$.

[1] **Note on Probability Theory.** Formally, a probability space is written as a triple (Ω, Σ, P), where Σ is the collection of "measurable" events. This level of mathematical detail goes beyond what we need in this course and will **not be assessed**. (See the Wikipedia article on probability spaces if you are curious.)

2. **Events** subsets of the sample space. We use capital letters $A, B, \ldots$ to denote events.
   - The **complement** of A is A^c, meaning A does *not* occur.
   - The **intersection** $A \cap B$ means A *and* B occur.
   - The **union** $A \cup B$ means A *or* B (or both) occur.
3. A **probability distribution** Pr. Assigns numbers between 0 and 1 to events, with $\Pr(\Omega) = 1$.

We can write the definitions of complement, intersection and union formally with sets:

Definition 1.1 (Complement, Intersection and Union)**.**

- **Complement:** $A^c = \{x \in \Omega \mid x \notin A\}$.
- **Intersection:** $A \cap B = \{x \in \Omega \mid x \in A \text{ and } x \in B\}$.
- **Union:** $A \cup B = \{x \in \Omega \mid x \in A \text{ or } x \in B\}$.

Definition 1.2 (Addition rule (inclusion–exclusion))**.** For any two events $A, B \subseteq \Omega$, we have

$$\Pr(A \cup B) = \Pr(A) + \Pr(B) - \Pr(A \cap B).$$

Example 1.1 (A quick inclusion–exclusion check)**.** Roll a fair six-sided die. Let $A = \{\text{even}\}$ and $B = \{\text{number} \geq 4\}$. Compute $\Pr(A \cup B)$.

We have

- $\Pr(A) = 3/6 = 1/2$ (even outcomes $2, 4, 6$).
- $\Pr(B) = 3/6 = 1/2$ (outcomes $4, 5, 6$).
- $\Pr(A \cap B) = 2/6 = 1/3$ (outcomes $4, 6$).

By inclusion–exclusion 1.2,

$$\Pr(A \cup B) = \tfrac{1}{2} + \tfrac{1}{2} - \tfrac{1}{3} = \tfrac{2}{3}.$$

Definition 1.3 (Mutually Exclusive, Exhaustive Events, and Partitions)**.**

- Two events A and B are **mutually exclusive** (or *disjoint*) if they cannot occur together:

$$\Pr(A \cap B) = 0.$$

- Two events A and B are **exhaustive** if at least one of them must occur:

$$\Pr(A \cup B) = 1.$$

  More generally, a collection of events $\{B_1, \dots, B_k\}$ is exhaustive if $\bigcup_{i=1}^{k} B_i = \Omega$, where Ω is the sample space.

- A collection of events $\{B_1, \dots, B_k\}$ forms a **partition** of Ω if they are both mutually exclusive and exhaustive. Equivalently,

  - $B_i \cap B_j = \emptyset$ for all $i \neq j$ (non-overlapping), and
  - $\bigcup_{i=1}^{k} B_i = \Omega$ (their union is the whole sample space).

Definition 1.4 (Conditional probability)**.** If $\Pr(B) > 0$, the probability of A given B is

$$\Pr(A \mid B) = \frac{\Pr(A \cap B)}{\Pr(B)}.$$

Definition 1.5 (Independence)**.** Two events A and B are **independent** if knowing that one occurs does not change the probability of the other:

$$\Pr(A \cap B) = \Pr(A)\Pr(B).$$

Equivalently,

$$\Pr(A \mid B) = \Pr(A) \quad \text{and} \quad \Pr(B \mid A) = \Pr(B)$$

Theorem 1.1 (Law of Total Probability)**.** *If $B_1, \dots, B_k$ form a partition 1.3 of the sample space (mutually exclusive and exhaustive events), then for any event A:*

$$\Pr(A) = \sum_{i=1}^{k} \Pr(A \mid B_i)\Pr(B_i).$$

Proof. Since $B_1 \dots, B_n$ form a partition of Ω, we can decompose any event A in a *disjoint* union of intersections:

$$A = \bigcup_{i=1}^{n} (A \cap B_i).$$

By induction on the addition rule 1.2, we thus have

$$\Pr(A) = \sum_{i=1}^{n} \Pr(A \cap B_i).$$

By the definition of conditional probability 1.4, we have $\Pr(A \cap B_i) = \Pr(A \mid B_i)\Pr(B_i)$. Therefore,

$$\Pr(A) = \sum_{i=1}^{n} \Pr(A \mid B_i)\Pr(B_i).$$

□

Theorem 1.2 (Bayes' Theorem)**.** *If* $\Pr(B) > 0$*, then*

$$\Pr(A \mid B) = \frac{\Pr(B \mid A)\Pr(A)}{\Pr(B)}.$$

Proof. From the definition of conditional probability 1.4,

$$\Pr(A \mid B) = \frac{\Pr(A \cap B)}{\Pr(B)}.$$

But the intersection $A \cap B$ can also be written the other way round:

$$\Pr(A \cap B) = \Pr(B \cap A).$$

Using the definition of conditional probability again,

$$\Pr(B \cap A) = \Pr(B \mid A)\Pr(A).$$

Substituting this into the first expression gives

$$\Pr(A \mid B) = \frac{\Pr(B \mid A)\Pr(A)}{\Pr(B)}.$$

This completes the proof.

□

Example 1.2 (Student Example)**.** Seventy percent of students are attentive in lectures and thirty percent are not. If a student is *attentive* the probability of passing the course is 0.8. If a student is *not attentive* the probability of passing the course is 0.1.

1. A student is selected at random. What is the probability that they pass the course?
2. Given that the student passed the course, what is the probability that they were attentive?

Let

- P denote the event of a student passing the course.
- A denote the event of a student being attentive.
- A^c denote the event of a student not being attentive.

1. Attentive and inattentive form a partition. By the law of total probability,

$$\Pr(P) = \Pr(P \mid A)\Pr(A) + \Pr(P \mid A^c)\Pr(A^c) = 0.8 \times 0.7 + 0.1 \times 0.3 = 0.59.$$

2. Using Bayes' theorem,

$$\Pr(A \mid P) = \frac{\Pr(P \mid A)\Pr(A)}{\Pr(P)} = \frac{0.8 \times 0.7}{0.59} = 0.949 \text{ (3 s.f.)}.$$

1.2 Random Variables

We use upper-case letters for random variables, e.g. $X, Y, X_1, X_2, \ldots$.

- **Cumulative distribution function (cdf).** For any (real-valued) random variable X,

$$F(x) = \Pr(X \leq x).$$

- **Discrete case (pmf).** If X is discrete, its *probability mass function* is

$$p(x) = \Pr(X = x), \qquad \sum_x p(x) = 1.$$

- **Continuous case (pdf).** If X is continuous, its *probability density function* satisfies

$$f(x) = \frac{\partial F(x)}{\partial x}, \qquad \Pr(a < X \leq b) = \int_a^b f(x)\,\mathrm{d}x, \qquad \int_{-\infty}^{\infty} f(x)\,\mathrm{d}x = 1.$$

Example 1.3 (Discrete CDF.)**.** Let X be the outcome of a fair six-sided die. Find $p(x)$ and $F(3)$.

$p(x) = \Pr(X = x) = 1/6$ for $x \in \{1, 2, 3, 4, 5, 6\}$, and 0 otherwise.
$F(3) = \Pr(X \leq 3) = \Pr(\{1, 2, 3\}) = 3 \times (1/6) = 1/2$.

Example 1.4. Let $X \sim \text{Uniform}(0, 1)$. Compute $\Pr(0.2 < X < 0.5)$.

Here $f(x) = 1$ for $0 < x < 1$ and 0 otherwise.

$$\Pr(0.2 < X < 0.5) = \int_{0.2}^{0.5} 1\,dx = 0.3.$$

1.2.1 Joint, Marginal, and Conditional Densities

Suppose X and Y are random variables with joint PDF/PMF $f_{XY}(x, y)$ (often written $f(x, y)$). From this joint distribution we can define:

- **Marginal distributions:** the distribution of each variable on its own.
- **Conditional distributions:** the distribution of one variable given the other.
- **Independence:** the case where the joint factorises.

Definition 1.6 (Marginal Distributions)**.** If we only care about X, regardless of Y, we integrate/sum out Y:

$$f_X(x) = \int f(x, y)\,\mathrm{d}y.$$

Similarly for $f_Y(y)$.
Note: knowing the marginals does **not** determine the joint.

Definition 1.7 (Conditional Densities)**.** For y in the support of Y with $f_Y(y) > 0$, we define:

$$f_{X|Y}(x \mid y) = \frac{f(x, y)}{f_Y(y)}.$$

This describes the distribution of X *given* $Y = y$.

Theorem 1.3 (Law of Total Probability)**.** *We can recover the marginal by "averaging" the conditional over Y:*

$$f_X(x) = \int f_{X|Y}(x \mid y) f_Y(y)\,\mathrm{d}y.$$

Definition 1.8 (Independence)**.** X and Y are independent if and only if

$$f(x, y) = f_X(x) f_Y(y).$$

In this case, $f_{X|Y}(x \mid y) = f_X(x)$: knowing Y tells us nothing about X.

1.3 Expectations, Variances and Covariances

Definition 1.9 (Expected Value)**.** The **expected value** (or **population mean**) of X is

$$\mathrm{E}[X] = \begin{cases} \displaystyle\sum_{\text{all } x} x\, p_X(x), & \text{if } X \text{ is discrete,} \\ \displaystyle\int_{-\infty}^{\infty} x\, f_X(x)\,\mathrm{d}x, & \text{if } X \text{ is continuous.} \end{cases}$$

We often write $\mu = \mathrm{E}[X]$. This is a measure of **location**.

Definition 1.10 (Expectation of $g(X)$)**.** Let $g(\cdot)$ be any function and X a random variable with PMF p_X (discrete) or PDF f_X (continuous). The **expected value** of $g(X)$ is

$$\mathrm{E}[g(X)] = \begin{cases} \displaystyle\sum_{\text{all } x} g(x)\, p_X(x), & \text{if } X \text{ is discrete,} \\ \displaystyle\int_{-\infty}^{\infty} g(x)\, f_X(x)\, \mathrm{d}x, & \text{if } X \text{ is continuous.} \end{cases}$$

Definition 1.11 (Expectation (Bivariate))**.** If (X, Y) is a pair of random variables with joint PMF $p_{X,Y}$ (discrete) or joint PDF $f_{X,Y}$ (continuous), then for any function $g(x, y)$

$$\mathrm{E}[g(X,Y)] = \begin{cases} \displaystyle\sum_{\text{all } x}\sum_{\text{all } y} g(x,y)\, p_{X,Y}(x,y), & \text{if } (X,Y) \text{ are discrete,} \\ \displaystyle\iint g(x,y)\, f_{X,Y}(x,y)\, \mathrm{d}x\, \mathrm{d}y, & \text{if } (X,Y) \text{ are continuous.} \end{cases}$$

Definition 1.12 (Moments)**.** The quantities $\mathrm{E}[X], \mathrm{E}[X^2], \mathrm{E}[X^3], ...$ are the (raw) **moments** of X.

Definition 1.13 (Variance)**.** The **population variance** of X is

$$\mathrm{Var}(X) = \mathrm{E}\big[(X - \mathrm{E}[X])^2\big] = \mathrm{E}[X^2] - \big(\mathrm{E}[X]\big)^2.$$

We often write $\sigma^2 = \mathrm{Var}(X)$. This is a measure of **spread**.

Definition 1.14 (Standard Deviation)**.** The **population standard deviation** is the positive square root of the variance:

$$\mathrm{SD}(X) = \sqrt{\mathrm{Var}(X)}.$$

We often write $\sigma = \mathrm{SD}(X)$. It shares the **same units** as X and measures spread.

Definition 1.15 (Covariance)**.** For two random variables X and Y, the **covariance** is

$$\mathrm{Cov}(X, Y) = \mathrm{E}[(X - \mathrm{E}[X])(Y - \mathrm{E}[Y])] = \mathrm{E}[XY] - \mathrm{E}[X]\mathrm{E}[Y].$$

It measures **linear association**: positive if Y tends to increase with X; negative if Y tends to decrease as X increases.

Definition 1.16 (Correlation)**.** The **correlation** (often denoted ρ) is the scaled covariance

$$\rho = \mathrm{Corr}(X, Y) = \frac{\mathrm{Cov}(X, Y)}{\mathrm{SD}(X)\, \mathrm{SD}(Y)}.$$

Correlation always satisfies $\rho \in [-1, 1]$.

Theorem 1.4 (Independence $\Rightarrow$ Zero Covariance)**.** *If X and Y are **independent**, then* $\mathrm{Cov}(X, Y) = 0$.

Note. *The converse need not hold: it is possible to have* $\mathrm{Cov}(X, Y) = 0$ *while X and Y are **not** independent.*

Example 1.5 (Variance of a Bernoulli Random Variable)**.** A **?@sec-bern** X takes only the values 0 and 1, with probabilities

$$\Pr(X = 1) = p, \qquad \Pr(X = 0) = 1 - p, \qquad 0 \le p \le 1.$$

For what value of p is the variance maximised?

We compute the mean and variance of X.

$$\mathrm{E}[X] = 1 \cdot p + 0 \cdot (1 - p) = p,$$

$$\mathrm{E}[X^2] = 1^2 \cdot p + 0^2 \cdot (1 - p) = p.$$

Therefore,

$$\mathrm{Var}(X) = \mathrm{E}[X^2] - \big(\mathrm{E}[X]\big)^2 = p - p^2 = p(1 - p).$$

To find the value of p that maximises this variance, consider the function

$$f(p) = p(1 - p), \quad 0 \le p \le 1.$$

Differentiate:

$$f'(p) = 1 - 2p.$$

Setting $f'(p) = 0$ gives $p = \frac{1}{2}$.

Second derivative:

$$f''(p) = -2 < 0,$$

so this is indeed a maximum.

Finally, check the boundary values:

$$f(0) = 0, \qquad f(1) = 0, \qquad f\big(\tfrac{1}{2}\big) = \tfrac{1}{4}.$$

Conclusion.
The variance of a Bernoulli random variable is maximised at

$$p = \tfrac{1}{2}, \quad \mathrm{Var}(X) = \tfrac{1}{4}.$$

1.4 Non-Examinable Material

> ⚠ Non-Examinable Content
>
> This section provides definitions and results that are used in **non-examinable proofs and remarks** elsewhere in the notes.
> None of the material in this section is examinable.

1.4.1 Convergence of Random Variables

Definition 1.17 (Convergence in Probability)**.** A sequence of random variables X_n is said to **converge in probability** to a random variable X if, for every $\varepsilon > 0$,

$$\Pr\left(|X_n - X| > \varepsilon\right) \longrightarrow 0 \quad \text{as } n \to \infty.$$

We write this as

$$X_n \xrightarrow{p} X.$$

Intuitively: as n grows, the probability that X_n is far from X becomes negligible.

Definition 1.18 (Convergence in Distribution)**.** A sequence of random variables X_n is said to **converge in distribution** to a random variable X if, for all continuity points x of F_X,

$$F_{X_n}(x) \longrightarrow F_X(x) \quad \text{as } n \to \infty,$$

where F_{X_n} and F_X are the cumulative distribution functions.

We write this as

$$X_n \xrightarrow{d} X.$$

Intuitively: the distribution of X_n "settles down" and approaches that of X.

Relationship

Convergence in probability is **stronger** than convergence in distribution:

$$X_n \xrightarrow{p} X \quad \Longrightarrow \quad X_n \xrightarrow{d} X.$$

But not the other way around.

1.4.2 Probability Inequalities

Proposition 1.1 (Chebyshev's Inequality)**.** *If a random variable X has mean μ and variance $\sigma^2 < \infty$, then for any $\varepsilon > 0$,*

$$\Pr\left(|X - \mu| \geq \varepsilon\right) \;\leq\; \frac{\sigma^2}{\varepsilon^2}.$$

This provides a bound on the probability of large deviations from the mean.

1.4.3 Factorising the Joint

Proposition 1.2 (The Chain Rule)**.** *Let $\underline{X} = (X_1, \dots, X_n)$ with joint PDF/PMF $f(\underline{x})$. Then*

$$f(\underline{x}) \;=\; f(x_1) \prod_{t=2}^{n} f(x_t \mid x_1, \dots, x_{t-1}).$$

Proof of **??**. We use the definition of conditional PDF/PMF (Definition 1.7) and induction.

- **Base case ($n = 2$).** By definition,

  $$f(x_2 \mid x_1) \;=\; \frac{f(x_1, x_2)}{f(x_1)} \quad (\text{for } f(x_1) > 0),$$

  hence $f(x_1, x_2) = f(x_1)\, f(x_2 \mid x_1)$.

- **Inductive step.** For $k \geq 2$,

$$f(x_1, \dots, x_k) \;=\; f(x_1, \dots, x_{k-1})\, f(x_k \mid x_1, \dots, x_{k-1}),$$

again by Definition 1.7:

$$f(x_k \mid x_1, \dots, x_{k-1}) \;=\; \frac{f(x_1, \dots, x_k)}{f(x_1, \dots, x_{k-1})}.$$

Iterating this factorisation from $k = 2$ up to $k = n$ yields

$$f(x_1, \dots, x_n) \;=\; f(x_1) \prod_{t=2}^{n} f(x_t \mid x_1, \dots, x_{t-1}).$$

Telescoping viewpoint. Equivalently,

$$\prod_{t=2}^{n} f(x_t \mid x_1, \dots, x_{t-1}) = \prod_{t=2}^{n} \frac{f(x_1, \dots, x_t)}{f(x_1, \dots, x_{t-1})} = \frac{f(x_1, \dots, x_n)}{f(x_1)},$$

so rearranging gives the same result.

Remarks.

- The argument applies to both PMFs and PDFs (assuming the relevant conditionals exist).
- The identity does **not** assume independence; it is always valid wherever the conditionals are defined.

□

2 What is Statistical Inference?

Learning objectives

- Define *population* and *sample*, and identify each in simple scenarios.
- State the *goal* of statistical inference: using a sample to learn about a population distribution.
- Recognise the idea of a *random sample* $X_1, \dots, X_n$ and the iid assumption.
- Define a *statistical model* as a set of distributions $\{P_\theta : \theta \in \Theta\}$.
- Identify the *parameter(s)* θ and the *parameter space* Θ for a given model.
- Write down a suitable statistical model for a short verbal problem.

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Statistical inference underpins experimental science, machine learning, clinical trials, and more. The overall aim is:

- Given a **sample** (the data you have) from a **population** (the broader group you care about),
- Draw a justified **conclusion about the population's distribution**.

Typical conclusions include estimating a proportion or mean, quantifying uncertainty, comparing groups, or predicting future outcomes.

Important 1: Notation: Distribution(θ)

In the following, we use Distribution(θ) as a **placeholder** to mean:

> "a distribution depending on parameter θ."

For example:

- $\text{Distribution}(\mu, \sigma^2) = \mathcal{N}(\mu, \sigma^2)$
- $\text{Distribution}(p) = \text{Bernoulli}(p)$

This allows us to keep the definitions *general*!

Important 2: Terminology: Population, Sample, Data

- **Population:** the entire group we want to study (conceptual, often infinite).
- **Sample:** the n random variables we actually collect from the population $X_1, X_2, \dots, X_n$.
- **Data:** the realised values of the sample once observed $x_1, x_2, \dots, x_n$.
- **Observation / data point:** a single element of the sample, e.g. X_i (random) or x_i (observed).

Note: In practice, the terms *data*, *observations*, and *data points* are often used interchangeably. In these notes, we'll do the same unless we need to be precise about whether something is random (sample) or fixed (data).

2.1 Statistical Inference as "Inverse Probability"

In **probability theory**, the usual direction is:

- Start with a distribution $f(x \mid \theta)$ (the model, with known parameter θ).
- Use it to generate random samples, or calculate probabilities of events involving the data.

In **statistical inference**, we reverse the problem:

- We observe data $\underline{x} = (x_1, \dots, x_n)$.
- The distribution $f(x \mid \theta)$ is only partially known, through the unknown parameter θ.
- Our task is to use the sample to learn about θ, and hence about the population distribution.

Formally, we (*usually*) assume our data are an **i.i.d. sample**:

$$X_1, \dots, X_n \overset{\text{iid}}{\sim} \text{Distribution}(\theta).$$

- *Independent*: knowing the value of one X_i gives no information about any other.
- *Identically distributed*: each X_i comes from the same population distribution $f(x \mid \theta)$.

The goal of *statistical inference* is to use the observed sample to learn about θ, and hence about the underlying population.

Important 3: Terminology: Random Sample

- A sample 2 means the n random variables we collect from the population, $X_1, \dots, X_n$. These need not be independent or identically distributed.
- A **random sample** is the special case where the X_i are assumed *independent and identically distributed (i.i.d.)* from the same distribution:

$$X_1, \dots, X_n \overset{\text{iid}}{\sim} \text{Distribution}(\theta).$$

Most of the inference procedures in this course are developed under the random-sample (iid) assumption.

2.1.1 Statistical Inference by Eye

As an informal starting point, suppose $X_1, \dots, X_n$ are sampled from a $\mathcal{N}(\mu, \sigma^2)$ distribution, with μ and σ^2 unknown.

Try to "guess" μ and σ^2 by eye from the data:

2.2 Types of Data

Before we build **statistical models**, it helps to think about the kinds of data we might collect. Different data types naturally lead to different models.

2.2.1 Univariate vs. Multivariate data

> **Important 4: Terminology: Unit**
>
> An **observational unit** (or simply *unit*) is a single element of the population on which we make a measurement.
>
> - In a survey of students: each *student* is a unit.
> - In a clinical trial: each *patient* is a unit.
> - In a manufacturing study: each *product* is a unit.
> - In an image analysis study: each *image* is a unit.

- **Univariate** data: a single measurement per observational unit.

  Example: the height of each student in a class.

- **Multivariate** data: two or more measurements on each unit.

  Example: the height and weight of each student (bivariate); or height, weight, and age (multivariate).

Data are often represented in **tables**.

2.2.2 Univariate data

Each unit has **one measurement**.

Table 2.1: **Earthquake counts.** Table shows the number of UK earthquakes of magnitude ≥ 2 on the Richter scale each year.

Year	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020
Earthquakes	4	0	2	6	4	0	2	8	5	7

Here, the **units** are the years (2011–2020), and the (single) **variable** is the annual count of UK earthquakes.

2.2.3 Multivariate data

Each unit has **two or more measurements**.

Table 2.2: **Weather data.** Average rainfall, wind speed, and air pressure recorded at weather stations during a storm.

Station	Rainfall (mm)	Wind speed (km/h)	Pressure (hPa)
A	214.8	87	995
B	196.3	91	992
C	180.2	79	1001
D	205.5	94	998
E	209.0	89	997
...	...	...	...

Here, the **units** are the weather **stations** (during the same storm), and the **variables** are rainfall (mm), wind speed (km/h), and pressure (hPa).

Some data is **extremely high-dimensional**:

Example 2.1 (Image Data)**.** An image is made of a grid of **pixels**, where each pixel is a **colour**.

A colour can be represented by three channels: red, green, and blue[1].

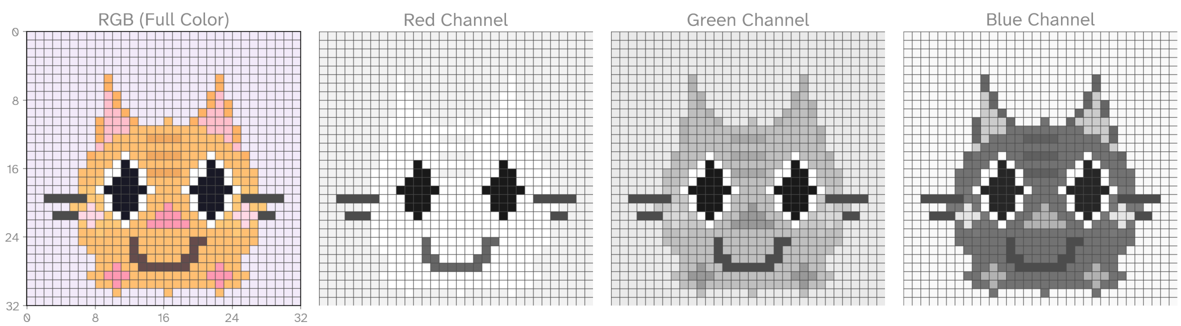

Figure 2.1: A 32×32 image of a cat, broken into its three colour channels (red, green, blue). Each pixel has three values (one per channel), making the image a high-dimensional observation.

Each pixel has **multiple measurements** (one per channel), so images are an example of **high-dimensional multivariate data**. To see why, consider the size of real images:

[1]**Note on images.** Some image formats, such as `.png`, include an *alpha channel* (the "A" channel), which specifies pixel transparency.

- A tiny image of size 32×32 pixels with 3 channels (RGB) already has $32 \times 32 \times 3 = 3072$ measurements.
- A small image of size 128×128 pixels with 3 channels (RGB) has $128 \times 128 \times 3 \approx 50{,}000$ measurements.

- A "HD" photo of size 1920×1080 (1080p) contains over $1920 \times 1080 \times 3 \approx 6.2$ million measurements.

So even a single image can be thought of as a *very high-dimensional observation*. And **videos** are higher still: sequences of such images evolving over time[2].

> ⚠ Non-Examinable Content: Multivariate Data
>
> In this course we focus **only on univariate data** (one measurement per observational unit).
> Although many real-world problems involve **multivariate data** (two or more measurements on the same unit), this topic will be studied in later statistics courses.

2.2.4 Numerical data

- **Continuous**: can take any real value in an interval.

  Examples: height, weight, reaction time.

  Models: Normal, Exponential, Uniform.

- **Discrete numeric**: counts, usually non-negative integers.

  Examples: number of accidents in a week, number of goals in a football match.

  Models: Binomial, Poisson, Geometric.

2.2.5 Continuous data

[2] **Note on compression.** In practice, images are often *compressed* (e.g. JPEG, PNG), which reduces file size by exploiting redundancy in pixel values. Statistically, though, the raw number of pixel measurements still reflects the dimensionality of the data.

Table 2.3: **Astronomy data.** The brightness of a star (its *magnitude*) is measured on a continuous scale. .

Star	Apparent magnitude
Sirius	–1.46
Vega	0.03
Altair	0.77
Deneb	1.25
...	...

2.2.6 Discrete data

Table 2.4: **Sports analytics.** Goals scored in football matches are discrete counts. Each match provides *multivariate discrete data* (two counts per unit).

Match	Home goals	Away goals
Newcastle United vs Arsenal	7	0
Manchester United vs Liverpool F.C.	0	1
Tottenham Hotspur vs Chelsea F.C.	1	3
Aston Villa vs Leeds United	3	2
West Ham United vs Manchester City	0	4
Brighton & Hove Albion vs Everton	2	2
Leicester City vs Southampton	1	0
Wolverhampton Wanderers vs Brentford	2	1
Crystal Palace vs Fulham	0	1
Nottingham Forest vs Bournemouth	1	1
...	...	...

2.2.7 Mixed data

In practice, most datasets are **mixed**

Table 2.5: **Medical study.** Here each patient (unit) has both continuous and discrete measurements.

Patient	Age (years)	Reaction time (ms)	Smoker (yes/no)
A	22	318	Yes
B	35	241	No
C	41	267	Yes

Patient	Age (years)	Reaction time (ms)	Smoker (yes/no)
D	29	452	No
...	...	...	...

2.2.8 Categorical data

- **Nominal**: categories with no natural order.

  Examples: hair colour, brand of cereal, country of birth.

- **Ordinal**: categories with a natural order, but not numerical spacing.

  Examples: customer satisfaction ratings ("poor", "fair", "good", "excellent").

Categorical data are often summarised in **contingency tables** (tables of counts).

2.2.9 One-way table

If we record the colours of sweets drawn from a bag:

Colour	Count
Red	R
Blue	B
Green	G
Total	n

This simple *one-way* table leads naturally to the **?@sec-multinomial**.

2.2.10 Two-way table

Suppose we also record the *shop* where each sweet was bought (Shop A or Shop B).
Now we cross-classify by both colour and shop:

Colour	Shop A	Shop B	**Total**
Red	R_A	R_B	R
Blue	B_A	B_B	B
Green	G_A	G_B	G
Total	n_A	n_B	n

This *two-way* contingency table allows us to ask questions like:

> *"Are colours distributed differently across shops?"*

Later in the course we will see this leads to the **chi-squared test of independence**.

2.3 Statistical Models

Whenever we perform statistical inference, we must specify a **model**.

In Section 2.1 we wrote

$$X_1, \dots, X_n \overset{\text{iid}}{\sim} \text{Distribution}(\theta)$$

using Distribution(θ) as a placeholder for "some distribution indexed by θ".

We now make this precise:

> A **statistical model** is a mathematical description of how we think the data are generated.
> Concretely, it is a **family of possible distributions**, one for each parameter value in a parameter space Θ.

Example 2.2 (Exponential model)**.** Suppose we are interested in modelling **waiting times** until some event occurs (e.g. the time between customer arrivals at a service desk).

A natural assumption is that the data are i.i.d. **?@sec-exp**:

$$X_1, \dots, X_n \overset{\text{iid}}{\sim} \text{Exponential}(\lambda).$$

- **Model:** each X_i is i.i.d. exponential with rate λ
- **Parameter:** $\theta = \lambda$
- **Parameter space**[3] **:** $\Theta = \{\lambda \mid \lambda > 0\} = (0, \infty)$

The exponential model often arises in *Poisson processes*, where events occur independently at a constant rate (e.g. radioactive decay, arrivals at a bus stop).

[3]**Set notation.** There are several equivalent ways to write sets:

- **Set-builder notation:** $\{\lambda \mid \lambda > 0\}$
- **Interval notation:** $(0, \infty)$
- **Named set:** $\mathbb{R}^+$

All three describe the same set. In this course, we recommend using **set-builder notation** when writing statistical models and their parameter spaces.

Example 2.3 (Normal model)**.** In Section 2.1.1 we assumed the data were i.i.d. Normal with unknown mean and variance:

$$X_1, \dots, X_n \overset{\text{iid}}{\sim} \mathcal{N}(\mu, \sigma^2).$$

- **Model:** each X_i is i.i.d. Normal with mean μ and variance σ^2
- **Parameters:** $\theta = (\mu, \sigma^2)$
- **Parameter space**[4] **:** $\Theta = \{(\mu, \sigma^2) \mid \mu \in \mathbb{R}, \sigma^2 > 0\} = \mathbb{R} \times (0, \infty)$

This model is widely used, e.g. for measurement error or traits like human height.

Common Misconceptions

- **Parameter $\neq$ statistic.**
  A *parameter* θ describes a feature of the population (e.g. p, μ, σ^2).
  A *statistic* (e.g. $\bar{X}$, a sample proportion) is calculated from the sample and used to learn about θ.

- **"Random sample" is not just "whatever data we found".**
  A random sample means data collected in a way that makes the i.i.d. assumption reasonable (e.g. independent trials, simple random sampling).

  > Example: If your population is *all humans on the planet*, then sampling only your family would not be random and could bias inference.

- **Models are approximations.**

[4]**Set notation (multiple parameters).** When a model has several parameters, it is best to use **set-builder notation** and collect the parameters into a vector

$$\underline{\theta} = (\theta_1, \theta_2, \dots, \theta_p).$$

The parameter space can then be written as

$$\Theta = \{(\theta_1, \dots, \theta_p) \mid \theta_1 \in S_1, \ \dots, \ \theta_p \in S_p\}.$$

Here each S_j is the set of allowed values for parameter θ_j. To construct Θ, simply **read off the constraints** on each parameter from the definition of the distribution (see the **?@sec-formula-sheet**).

For example, in the Normal model:

- $\mu \in \mathbb{R}$ (any real number),
- $\sigma^2 \in (0, \infty)$ (variance must be positive),

so the parameter space is

$$\Theta = \mathbb{R} \times (0, \infty).$$

This is called a **Cartesian product**: we form the parameter space by taking all possible pairs (μ, σ^2) where the first component is from $\mathbb{R}$ and the second from $(0, \infty)$.
Intuitively, it's just "all combinations" of the allowed values for each parameter.

> In practice you should treat models as *useful simplifications*, not as exact descriptions of reality. It's important to choose them sensibly, and be prepared to adjust if they don't fit the data well.

Example 2.4 (From words to a model)**.** A questionnaire records whether each respondent would recommend a service ("yes" or "no"). Write down a simple model and identify the parameter.

Let $X_i = 1$ if respondent i says "yes", and 0 otherwise.

Assume $X_1, \ldots, X_n \overset{\text{iid}}{\sim} \text{Bernoulli}(p)$.

- **Model:** $\mathcal{P} = \{\text{Bernoulli}(p) : 0 < p < 1\}$
- **Parameter:** $\theta = p$
- **Parameter space:** $\Theta = (0, 1)$

Here p is the long-run proportion of "yes" responses.

Example 2.5 (A slightly richer example)**.** A bag contains sweets of three colours $\{\text{red}, \text{blue}, \text{green}\}$. You draw n sweets *with replacement* and record counts (R, B, G). Propose a model and identify the parameters.

Each *draw* is an independent categorical outcome with probabilities (p_R, p_B, p_G) for red/blue/green. After n draws, we observe the *count vector* (R, B, G).

A natural model is the **?@sec-multinomial** distribution:

- **Model:** $(R, B, G) \sim \text{Multinomial}(n; p_R, p_B, p_G)$ with $p_R + p_B + p_G = 1$
  (equivalently, $X_1, \ldots, X_n \overset{iid}{\sim} \text{Categorical}(p_R, p_B, p_G)$ and (R, B, G) are the totals).
- **Parameters:** $\theta = (p_R, p_B, p_G)$
- **Parameter space:** the 2-simplex $\Theta = \{(p_R, p_B, p_G) \in [0, 1]^3 \mid p_R + p_B + p_G = 1\}$

If you only had two colours, the multinomial distribution would become the **?@sec-binom**.

Example 2.6 (Machine learning example)**.** In image classification, each "training" example is a pair (X_i, Y_i) where X_i is an image and $Y_i \in \{0, 1\}$. A generic probabilistic model specifies

$$\Pr_\theta(Y = 1 \mid X = x) = g_\theta(x),$$

for some family of functions g_θ (e.g. a *neural network*).

What are the *sample*, *population*, *model*, and *parameter*?

- **Sample:** the dataset $\{(X_i, Y_i)\}_{i=1}^n$.
- **Population:** the (conceptual) set of all image–label pairs of interest; equivalently, the joint distribution of (X, Y) in the real world we care about.
- **Model:** we specify the conditional distribution of Y given X,

  $$Y \mid X \sim \text{Bernoulli}(g_\theta(X)),$$

  i.e.

  $$\Pr_\theta(Y = 1 \mid X = x) = g_\theta(x).$$
- **Parameter:** θ indexes the function g_θ (e.g. regression coefficients or neural-network weights).
- **Parameter space:** $\Theta \subseteq \mathbb{R}^d$ (dimension d depends on the chosen model class).

Common misconception: confusing the *training sample* (finite) with the *population* (conceptual/infinite). The goal of inference is to use the sample to learn θ so that the model generalises to the population.

> ⚠ Non-Examinable Content: Specifying Statistical Models
>
> In this course we concentrate on **how to carry out statistical inference once a model has been chosen**.
> In all assessed questions the **statistical model will be provided**. Your task will be to perform inference by working with the parameter(s), using methods such as:
>
> - Parameter Estimation 4
> - Interval Estimation 7
> - Hypothesis Testing 9

The (non-examinable) **formal** definition of a statistical model is the following:

Definition 2.1 (Statistical Model (non-examinable))**.** A **statistical model**[5] is a *family of probability distributions* labelled by a parameter value:

$$\mathcal{P} = \{P_\theta \mid \theta \in \Theta\},$$

where:

[5]**Note on Statistical models.** Strictly speaking, the model refers to the joint distribution of the full dataset. If $X_1, \dots, X_n$ are i.i.d. from P_θ, then the model is

$$\mathcal{P} = \{P_\theta^n \mid \theta \in \Theta\}.$$

But for intuition, we usually just describe the distribution of a single observation, $X_i \sim P_\theta$, and keep in mind that the whole dataset follows from this.

- P_θ is one possible distribution the data might follow,
- θ is the **parameter**,
- Θ is the **parameter space** (the set of all possible parameter values).

3 Types of Inference

Statistical inference is about using a **sample** to learn about a **population**. There are two fundamentally different philosophies of inference, and several types of inferential tasks that appear in both.

3.1 Philosophies of Inference

3.1.1 Frequentist Inference

Frequentist inference is based on the idea of **long-run frequency**.

- The **data** are considered random, as they represent one possible outcome of repeated sampling from the population.
- The **parameter** θ is fixed but unknown.
- Inference is about constructing procedures (estimators, tests, intervals) that perform well under repeated sampling.

Preview examples (we will return to these later): the sample mean $\hat{\mu}$ as an estimator of μ, a t-test for comparing means, or a 95% confidence interval.

3.1.2 Bayesian Inference

Bayesian inference is based on **subjective probability as degree of belief**.

- The **data** are fixed once observed.
- The **parameter** θ is treated as a random variable, with uncertainty described by a **prior distribution** $\pi(\theta)$.
- Inference updates this prior in light of the data, giving a **posterior distribution** $\pi(\theta \mid \text{data})$.

Preview examples (we will return to these later): a posterior mean as an estimator of μ, a 95% credible interval, or comparing hypotheses via posterior probabilities.

3.1.3 Key Differences

Perspective	Frequentist Inference	Bayesian Inference
Parameter	Fixed but unknown	Random variable with a prior
Sample	Random (from repeated sampling)	Fixed (once observed)
Probability	Long-run frequency	Degree of belief
Goal	Construct estimators, tests, CIs	Update beliefs via posterior
Prior Knowledge	Ignored or minimal (e.g. non-informative)	Explicitly modelled via prior distribution
Output	Estimators, confidence intervals, p-values	Posterior distribution, credible intervals

3.2 Inference Tasks

Regardless of philosophy, the core tasks of inference are similar. Suppose we want to infer a parameter θ from data:

1. **Point Estimation** Give a single "best guess" $\hat{\theta}$ for the parameter.

   - Frequentist: choose an estimator with good properties (e.g. unbiasedness).
   - Bayesian: often the posterior mean, median, or mode.

2. **Interval Estimation** Quantify uncertainty in θ by reporting a range.

   - Frequentist: **confidence interval**

     $$L \leq \theta \leq U$$

     with a given long-run coverage probability.
   - Bayesian: **credible interval**, the posterior probability that θ lies in $[L, U]$ is, say, 95%.

3. **Hypothesis Testing** Ask a **yes/no question** about the parameter θ. This is done by formalising a claim (the *null hypothesis*) such as

   $$H_0 : \theta \in \Theta_0,$$

   and then using the data to judge whether this claim is **compatible with the evidence**.

   - **Frequentist approach:** base the decision on a **test statistic** and its distribution under H_0, leading to a p**-value** or a comparison with a **significance level**.

- **Bayesian approach:** compare hypotheses using **posterior probabilities** or **Bayes factors**, quantifying how much the data shift our beliefs.

4. **Prediction** Use data to predict future or unobserved outcomes.

   - Frequentist: predictive distributions based on sampling models.
   - Bayesian: posterior predictive distribution, integrating over parameter uncertainty.

Part II

Part II — Point Estimation

4 Frequentist Point Estimation

```
bodyFont = getComputedStyle(document.body).fontFamily
```

A **statistic** is any function of the sample

$$T = T(X_1, \dots, X_n).$$

Because it is built from the random variables $X_1, \dots, X_n$, the statistic T itself is a **random quantity**: its value varies from sample to sample. Once the data have been observed $(x_1, \dots, x_n)$, plugging them into the same formula produces a single **non-random number**, which we denote by t_{obs} (or simply t).

When we pick a statistic for the specific purpose of approximating a parameter θ, we call it an **estimator**. Its realised value is the **estimate**.

> **! Important 5: Notation: Random vs. Non-Random statistics**
>
> - **Upper-case** letters $(T, \bar{X}, S^2)$ denote random statistics.
> - **Lower-case** letters with a subscript "obs" $(t_{\text{obs}}, \bar{x}, s^2)$ denote the observed values of those statistics.

4.1 Estimators vs. Estimates

Concept	Notation	Description
Estimator	$\hat{\theta} = \hat{\theta}(X_1, \dots, X_n)$	A **function** of the data. A rule for producing an estimate.
Estimate	$\hat{\theta}_{\text{obs}} = \hat{\theta}(x_1, \dots, x_n)$	The **numerical value** you get after plugging in your data.

An estimator is a random variable; an estimate is its realised value.

4.1.1 Sample Mean and Variance

Two of the most important estimators are the sample mean and sample variance.

Definition 4.1 (Sample Mean)**.** The sample mean of a **random sample** $X_1, \dots, X_n$ is

$$\bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i.$$

For observed data $x_1, \dots, x_n$, the corresponding **estimate** is

$$\bar{x} \; = \; \frac{1}{n}\sum_{i=1}^{n} x_i,$$

and we will frequently use the shorthand

$$\sum_{i=1}^{n} x_i \; = \; n\,\bar{x}.$$

Definition 4.2 (Sample Variance)**.** For a **random sample** $X_1, \dots, X_n$, the (unbiased) sample variance is

$$S^2 \; = \; \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X})^2.$$

For observed data $x_1, \dots, x_n$, the corresponding estimate is

$$s^2 \; = \; \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2.$$

Remark 4.1 (Why $n-1$?)*.* The factor $\frac{1}{n-1}$ in the definition of S^2 is called **Bessel's correction**.

It accounts for the fact that the deviations $(X_i - \bar{X})$ satisfy the constraint $\sum_{i=1}^{n}(X_i - \bar{X}) = 0$, leaving only $n-1$ degrees of freedom.

Under an i.i.d. model with finite variance σ^2, this choice makes S^2 *unbiased*:

$$\mathrm{E}[S^2] = \sigma^2.$$

See Example 4.3 for more details.

The following representation of the **sample variance** is convenient in *theoretical derivations* (e.g. the unbiasedness of S^2) and in *practice*, where precomputed totals $\sum_i X_i$ and $\sum_i X_i^2$ allow immediate evaluation of S^2.

Proposition 4.1 (Alternative Form of the Sample Variance)**.** *For a **random sample** $X_1, \dots, X_n$ with sample mean $\bar{X}$, the sample variance 4.2 admits the* computational identity

$$S^2 \;=\; \frac{1}{n-1}\left(\sum_{i=1}^{n} X_i^2 \;-\; n\,\bar{X}^{\,2}\right).$$

This is often convenient in deriving properties of the sample variance and, in practice, when sums of squares are already available.

Proof. Starting from Definition 4.2 and expanding the quadratic, we have

$$S^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X})^2,$$

expand the square and sum term-by-term:

$$\sum_{i=1}^{n}(X_i - \bar{X})^2 = \sum_{i=1}^{n}(X_i^2 - 2\bar{X}\,X_i + \bar{X}^{\,2}) = \sum_{i=1}^{n} X_i^2 \;-\; 2\bar{X}\sum_{i=1}^{n} X_i \;+\; \sum_{i=1}^{n}\bar{X}^{\,2}.$$

Since $\sum_{i=1}^{n} X_i = n\bar{X}$ and $\sum_{i=1}^{n}\bar{X}^{\,2} = n\bar{X}^{\,2}$, this becomes

$$\sum_{i=1}^{n}(X_i - \bar{X})^2 = \sum_{i=1}^{n} X_i^2 \;-\; 2n\bar{X}^{\,2} \;+\; n\bar{X}^{\,2} = \sum_{i=1}^{n} X_i^2 \;-\; n\bar{X}^{\,2}.$$

Dividing by $n-1$ yields the stated identity.

□

4.1.2 The Randomness of Estimators

An **estimator** is a rule that turns a random sample into a number:

$$\hat{\theta} \;=\; \hat{\theta}(X_1, \dots, X_n).$$

Since $X_1, \dots, X_n$ are random variables, the output $\hat{\theta}$ is **also a random variable**. Its value changes from sample to sample; only after we observe data $(x_1, \dots, x_n)$ do we get the **estimate** $\hat{\theta}_{\text{obs}} = \hat{\theta}(x_1, \dots, x_n)$, a fixed number.

Definition 4.3 (Sampling Distribution)**.** The **sampling distribution** of an estimator $\hat{\theta} = \hat{\theta}(X_1, \dots, X_n)$ is the probability distribution of $\hat{\theta}$ induced by the joint distribution of $(X_1, \dots, X_n)$.

The properties 4.2 of an estimator, such as *bias* and *standard error*, are defined via this distribution.

In general, however, deriving an exact sampling distribution is **hard**, because:

1. It depends on the chosen **statistical model** for the data (the distribution of X_i).

2. The estimator $\hat{\theta}$ may be a **complicated/nonlinear function** of the sample.

In a few special cases, however, we can derive the full sampling distribution of the estimator.

Proposition 4.2 (Normal Random Sample: Sampling Distribution of $\bar{X}$)**.** *If*

$$X_1, \dots, X_n \overset{iid}{\sim} \mathcal{N}(\mu, \sigma^2),$$

then

$$\bar{X} \;\sim\; \mathcal{N}\left(\mu,\; \frac{\sigma^2}{n}\right).$$

Proof. Averages are linear combinations: $\bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i$. By the **?@sec-normal-properties**, a linear combination of independent Normal variables is Normal. Moreover,

$$\mathrm{E}[\bar{X}] = \tfrac{1}{n}\sum \mathrm{E}[X_i] = \mu, \qquad \mathrm{Var}(\bar{X}) = \tfrac{1}{n^2}\sum \mathrm{Var}(X_i) = \frac{\sigma^2}{n}.$$

So $\bar{X} \sim \mathcal{N}(\mu, \sigma^2/n)$.

□

Proposition 4.3 (Normal Random Sample: Sampling Distribution of S^2)**.** *If*

$$X_1, \dots, X_n \overset{iid}{\sim} \mathcal{N}(\mu, \sigma^2),$$

then

1. *The scaled sample variance follows a chi-squared distribution:*

$$\frac{(n-1)S^2}{\sigma^2} \;\sim\; \chi^2_{n-1},$$

2. *The sample mean $\bar{X}$ and the sample variance S^2 are independent random variables.*

Proof of **??**. The proof relies on the following *property* of the χ^2_p distribution: For p i.i.d. standard normal random variables

$$Z_1, \dots, Z_p \overset{\text{iid}}{\sim} \mathcal{N}(0,1)$$

their *squared* sum is χ^2_p distributed:

$$\sum_{i=1}^{p} Z_i^2 \sim \chi^2_p.$$

The key idea in the following proof to express $\dfrac{(n-1)S^2}{\sigma^2}$ as a sum of $n-1$ i.i.d. standard normal random variables.

Step 1 (Standardise and express S^2 as a residual sum of squares). Define the standardised variables

$$Y_i \;=\; \frac{X_i - \mu}{\sigma} \quad (i = 1, \dots, n), \qquad \text{so } Y_1, \dots, Y_n \overset{\text{iid}}{\sim} \mathcal{N}(0,1).$$

Let $\bar{Y} = \frac{1}{n}\sum_{i=1}^{n} Y_i = \dfrac{\bar{X} - \mu}{\sigma}$. Then

$$\sum_{i=1}^{n} (X_i - \bar{X})^2 = \sigma^2 \sum_{i=1}^{n} \big(Y_i - \bar{Y}\big)^2,$$

so

$$\frac{(n-1)S^2}{\sigma^2} = \sum_{i=1}^{n} \big(Y_i - \bar{Y}\big)^2.$$

Thus $\dfrac{(n-1)S^2}{\sigma^2}$ is the **sum of squares of residuals** after projecting Y onto its mean.

The remaining steps are to manipulate $\sum_{i=1}^{n} \big(Y_i - \bar{Y}\big)^2$ as a sum of $n-1$ i.i.d. squared standard Normal random variables.

Step 2 (Define $n-1$ "difference of mean vs next observation" variables). For $k = 1, \dots, n-1$, let $\bar{Y}_k = \frac{1}{k}\sum_{i=1}^{k} Y_i$ and define

$$Z_k \;=\; \sqrt{\frac{k}{k+1}}\, (\bar{Y}_k - Y_{k+1}).$$

Each Z_k is a linear combination of independent $\mathcal{N}(0,1)$, hence Normal. Also,

$$
\mathrm{E}[Z_k]=0, \qquad \mathrm{Var}(Z_k)=\frac{k}{k+1}\Big(\mathrm{Var}(\bar{Y}_k)+\mathrm{Var}(Y_{k+1})\Big)=\frac{k}{k+1}\Big(\frac{1}{k}+1\Big)=1,
$$

so $Z_k \sim \mathcal{N}(0,1)$.

Step 3 (The Z_k are independent).

Fix $1 \le j < k \le n-1$. Write $Z_j = a_j(\bar{Y}_j - Y_{j+1})$ with $a_j = \sqrt{j/(j+1)}$, and

$$
Z_k = a_k(\bar{Y}_k - Y_{k+1}), \qquad a_k = \sqrt{\frac{k}{k+1}}, \qquad \bar{Y}_k = \frac{1}{k}\sum_{i=1}^{k} Y_i.
$$

View each Z as a linear combination of the Y_i. The coefficients are:

- for Z_j: $a_j \cdot \frac{1}{j}$ on $Y_1, \dots, Y_j$; $-a_j$ on Y_{j+1}; 0 otherwise;
- for Z_k: $a_k \cdot \frac{1}{k}$ on $Y_1, \dots, Y_k$; $-a_k$ on Y_{k+1}; 0 otherwise.

Using independence and $\mathrm{Var}(Y_i) = 1$,

$$
\mathrm{Cov}(Z_j, Z_k) = \sum_{i=1}^{n} (\text{coeff of } Y_i \text{ in } Z_j) \times (\text{coeff of } Y_i \text{ in } Z_k).
$$

Only $i = 1, \dots, j$ and $i = j+1$ overlap. Hence

$$
\mathrm{Cov}(Z_j, Z_k) = \sum_{i=1}^{j}\Big(a_j\frac{1}{j}\Big)\Big(a_k\frac{1}{k}\Big) \; + \; \Big(-a_j\Big)\Big(a_k\frac{1}{k}\Big) = a_j a_k\Big(\frac{j}{jk} - \frac{1}{k}\Big) = 0.
$$

Thus the Z_k are pairwise uncorrelated. Since they are all Normal (linear combos of independent Normals), *uncorrelated* $\Rightarrow$ *independent*. So $Z_1, \dots, Z_{n-1}$ are **independent** $\mathcal{N}(0,1)$.

Step 4 ($\frac{(n-1)S^2}{\sigma^2} = \sum_{k=1}^{n-1} Z_k^2$).

Define $S_k = \sum_{i=1}^{k}(Y_i - \bar{Y}_k)^2$ (so $S_n = \sum_{i=1}^{n}(Y_i - \bar{Y})^2$). Write $Y_i - \bar{Y}_{k+1} = (Y_i - \bar{Y}_k) + (\bar{Y}_k - \bar{Y}_{k+1})$, use $\sum_{i=1}^{k}(Y_i - \bar{Y}_k) = 0$ to kill the cross-term, and note $\bar{Y}_{k+1} - \bar{Y}_k = \frac{Y_{k+1} - \bar{Y}_k}{k+1}$.

Therefore, we have the recursive formula for S_k:

$$
S_{k+1} \;=\; S_k \;+\; \frac{k}{k+1}\,(\bar{Y}_k - Y_{k+1})^2 \qquad (k = 1, \dots, n-1).
$$

By induction,

$$
S_n = \sum_{k=1}^{n-1} \frac{k}{k+1}\,(\bar{Y}_k - Y_{k+1})^2 = \sum_{k=1}^{n-1} Z_k^2.
$$

Conclusion From the previous step, we have shown

$$\frac{(n-1)S^2}{\sigma^2} = \sum_{i=1}^{n}(Y_i - \bar{Y})^2 = \sum_{k=1}^{n-1} Z_k^2,$$

a sum of $n-1$ **independent standard Normal squares**, hence

$$\boxed{\frac{(n-1)S^2}{\sigma^2} \sim \chi^2_{n-1}}.$$

□

4.2 Properties of Estimators

When we use a statistic to estimate a parameter, we want to judge how *good* an estimator is.

Three fundamental properties guide this assessment:

1. **Bias:** Does the estimator hit the right target on average?
2. **Variance/Standard Error:** How much does it fluctuate from sample to sample?
3. **Mean Squared Error (MSE):** A combined measure that balances both bias and variance.

In this section we first look at bias 4.4, variance 4.5, and standard error 4.6, then bring them together through the MSE 4.8.

Remark 4.2 (Properties via Moments). In the previous section we introduced the sampling distribution 4.3 of an estimator.

The key idea for what follows is that we don't need the entire distribution to study how *good* an estimator is. The main properties — bias, variance, standard error, and MSE — can all be expressed in terms of expectations (i.e. moments 1.12).

So in practice, we will usually only need to work with the first and second moments of $\hat{\theta}$, rather than its full sampling distribution.

4.2.1 Bias, Variance, and Standard Error

Definition 4.4 (Bias of an Estimator)**.** The **bias** of an estimator $\hat{\theta}$ for a parameter θ is the difference between its expected value and the true parameter value:

$$\mathrm{Bias}(\hat{\theta}) = \mathrm{E}[\hat{\theta}] - \theta.$$

- If $\mathrm{Bias}(\hat{\theta}) = 0$, the estimator is called **unbiased**.
- Otherwise, it is **biased**: it systematically over- or under-estimates θ.

Example 4.1 (Bias of the Sample Mean)**.** Suppose $X_1, \dots, X_n \overset{\text{iid}}{\sim} \text{Distribution}(\mu, \sigma^2)$ with mean μ.
Show that the sample mean

$$\bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i$$

is an **unbiased estimator** of μ.

We compute the expectation of $\bar{X}$:

$$\mathrm{E}[\bar{X}] = \mathrm{E}\left[\frac{1}{n}\sum_{i=1}^{n} X_i\right] = \frac{1}{n}\sum_{i=1}^{n} \mathrm{E}[X_i].$$

Since each X_i has mean μ,

$$\mathrm{E}[\bar{X}] = \frac{1}{n}(n\mu) = \mu.$$

Therefore,

$$\mathrm{Bias}(\bar{X}) = \mathrm{E}[\bar{X}] - \mu = \mu - \mu = 0.$$

So the sample mean $\bar{X}$ is an **unbiased** estimator of μ.

Definition 4.5 (Variance of an Estimator)**.** The variance measures how much the estimator fluctuates around its expectation:

$$\mathrm{Var}(\hat{\theta}) = \mathrm{E}\big[(\hat{\theta} - \mathrm{E}[\hat{\theta}])^2\big].$$

A smaller variance means the estimator is more **concentrated** and less "noisy."

Example 4.2 (Variance of the Sample Mean)**.** Suppose $X_1, \ldots, X_n$ is a *random sample* such that $\mathrm{Var}(X_i) = \sigma^2$.

Show that

$$\mathrm{Var}(\bar{X}) = \frac{\sigma^2}{n}.$$

Since the data X_i are *independent* and have common variance σ^2, we have

$$\mathrm{Var}(\bar{X}) = \frac{1}{n^2}\mathrm{Var}\left[\sum_{i=1}^{n} X_i\right] = \frac{1}{n^2}\sum_{i=1}^{n}\mathrm{Var}(X_i) = \frac{\sigma^2}{n}.$$

So as n increases, $\bar{X}$ becomes more tightly clustered around μ.

Example 4.3 (Bias of the Sample Variance)**.** Suppose $X_1, \ldots, X_n$ is a *random sample* such that

$$\mathrm{E}[X_i] = \mu \quad \text{and} \quad \mathrm{Var}(X_i) = \sigma^2 < \infty.$$

Consider the class of estimators

$$S_c^2 \;=\; \frac{1}{c}\sum_{i=1}^{n}(X_i - \bar{X})^2.$$

Find c such that S_c^2 is **unbiased** for σ^2.

Note: In the following, we use the fact that, for any random variable Y,

$$\mathrm{Var}(Y) = \mathrm{E}[Y^2] - \mathrm{E}[Y]^2 \Rightarrow \mathrm{E}[Y^2] = \mathrm{Var}(Y) + \mathrm{E}[Y]^2.$$

We seek c such that $\mathrm{E}[S_c^2] = \sigma^2$.

Using the computational identity 4.1,

$$\sum_{i=1}^{n}(X_i - \bar{X})^2 \;=\; \sum_{i=1}^{n} X_i^2 \;-\; n\,\bar{X}^{\,2}.$$

Hence, taking expectations termwise,

$$\mathrm{E}[X_i^2] = \sigma^2 + \mu^2, \qquad \mathrm{E}[\bar{X}^{\,2}] = \mathrm{Var}(\bar{X}) + \{\mathrm{E}[\bar{X}]\}^2 = \frac{\sigma^2}{n} + \mu^2.$$

Therefore

$$\mathrm{E}\left[\sum_{i=1}^{n}(X_i - \bar{X})^2\right] = n(\sigma^2 + \mu^2) - n\left(\frac{\sigma^2}{n} + \mu^2\right) = (n-1)\sigma^2.$$

It follows that

$$\mathrm{E}[S_c^2] = \frac{(n-1)\sigma^2}{c}$$

Unbiasedness $\mathrm{E}[S_c^2] = \sigma^2$ gives $c = n - 1$.

Conclusion. The unbiased estimator is

$$S^2 \;=\; \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X})^2.$$

In the next visualisation we look at the *bias* of the *naive sample variance*:

$$S_n^2 = \frac{1}{n}\sum_{i=1}^{n}(X_i - \bar{X})^2.$$

From Example 4.3, the bias is

$$\mathrm{Bias}(S_n^2) = -\frac{\sigma^2}{n}.$$

This means S_n^2 **systematically underestimates** the true population variance σ^2.

Remark 4.3 (Notes on Example 4.3).

- **Distributional assumptions.** No Normality is required: the conclusion holds for i.i.d. observations with finite variance, $\mathrm{Var}(X_i) = \sigma^2 < \infty$.
- **MLE under Normality.** If the data are Normal, the maximum-likelihood estimator is

  $$\hat{\sigma}^2_{\mathrm{MLE}} = \frac{1}{n}\sum_{i=1}^{n}(X_i - \bar{X})^2,$$

  which is **biased downward**:

  $$\mathrm{E}[\hat{\sigma}^2_{\mathrm{MLE}}] = \frac{n-1}{n}\,\sigma^2 \quad (\text{bias} = -\sigma^2/n).$$

  See Example 5.12 for full details.

Definition 4.6 (Standard Error)**.** The **standard error** of an estimator $\hat{\theta}$ of θ, is just the *standard deviation* of the estimator:

$$\mathrm{S.E.}(\hat{\theta}) = \sqrt{\mathrm{Var}(\hat{\theta})}.$$

It gives a scale for how much we expect the estimator to vary from sample to sample.

4.2.2 Mean Squared Error (MSE)

So far we have seen that estimators can have **bias** and **variance**. Both are important, but it is often useful to combine them into a *single* overall measure of accuracy.

To get there, we start with the basic idea of **error**.

Definition 4.7 (Error)**.** The **error** of an estimator is the difference between the estimate and the true parameter:

$$\text{Error}(\hat{\theta}) = \hat{\theta} - \theta.$$

This captures exactly "how far off" our estimate is. But there are two problems:

1. The error is a **random variable** (it depends on the sample we happened to draw).
2. The error can be **positive or negative** — so if we just averaged it across many samples, the positives and negatives might cancel out, misleading us into thinking the estimator is more accurate than it really is.

To fix this, we square the error (so it's always positive) and then take its expectation (so we measure its *typical size* across many samples).

Definition 4.8 (Mean Squared Error (MSE))**.** The **MSE** of an estimator $\hat{\theta}$ of θ is

$$\text{MSE}(\hat{\theta}) = \text{E}\big[(\hat{\theta} - \theta)^2\big].$$

This measures the *average squared distance* between the estimator and the true parameter value.

Remark 4.4 (Why Squared Error?)*.* The MSE is built to solve the two problems with using the error 4.7:

- **Squaring** makes all errors positive (so no cancellation).
- **Expectation** removes randomness and shows the long-run behaviour of the estimator.

But squaring isn't the only possible choice. For example, we could use absolute error and define the **Mean Absolute Error (MAE):**

$$\text{MAE}(\hat{\theta}) = \text{E}\big[|\hat{\theta} - \theta|\big].$$

This is also a sensible measure. In fact, in practice people sometimes prefer MAE because it is less sensitive to occasional very large errors.

So why is **MSE** so common in theory?

1. It penalises **large deviations more strongly**, which can be useful.
2. It is mathematically **easier to work with**. Many nice algebraic decompositions (like the bias-variance decomposition 4.4) only hold neatly for squared error.

Directly using the definition 4.8 of MSE can be awkward. A more useful identity decomposes MSE into **bias** and **variance**:

Proposition 4.4 (MSE: Bias–Variance Decomposition)**.** *For any estimator $\hat{\theta}$,*

$$\mathrm{MSE}(\hat{\theta}) = \mathrm{Bias}(\hat{\theta})^2 + \mathrm{Var}(\hat{\theta}).$$

Proof. Start from the definition:

$$\mathrm{MSE}(\hat{\theta}) = \mathrm{E}\big[(\hat{\theta} - \theta)^2\big].$$

Step 1 (add–and–subtract the mean). Let $m = \mathrm{E}[\hat{\theta}]$. Then

$$\hat{\theta} - \theta = (\hat{\theta} - m) + (m - \theta).$$

Step 2 (expand the square).

$$(\hat{\theta} - \theta)^2 = \left(\hat{\theta} - m\right)^2 + 2(\hat{\theta} - m)(m - \theta) + \left(m - \theta\right)^2.$$

Step 3 (take expectations; use linearity).

$$\mathrm{E}\big[(\hat{\theta} - \theta)^2\big] = \underbrace{\mathrm{E}\big[(\hat{\theta} - m)^2\big]}_{\mathrm{Var}(\hat{\theta})} + 2\,(m - \theta)\,\underbrace{\mathrm{E}\big[\hat{\theta} - m\big]}_{=0} + \underbrace{(m - \theta)^2}_{\mathrm{Bias}(\hat{\theta})^2}.$$

The middle term vanishes because $\mathrm{E}[\hat{\theta} - m] = \mathrm{E}[\hat{\theta}] - m = 0$. Hence

$$\mathrm{MSE}(\hat{\theta}) = \mathrm{Var}(\hat{\theta}) + \left(\mathrm{E}[\hat{\theta}] - \theta\right)^2 = \mathrm{Var}(\hat{\theta}) + \mathrm{Bias}(\hat{\theta})^2.$$

□

Remark 4.5 (Why this helps)*.*

- If $\hat{\theta}$ is **unbiased**, then $\mathrm{MSE}(\hat{\theta}) = \mathrm{Var}(\hat{\theta})$.
- In general there is a **bias–variance trade-off**: a small intentional bias can reduce variance enough to lower MSE (common in shrinkage/regularised estimators).

Example 4.4 (MSE of Sample Mean)**.** Suppose we have a random sample

$$X_1, \ldots, X_n \overset{\text{iid}}{\sim} \text{Distribution}(\underline{\theta})$$

and we wish to estimate its *population mean* μ with $\hat{\mu} = c\bar{X}$, where c is a constant.

Compute the **MSE** of this estimator.

- Bias:

$$\text{Bias}(\hat{\mu}) = (c-1)\mu.$$

- Variance:

$$\text{Var}(\hat{\mu}) = c^2 \frac{\sigma^2}{n}.$$

- MSE:

$$\text{MSE}(\hat{\mu}) = \frac{c^2\sigma^2}{n} + (c-1)^2\mu^2.$$

This shows how a biased estimator can still have low MSE if variance is reduced.

Example 4.5 (MSE of Three Estimators)**.** Suppose that a random variable X has expected value $\text{E}[X] = 2\theta$ and variance $\text{Var}(X) = \theta^2$, where $\theta > 0$ is an unknown parameter.

Another random variable Y, independent of X, has $\text{E}[Y] = 3\theta$ and $\text{Var}(Y) = 2\theta^2$.

Consider the following three estimators of θ:

1. $\hat{\theta}_1 = X/2$,
2. $\hat{\theta}_2 = Y/3$,
3. $\hat{\theta}_3 = (3X + 2Y)/12$.

Compute the **MSE** of each estimator, and recommend the best one.

We use the rules of expectation and variance for linear functions of random variables (independence of X and Y is needed for the variance of $\hat{\theta}_3$).

Estimator (i): $\hat{\theta}_1 = X/2$

- Mean:

$$\mathrm{E}[\hat{\theta}_1] = \frac{\mathrm{E}[X]}{2} = \frac{2\theta}{2} = \theta.$$

- Variance:

$$\mathrm{Var}(\hat{\theta}_1) = \frac{\mathrm{Var}(X)}{4} = \frac{\theta^2}{4}.$$

- Since unbiased, $\mathrm{MSE}(\hat{\theta}_1) = \frac{\theta^2}{4}$.

Estimator (ii): $\hat{\theta}_2 = Y/3$

- Mean:

$$\mathrm{E}[\hat{\theta}_2] = \frac{\mathrm{E}[Y]}{3} = \frac{3\theta}{3} = \theta.$$

- Variance:

$$\mathrm{Var}(\hat{\theta}_2) = \frac{\mathrm{Var}(Y)}{9} = \frac{2\theta^2}{9}.$$

- Since unbiased, $\mathrm{MSE}(\hat{\theta}_2) = \frac{2\theta^2}{9}$.

Estimator (iii): $\hat{\theta}_3 = (3X + 2Y)/12$

- Mean:

$$\mathrm{E}[\hat{\theta}_3] = \frac{3\,\mathrm{E}[X] + 2\,\mathrm{E}[Y]}{12} = \frac{3(2\theta) + 2(3\theta)}{12} = \theta.$$

- Variance (using independence):

$$\mathrm{Var}(\hat{\theta}_3) = \frac{1}{144}\Big(9\,\mathrm{Var}(X) + 4\,\mathrm{Var}(Y)\Big) = \frac{1}{144}(9\theta^2 + 8\theta^2) = \frac{17}{144}\theta^2.$$

- Since unbiased, $\mathrm{MSE}(\hat{\theta}_3) = \frac{17}{144}\theta^2$.

Comparison:
- $\text{MSE}(\hat{\theta}_1) = 0.25\theta^2$
- $\text{MSE}(\hat{\theta}_2) \approx 0.222\theta^2$
- $\text{MSE}(\hat{\theta}_3) \approx 0.118\theta^2$

Thus, $\hat{\theta}_3$ has the smallest MSE and is the best choice.

4.3 Central Limit Theorem (CLT)

Theorem 4.1 (Central Limit Theorem (CLT))**.** *Let $X_1, X_2, \ldots$ be an infinite sequence of **independent and identically distributed** random variables with finite mean $\mu = \mathrm{E}[X_1]$ and finite variance $\sigma^2 = \mathrm{Var}(X_1) < \infty$.*
Define the sample mean

$$\bar{X}_n \;=\; \frac{1}{n}\sum_{i=1}^{n} X_i.$$

*Then, **regardless of the original distribution of the** X_i,*

$$\frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} \;\xrightarrow{d}\; \mathcal{N}(0,1) \quad \textit{as } n \to \infty.$$

Proof of **??***.* We do not give a proof here. An outline is provided in separate practice exercises (not examinable).

Note. The notation $\xrightarrow{d}$ means **convergence in distribution**: the distribution of the random variable on the left approaches the distribution on the right as n grows.

□

The CLT is remarkable: it tells us that **whatever the original distribution** of the data, the **sample mean** is approximately Normal for sufficiently large n, and the approximation **improves** as n increases.

In practice, for a *reasonable* sample size we may write

$$\bar{X}_n \overset{\text{approx.}}{\sim} \mathcal{N}\left(\mu,\ \frac{\sigma^2}{n}\right),$$

not only for Normal samples but for many non-Normal populations as well. What counts as "reasonable" depends on how skewed/heavy-tailed the original distribution is; often $n > 20$ suffices, and in many settings even smaller n will do.

Because $\bar{X}_n = \frac{1}{n}\sum_{i=1}^{n} X_i$, the CLT also yields an approximation for **sums**:

$$\sum_{i=1}^{n} X_i \overset{\text{approx.}}{\sim} \mathcal{N}(n\mu,\ n\sigma^2)\,.$$

Common misconceptions

The CLT does **not** say the *data* become Normal; it says the **distribution of the sample mean** becomes approximately Normal as n grows.

4.4 Asymptotic Properties of Estimators

Important 6: Terminology: *Asymptotic* in Statistics

An **asymptotic property** of an estimator is a property that emerges in the limit as the number of observations $n \to \infty$.
In other words, it describes the *large-sample behaviour* of an estimator.

So far, we have judged estimators by properties that hold for a *fixed* sample size n, such as bias 4.4, variance 4.5, and MSE 4.8.

It is also useful to consider how estimators behave when the sample size grows very large. These are known as the **asymptotic (large-sample) properties** of estimators.

Formally, suppose we have a random sample

$$X_1, X_2, \dots, X_n \overset{\text{iid}}{\sim} \text{Distribution}(\theta),$$

and a *sequence* of estimators of θ given by

$$\hat{\theta}_n = g(X_1, X_2, \dots, X_n).$$

We are interested in the properties of $\hat{\theta}_n$ as $n \to \infty$. Two fundamental criteria are *asymptotic unbiasedness* and *consistency*.

Definition 4.9 (Asymptotic Unbiasedness)**.** We say that $\hat{\theta}_n$ is **asymptotically unbiased** for θ if

$$\lim_{n\to\infty} \mathrm{E}[\hat{\theta}_n] = \theta.$$

If an estimator is unbiased for every n, then it is automatically asymptotically unbiased. However, some biased estimators may still "correct themselves" in the limit and become asymptotically unbiased.

Definition 4.10 (Consistency)**.** A sequence of estimator $\hat{\theta}_n$ is **consistent** for θ if

$$\hat{\theta}_n \xrightarrow{p} \theta \quad \text{as } n \to \infty,$$

that is, it converges in probability 1.17 to the true parameter value.

Intuitively, a consistent estimator will "home in" on the true parameter as the sample size grows.

Using Definition 4.10 directly to prove an estimator is consistent can be difficult. The following gives a **sufficient condition** that is *usually* easier to verify in practice.

Proposition 4.5 (Sufficient Condition for Consistency)**.** *An estimator $\hat{\theta}_n$ of θ is consistent **if** both the following hold:*

1. $\hat{\theta}_n$ *is* asymptotically unbiased*, i.e.*

$$\text{Bias}(\hat{\theta}_n) \;\to\; 0 \quad \textit{as } n \to \infty,$$

2. *Its variance shrinks to zero:*

$$\text{Var}(\hat{\theta}_n) \;\to\; 0 \quad \textit{as } n \to \infty.$$

Proof of **??**. Suppose $\hat{\theta}_n$ satisfies conditions (1) and (2).
For any $\varepsilon > 0$, by Chebyshev's inequality 1.1:

$$\Pr\left(|\hat{\theta}_n - \text{E}[\hat{\theta}_n]| \geq \varepsilon\right) \;\leq\; \frac{\text{Var}(\hat{\theta}_n)}{\varepsilon^2}.$$

As $n \to \infty$, the variance term tends to 0, so

$$\hat{\theta}_n - \text{E}[\hat{\theta}_n] \;\xrightarrow{p}\; 0.$$

Meanwhile, $\text{E}[\hat{\theta}_n] \to \theta$ by assumption of asymptotic unbiasedness.
Hence,

$$\hat{\theta}_n \;\xrightarrow{p}\; \theta,$$

which proves consistency.

□

Example 4.6 (Consistency of the Sample Mean)**.** Suppose $X_1, \dots, X_n$ are i.i.d. with mean μ and variance $\sigma^2 < \infty$.
Show that $\bar{X}_n$ is a **consistent estimator** of μ.

Recall that

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^{n} X_i.$$

- **Bias:**

  $$\mathrm{E}[\bar{X}_n] = \frac{1}{n} \sum_{i=1}^{n} \mathrm{E}[X_i] = \mu.$$

  Thus $\bar{X}_n$ is unbiased for all n, hence asymptotically unbiased.

- **Variance:**
  Since the X_i are independent,

  $$\mathrm{Var}(\bar{X}_n) = \frac{1}{n^2} \sum_{i=1}^{n} \mathrm{Var}(X_i) = \frac{\sigma^2}{n}.$$

  As $n \to \infty$, this variance tends to 0.

Therefore, $\bar{X}_n$ is both unbiased and has variance vanishing as n grows.
By Proposition 4.5, it is a **consistent** estimator of μ.

In some cases, applying Proposition 4.5 is *much harder* to do:

Proposition 4.6 (Consistency of the Sample Variance)**.** *Suppose $X_1, \dots, X_n$ are i.i.d. with mean μ and variance $\sigma^2 < \infty$.*

Then

$$S_n^2 \; = \; \frac{1}{n-1} \sum_{i=1}^{n} (X_i - \bar{X}_n)^2$$

*is a **consistent estimator** of σ^2.*

Proof of **??**. Let's first try to use the bias-variance approach of Proposition 4.5.

Bias. From Example 4.3, $\mathrm{Bias}(S_n^2) = 0$ for all n, i.e. $\mathrm{E}[S_n^2] = \sigma^2$.

Variance. Using Proposition 4.1,

$$S_n^2 \; = \; \frac{1}{n-1} \Big(\sum_{i=1}^{n} X_i^2 \; - \; n \, \bar{X}_n^2 \Big),$$

so

$$\mathrm{Var}(S_n^2) = \mathrm{E}[(S_n^2)^2] - \sigma^4 = \frac{1}{(n-1)^2}\Big\{\mathrm{E}\big[\big(\sum X_i^2 - n\bar{X}_n^2\big)^2\big]\Big\} - \sigma^4.$$

Expanding the square gives

$$\frac{1}{(n-1)^2}\left[\mathrm{E}\left(\sum_{i=1}^n X_i^2\right)^2 - 2n\,\mathrm{E}\left(\bar{X}_n^2\sum_{i=1}^n X_i^2\right) + n^2\,\mathrm{E}(\bar{X}_n^4)\right] - \sigma^4 \;=\; \cdots$$

This leads to **non-trivial** moment algebra (4th moments and many cross-terms). If one carries it through (assuming the 4th central moment $\mu_4 = \mathrm{E}[(X-\mu)^4]$ is finite), the result is

$$\mathrm{Var}(S_n^2) \;=\; \frac{1}{n}\left(\mu_4 - \frac{n-3}{n-1}\sigma^4\right),$$

so in particular $\mathrm{Var}(S_n^2) \le \frac{1}{n}(\mu_4 + \sigma^4) \to 0$.

A Different Approach.

Using Proposition 4.1 in **centred** form,

$$S_n^2 \;=\; \frac{n}{n-1}\left(\frac{1}{n}\sum_{i=1}^n (X_i-\mu)^2\right) \;-\; \frac{n}{n-1}\,(\bar{X}_n-\mu)^2.$$

By the Law of Large Numbers, $\frac{1}{n}\sum_{i=1}^n (X_i-\mu)^2 \xrightarrow{p} \mathrm{E}[(X-\mu)^2] = \sigma^2$ (finite **second** moment suffices), and $(\bar{X}_n-\mu)^2 \xrightarrow{p} 0$. Since $\frac{n}{n-1} \to 1$,

$$S_n^2 \xrightarrow{p} \sigma^2.$$

Thus S_n^2 is **consistent**.

□

Remark 4.6 (Consistency).

- Consistency requires convergence in probability 1.17, not necessarily unbiasedness at every finite n.
- In this course, you can always use the sufficient condition (Proposition 4.5) to prove consistency.

5 Likelihood

```
gap = 20
containerW = width
plotW = Math.round(Math.max(240, Math.min(600, (containerW - gap) / 2 - 15)))
plotH = Math.round(plotW * 0.8)

marginTop = 20
marginRight = 35
marginBottom = 35
marginLeft = 35
innerW = plotW - marginLeft - marginRight
innerH = plotH - marginTop - marginBottom

bodyFont = getComputedStyle(document.body).fontFamily
```

The **likelihood** is simply the **joint density (or probability mass function)** of the observed data:

$$f(\underline{x} \mid \theta), \quad \underline{x} = (x_1, \dots, x_n).$$

It plays a central role in both **frequentist** and **Bayesian** statistics.

Important 7: Notation Convention

- When **specifying the distribution of the data** in the frequentist setting, we write

$$X_i \sim \text{Distribution}(\theta),$$

to emphasise that θ is fixed (but unknown).

- When writing **densities**, we will always use

$$f(\underline{x} \mid \theta).$$

  - In the **frequentist** context: read this as *"the* **joint density** *of the data when the parameter takes the value θ."*
  - In the **Bayesian** context: it is literally the **conditional density** of the data given the **random** parameter θ.

- Later, in the **Bayesian** parts of the course, we will also use

$$X_i \mid \theta \sim \text{Distribution}(\theta),$$

where the bar $\mid \theta$ reflects its probabilistic meaning: *conditioning on random parameters*.

5.1 Likelihoods Under Independence and Dependence

The form of the likelihood depends crucially on whether our observations are independent and identically distributed (i.i.d.), merely independent, or dependent. These three cases look similar but lead to different joint densities.

5.1.1 i.i.d. data

This is the default case in most of this course.

Suppose $X_1, \dots, X_n$ are **independent and identically distributed (i.i.d.)** random variables:

$$X_i \overset{\text{iid}}{\sim} \text{Distribution}(\theta), \quad i = 1, \dots, n.$$

Each X_i has density (or PMF)

$$f(x \mid \theta).$$

Then the **joint density** of the full sample is

$$f(x_1, \dots, x_n \mid \theta) = \prod_{i=1}^{n} f(x_i \mid \theta).$$

5.1.2 Independent (not identical) data

If instead the samples are **independent but not identically distributed**:

$$X_i \sim \text{Distribution}_i(\theta),$$

with individual densities $f_{X_i}(x_i \mid \theta)$, then the joint density becomes

$$f(x_1, \dots, x_n \mid \theta) = \prod_{i=1}^{n} f_{X_i}(x_i \mid \theta).$$

5.1.3 Dependent data

In many real-world situations, data are **dependent**.

Example: Suppose our data are daily temperatures in Newcastle upon Tyne. Yesterday's temperature will influence today's temperature, and today's temperature will influence tomorrow's.

In such cases, we must model the dependencies explicitly. For instance,

$$X_{t+1} \mid X_t = x_t \sim \mathcal{N}(x_t, \sigma^2).$$

The likelihood is still the **joint distribution**, but expressed as a product of **conditional densities**:

$$f(x_1, \dots, x_n \mid \sigma^2) = \prod_{t=1}^{n} f(x_t \mid x_{t-1}, \sigma^2).$$

The example above assumed a Markov property (each observation only depends on the previous one). In general, observations may depend on the entire history, in which case the joint density factorises via the chain rule 1.2:

$$f(x_1, \dots, x_n \mid \theta) = f(x_1 \mid \theta) \prod_{t=2}^{n} f(x_t \mid x_1, \dots, x_{t-1}, \theta)$$

> ⚠ Non-Examinable Content
>
> This course focuses only on the **i.i.d.** or **independent** case.
> Dependent data will be covered in later modules and is **not examinable**.

Case	Assumption	Factorisation of Joint Density
i.i.d. data	Independent and identically distributed; each with density $f(x \mid \theta)$	$f(\underline{x} \mid \theta) = \prod_{i=1}^{n} f(x_i \mid \theta)$
Independent (not identical) data	Independent; each X_i may have its own density $f_{X_i}(x_i \mid \theta)$	$f(\underline{x} \mid \theta) = \prod_{i=1}^{n} f_{X_i}(x_i \mid \theta)$
Dependent data (general)	Dependent; factorise using the chain rule 1.2	$f(\underline{x} \mid \theta) = f(x_1 \mid \theta) \prod_{t=2}^{n} f(x_t \mid x_1, \dots, x_{t-1}, \theta)$

> ⚠ Reminder: Non-Examinable Content
>
> The **dependent case** in the above table is shown for completeness.
> This course focuses only on the **i.i.d.** or **independent** case, and the dependent case is **not examinable**.

Remark 5.1 (Why products?).

- Independence means joint probabilities factorise:

$$\Pr(A \cap B) = \Pr(A)\Pr(B).$$

- By the same rule 1.8, the joint density of independent random variables is the product of their individual densities.
- This is a common **misconception**: without independence, you **cannot** just multiply.

5.2 Likelihood Function

So far we have described the likelihood informally as "the joint density of the data given the parameter."

We now give a **formal definition** of the *likelihood function* and introduce the notation we will use in the **frequentist parts** of the course.

Definition 5.1. Likelihood Function. For fixed observed data $\underline{x}$, the **likelihood function** is defined as

$$L(\theta; \underline{x}) := f(\,\underline{x} \mid \theta).$$

Here we treat θ as the **variable**, and the data $\underline{x}$ as fixed.

5.2.1 Illustration (Likelihood with a coin)

Suppose we toss a possibly biased coin twice, where

$$\Pr(\text{Head}) = \theta, \qquad \Pr(\text{Tail}) = 1 - \theta.$$

Let X be the number of heads in two tosses. Then, since this is a **?@sec-binom** observation:

$$f(x \mid \theta) = \Pr(X = x \mid \theta) = \binom{2}{x} \theta^x (1-\theta)^{2-x}, \qquad x = 0, 1, 2.$$

There are *two* interpretations of this:

1. **Probability view (before observing data).**
   For fixed θ, this is a *probability mass function* (PMF) in x. It satisfies

   $$f(x = 0 \mid \theta) + f(x = 1 \mid \theta) + f(x = 2 \mid \theta) = 1.$$

   It tells us how likely different data values x are under parameter θ.

2. **Likelihood view (after observing data).**
   Now suppose we observe $x = 2$. Now, since $x = 2$ is fixed, the same formula becomes

   $$L(\theta; x = 2) = f(2 \mid \theta) = \theta^2.$$

   This function tells us how plausible different values of θ are, given the observed data.

Interpretation

Consider two **competing** values of θ for observing $x = 2$ heads in $n = 2$ tosses:

1. If $\theta = 0.01$, the probability of observing $X = 2$ is

$$L(0.01; 2) = 0.0001.$$

   For such a coin, two heads would be highly unusual.

2. If $\theta = 0.99$, the probability of observing $X = 2$ is

$$L(0.99; 2) = 0.9801 \approx 0.98.$$

   For such a coin, observing $X = 2$ is exactly what we'd expect.

Given $x = 2$, values of θ close to 1 are much more plausible than values near 0.

Remark 5.2 (Likelihood vs. Probability). It is important to distinguish between the **joint density** of the data and the **likelihood function**:

- **Joint density (PDF/PMF).** $f(\underline{x} \mid \theta)$ is viewed as a function of the *data* $\underline{x}$, with the parameter θ fixed. This describes the probability distribution of the sample under θ.
- **Likelihood function.** The same expression, written as

$$L(\theta; \underline{x}) = f(\underline{x} \mid \theta),$$

  is instead viewed as a function of the *parameter* θ, with the observed data $\underline{x}$ held fixed. This is the function we use to make inferences about θ.

In short:

- **Joint density**: given θ, how *likely* are different data values $\underline{x}$?
- **Likelihood**: given observed $\underline{x}$, how *plausible* are different values of θ?

5.2.2 Computing Likelihood Functions

This is a **very important skill** in the course: you will use it often in the *frequentist* parts, and it is **essential** for the *Bayesian* part.

Example 5.1 (Bernoulli Likelihood)**.** Suppose we have n i.i.d. data distributed according to the **?@sec-bern**:

$$Y_i \sim \text{Bernoulli}(p).$$

Derive the **likelihood function**.

Because the Y_i are i.i.d., the joint PMF factorises:

$$L(p;\underline{y}) = f(\underline{y} \mid p) = \prod_{i=1}^{n} f(y_i \mid p).$$

For the Bernoulli distribution,

$$f(y_i \mid p) = p^{y_i}(1-p)^{1-y_i}.$$

Hence,

$$\begin{aligned}
L(p;\underline{y}) &= \prod_{i=1}^{n} p^{y_i}(1-p)^{1-y_i} \\
&= p^{y_1}(1-p)^{1-y_1} \times p^{y_2}(1-p)^{1-y_2} \times ... \times p^{y_n}(1-p)^{1-y_n} \\
&= p^{y_1+...+y_n}(1-p)^{(1-y_1)+...+(1-y_n)} \\
&= p^{\sum_{i=1}^{n} y_i}(1-p)^{\sum_{i=1}^{n} 1-y_i}.
\end{aligned}$$

Writing $\sum_{i=1}^{n} y_i = n\bar{y}$ gives

$$L(p;\underline{y}) = p^{n\bar{y}}(1-p)^{n(1-\bar{y})}.$$

In practice, and in **assessed questions** in this course, it is important to be comfortable with computing likelihood functions for actual observed data. The next example does this:

Example 5.2 (Bernoulli Likelihood with Data)**.** Suppose $n = 5$ i.i.d. observations $X_i \sim$ Bernoulli(p) are observed as

$$\underline{x} = (1, 0, 1, 1, 0).$$

Derive the **likelihood function** and evaluate it for a few values of p.

Since $\underline{x} = (1, 0, 1, 1, 0)$, we have $\sum_i x_i = 3$. Therefore:

$$L(p; \underline{x}) = p^3(1-p)^2.$$

- If $p = 0.5$, $L(0.5) = 0.5^3 \cdot 0.5^2 = 0.03125$.
- If $p = 0.7$, $L(0.7) = 0.7^3 \cdot 0.3^2 \approx 0.0309$.
- If $p = 0.8$, $L(0.8) = 0.8^3 \cdot 0.2^2 = 0.0205$.

So the data are slightly more likely under $p = 0.5$ than $p = 0.8$.

Example 5.3 (Normal Likelihood)**.** Suppose we have n i.i.d. data distributed according to the **?@sec-normal**:

$$T_i \sim \mathcal{N}(\mu, \gamma).$$

Derive the **likelihood function**.

Because the T_i are i.i.d., the joint PDF factorises:

$$L(\mu, \gamma; \underline{t}) = f(\underline{t} \mid \mu, \gamma) = \prod_{i=1}^{n} f(t_i \mid \mu, \gamma).$$

For the Normal distribution (with variance $\gamma = \sigma^2$),

$$f(t_i \mid \mu, \gamma) = \frac{1}{\sqrt{2\pi\gamma}} \exp\left\{-\frac{1}{2\gamma}(t_i - \mu)^2\right\}.$$

Hence,

$$\begin{aligned} L(\mu, \gamma; \underline{t}) &= \prod_{i=1}^{n} \frac{1}{\sqrt{2\pi\gamma}} \exp\left\{-\frac{1}{2\gamma}(t_i - \mu)^2\right\} \\ &= (2\pi\gamma)^{-\frac{n}{2}} \exp\left\{\sum_{i=1}^{n} -\frac{1}{2\gamma}(t_i - \mu)^2\right\}. \end{aligned}$$

Therefore,

$$L(\mu, \gamma; \underline{t}) = (2\pi\gamma)^{-\frac{n}{2}} \exp\left\{-\frac{1}{2\gamma}\sum_{i=1}^{n}(t_i - \mu)^2\right\}.$$

Example 5.4 (Poisson Likelihood)**.** Suppose we have n i.i.d. data distributed according to the **?@sec-pois**:

$$V_i \sim \text{Poisson}(\alpha)$$

Derive the **likelihood function**.

Because the V_i are i.i.d., the joint PMF factorises:

$$L(\alpha; \underline{v}) = f(\underline{v} \mid \alpha) = \prod_{i=1}^{n} f(v_i \mid \alpha).$$

For the Poisson distribution

$$f(v_i \mid \alpha) = \frac{\alpha^{v_i} e^{-\alpha}}{v_i!}$$

Hence,

$$\begin{aligned} L(\alpha; \underline{v}) &= \prod_{i=1}^{n} \frac{\alpha^{v_i} e^{-\alpha}}{v_i!} \\ &= \alpha^{n\bar{v}} e^{-n\alpha} \prod_{i=1}^{n} \frac{1}{v_i!} \end{aligned}$$

Therefore,

$$L(\alpha; \underline{v}) = \alpha^{n\bar{v}} e^{-n\alpha} \prod_{i=1}^{n} \frac{1}{v_i!}.$$

5.2.3 Data Reduction and Sufficient Statistics

When we write down a likelihood, the parameter often only depended on the sample through a few summaries (e.g. a sum or a mean).

Sufficiency formalises when such a summary contains *all the information in the sample about* the parameter θ.

Definition 5.2 (Sufficient Statistic)**.** Let $\underline{X} = (X_1, \dots, X_n)$ have joint PDF/PMF $f(\underline{x} \mid \theta)$.

A statistic $T(\underline{X})$ is **sufficient** for θ if the conditional distribution of the data given the statistic does **not** depend on θ, i.e.

$$f(\underline{x} \mid T(\underline{x}) = t,\ \theta) \text{ does not depend on } \theta.$$

Theorem 5.1 (Neyman–Fisher Factorisation)**.** *$T(\underline{X})$ is sufficient for θ* ***if and only if*** *there exist functions $g(\theta, T(\underline{x}))$ and $h(\underline{x})$ such that*

$$f_{\underline{X}}(\underline{x} \mid \theta) = g(\theta, T(\underline{x}))\, h(\underline{x}).$$

Example 5.5 (Sufficiency: Bernoulli model)**.** If $X_i \sim \text{Bern}(p)$ i.i.d. and $y = \sum_{i=1}^n x_i$, then

$$L(p; \underline{x}) = f_{\underline{X}}(\underline{x} \mid p) = \prod_{i=1}^n p^{x_i}(1-p)^{1-x_i} = p^{\sum x_i}(1-p)^{n-\sum x_i} = \underbrace{p^y(1-p)^{n-y}}_{g(p,y)} \underbrace{1}_{h(\underline{x})}.$$

By factorisation, $T(\underline{X}) = \sum_{i=1}^n X_i$ is sufficient for p.

Example 5.6 (Sufficiency: Poisson model)**.** If $X_i \sim \text{Poisson}(\lambda)$ iid and $y = \sum_{i=1}^n x_i$, then

$$L(\lambda; \underline{x}) = f_{\underline{X}}(\underline{x} \mid \lambda) = \prod_{i=1}^n \frac{e^{-\lambda}\lambda^{x_i}}{x_i!} = \underbrace{e^{-n\lambda}\lambda^{\sum x_i}}_{g(\lambda, y)} \underbrace{\prod_{i=1}^n \frac{1}{x_i!}}_{h(\underline{x})},$$

so $T(\underline{X}) = \sum_{i=1}^n X_i$ is sufficient for λ.

Example 5.7 (Sufficiency: Normal model)**.** If $X_i \sim \mathcal{N}(\mu, \sigma^2)$ iid:

- **Known σ^2.** With $\bar{x} = \frac{1}{n}\sum x_i$,

  $$f_{\underline{X}}(\underline{x} \mid \mu) = (2\pi\sigma^2)^{-n/2} \exp\left[-\frac{1}{2\sigma^2}\sum_{i=1}^n (x_i - \mu)^2\right] = (2\pi\sigma^2)^{-n/2} \underbrace{\exp\left[-\frac{1}{2\sigma^2}\sum_{i=1}^n (x_i - \bar{x})^2\right]}_{h(\underline{x})} \underbrace{\exp\left[-\frac{n}{2\sigma^2}(\bar{x} - \right.}_{g(\mu, \bar{x})}$$

  Hence $T(\underline{X}) = \bar{X}$ is sufficient for μ.

- **Unknown σ^2.** Write $S^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X})^2$. Then

$$f_{\underline{X}}(\underline{x} \mid \mu, \sigma^2) = (2\pi\sigma^2)^{-n/2} \exp\left[-\frac{1}{2\sigma^2}\left\{ \underbrace{\sum_{i=1}^{n}(x_i - \bar{x})^2}_{(n-1)s^2} + n(\bar{x} - \mu)^2 \right\} \right] = \underbrace{(2\pi)^{-n/2}\exp(0)}_{h(\underline{x})} \underbrace{(\sigma^2)^{-n/2}\exp\Big[-\frac{(n -}{}}_{g\big((\mu,\sigma^2}$$

  Thus $T(\underline{X}) = (\bar{X}, S^2)$ is jointly sufficient for (μ, σ^2).

Common misconceptions

- Sufficiency is **model-dependent**: change the sampling model, and what's sufficient may change.
- *Sufficient*   *efficient*: it tells us what data features matter for θ, not how variable an estimator is.
- Although not explicitly mentioned, we will **reuse** these T when we derive MLEs and, later in the Bayesian chapter, when posteriors update via the same statistics.

5.3 Maximum Likelihood Estimation

Definition 5.3 (Maximum Likelihood Estimator (MLE))**.** The **maximum likelihood estimator (MLE)** is the parameter value that makes the observed data most likely:

$$\hat{\theta}_{\text{MLE}} = \arg\max_{\theta \in \Theta} L(\theta; \underline{X}),$$

where $L(\theta; \underline{X})$ is the likelihood function.

❗ **Important 8: Terminology:** $\arg\max$ **vs.** $\max$

The max gives the **largest value** attained by a function.

$$\max_{\theta \in \Theta} L(\theta; \underline{X})$$

means "the maximum likelihood value."
The $\arg\max$ gives the **argument (input)** where the maximum occurs.

$$\arg\max_{\theta \in \Theta} L(\theta; \underline{X})$$

means "the parameter value that maximises the likelihood."
So: max is the **height**, $\arg\max$ is the **location**.

Because the logarithm is a **monotonically increasing function**, the value of θ that maximises the likelihood will also maximise the **log-likelihood**.

In practice, we almost always work with the log-likelihood because it is **simpler to handle mathematically**: it turns products into sums, and derivatives are easier to compute.

Definition 5.4 (Log-Likelihood)**.** The **log-likelihood** is defined as

$$\ell(\theta; \underline{x}) = \log L(\theta; \underline{x}).$$

- This is literally just the logarithm of the likelihood.

- Because log is increasing, maximising $\ell(\theta; \underline{x})$ gives the same $\hat{\theta}$ as maximising $L(\theta; \underline{x})$.

✨ **Big picture:**

- The likelihood function 5.1 tells us how **plausible** each parameter value is, given the data.
- The MLE 5.3 chooses the parameter value that makes the observed data most likely.
- Later, in Bayesian inference, we will combine the likelihood with a **prior distribution** on θ to obtain a **posterior distribution**.

5.3.1 Computing Univariate MLEs

Example 5.8 (Bernoulli MLE)**.** From Example 5.1, we saw that the likelihood function for n i.i.d. $Y_i \sim \text{Bernoulli}(p)$ observations was

$$L(p; \underline{y}) = p^{n\bar{y}}(1-p)^{n(1-\bar{y})}.$$

Compute the MLE of p.

From the provided likelihood, we first compute the **log-likelihood**:

$$\ell(p \mid \underline{x}) = \Big(\sum_i x_i\Big) \log p + \Big(n - \sum_i x_i\Big) \log(1-p).$$

Differentiate and solve:

$$\frac{\partial}{\partial p} \ell(p \mid \underline{x}) = \frac{\sum_i x_i}{p} - \frac{n - \sum_i x_i}{1-p} = 0.$$

This gives

$$\hat{p}_{\text{MLE}} = \frac{1}{n} \sum_{i=1}^{n} x_i,$$

the **sample proportion of successes**.

Example 5.9 (Binomial MLE)**.** Suppose we have a **single ?@sec-binom** observation

$$X \sim \text{Bin}(n, \theta).$$

Derive the **maximum likelihood estimator** of θ.

Since we only have **one** observation, the likelihood function is just the distribution of the observation:

$$L(\theta; x) = f(x \mid \theta) = \binom{n}{x} \theta^x (1-\theta)^{n-x}.$$

The log-likelihood is thus

$$\ell(\theta; x) = \log f(x \mid \theta) = \log\left(\binom{n}{x}\right) + x \log\theta + (n-x)\log(1-\theta).$$

Differentiating, we obtain

$$\frac{\partial}{\partial\theta}\,\ell(\theta;x) = \frac{x}{\theta} - \frac{n-x}{1-\theta}.$$

Setting the derivative to zero and solving,

$$\frac{x}{\theta} = \frac{n-x}{1-\theta} \implies x(1-\theta) = (n-x)\theta \implies x = n\theta \implies \hat{\theta} = \frac{x}{n}.$$

To confirm it is a maximum, the second derivative is

$$\frac{\partial^2}{\partial\theta^2}\,\ell(\theta;x) = -\frac{x}{\theta^2} - \frac{n-x}{(1-\theta)^2} < 0 \quad \text{for } \theta \in (0,1),$$

so $\hat{\theta} = x/n$ maximises the likelihood. (At the boundaries, if $x = 0$ then $\hat{\theta} = 0$; if $x = n$ then $\hat{\theta} = 1$.)

Example 5.10 (Another Binomial MLE)**.** Suppose we have n **independent** Binomial data:

$$X_i \sim \text{Bin}(k_i, \theta).$$

Note here that each Binomial data has a different number of Bernoulli trials k_i, which are known.

Derive the **maximum likelihood estimator** of θ.

The likelihood function is the joint distribution of the n observations

$$f(\underline{x} \mid \theta) = \prod_{i=1}^{n} \binom{k_i}{x_i} \theta^{x_i}(1-\theta)^{k_i - x_i}.$$

The log-likelihood is

$$\ell(\theta;\underline{x}) = \sum_{i=1}^{n} \log \binom{k_i}{x_i} + \Big(\sum_{i=1}^{n} x_i\Big)\log\theta + \Big(\sum_{i=1}^{n}(k_i - x_i)\Big)\log(1-\theta).$$

Let $S = \sum_{i=1}^{n} x_i$ (total successes) and $K = \sum_{i=1}^{n} k_i$ (total trials). Then

$$\frac{\partial}{\partial\theta}\ell(\theta;\underline{x}) = \frac{S}{\theta} - \frac{K-S}{1-\theta}.$$

Setting to zero and solving:

$$\frac{S}{\theta} = \frac{K-S}{1-\theta} \implies S(1-\theta) = (K-S)\theta \implies S = K\theta \implies \hat{\theta} = \frac{S}{K} = \frac{\sum_{i=1}^n x_i}{\sum_{i=1}^n k_i}.$$

The second derivative is

$$\frac{\partial^2}{\partial\theta^2}\ell(\theta;\underline{x}) = -\frac{S}{\theta^2} - \frac{K-S}{(1-\theta)^2} < 0 \quad \text{for } \theta \in (0,1),$$

so this is a (strict) maximum. Boundary cases: if $S = 0$ then $\hat{\theta} = 0$; if $S = K$ then $\hat{\theta} = 1$.

Example 5.11 (Poisson MLE)**.** From Example 5.4, we saw that the likelihood function for n i.i.d. $V_i \sim \text{Poisson}(\alpha)$ observations was

$$L(\alpha;\underline{v}) = \alpha^{n\bar{v}} e^{-n\alpha} \prod_{i=1}^n \frac{1}{v_i!}.$$

Derive the **maximum likelihood estimator** of α and determine its **expected value** and **variance**.

Step 1: Computing MLE

The log-likelihood is

$$\ell(\alpha;\underline{v}) = n\bar{v}\log\alpha - n\alpha - \sum_{i=1}^n \log(v_i!), \qquad \alpha \geq 0.$$

Differentiate w.r.t. α:

$$\frac{\partial}{\partial\alpha}\ell(\alpha;\underline{v}) = \frac{n\bar{v}}{\alpha} - n.$$

Set to zero and solve:

$$\frac{n\bar{v}}{\alpha} - n = 0 \implies \hat{\alpha} = \bar{v} = \frac{1}{n}\sum_{i=1}^n v_i.$$

Second derivative:

$$\frac{\partial^2}{\partial\alpha^2}\ell(\alpha;\underline{v}) = -\frac{n\bar{v}}{\alpha^2} < 0 \quad (\alpha > 0),$$

so this is a maximum. Boundary case: if all $v_i = 0$ (so $\bar{v} = 0$), the likelihood $L(\alpha) = e^{-n\alpha}$ is maximised at $\hat{\alpha} = 0$.

Therefore, the maximum likelihood *estimator* is:

$$\hat{\alpha} = \bar{V}$$

Step 2: Mean and Variance of $\hat{\alpha}$

Recall $\hat{\alpha} = \bar{V} = \frac{1}{n}\sum_{i=1}^{n} V_i$ with $V_i \overset{\text{iid}}{\sim} \text{Poisson}(\alpha)$.

Mean:

$$\mathrm{E}[\hat{\alpha}] = \frac{1}{n}\sum_{i=1}^{n} \mathrm{E}[V_i] = \frac{1}{n} \cdot n\alpha = \alpha.$$

Variance (using independence):

$$\mathrm{Var}[\hat{\alpha}] = \mathrm{Var}\left(\frac{1}{n}\sum_{i=1}^{n} V_i\right) = \frac{1}{n^2}\sum_{i=1}^{n} \mathrm{Var}(V_i) = \frac{1}{n^2} \cdot n\alpha = \frac{\alpha}{n}.$$

Here we used $\mathrm{Var}(V_i) = \mathrm{E}[V_i] = \alpha$ for the Poisson(α) distribution.

Example 5.12 (Normal MLE for μ)**.** From Example 5.3, we saw that the likelihood function for n i.i.d. $T_i \sim \mathcal{N}(\mu, \gamma)$ observations was

$$L(\mu, \gamma; \underline{t}) = (2\pi\gamma)^{-\frac{n}{2}} \exp\left\{-\frac{1}{2\gamma}\sum_{i=1}^{n}(t_i - \mu)^2\right\}.$$

Suppose that the variance γ is *known*. Determine the **MLE** of μ.

The log-likelihood (as a function of μ) is

$$\ell(\mu; \underline{t}) = -\frac{n}{2}\log(2\pi\gamma) - \frac{1}{2\gamma}\sum_{i=1}^{n}(t_i - \mu)^2.$$

Differentiating w.r.t. μ,

$$\frac{\partial}{\partial\mu}\,\ell(\mu; \underline{t}) = -\frac{1}{2\gamma} \cdot 2\sum_{i=1}^{n}(t_i - \mu)(-1) = \frac{1}{\gamma}\sum_{i=1}^{n}(t_i - \mu).$$

Setting equal to 0 and solve:

$$\sum_{i=1}^{n}(t_i - \mu) = 0 \implies n\mu = \sum_{i=1}^{n} t_i \implies \hat{\mu} = \bar{t} = \frac{1}{n}\sum_{i=1}^{n} t_i.$$

The second derivative is

$$\frac{\partial^2}{\partial \mu^2}\, \ell(\mu; \underline{t}) = -\frac{n}{\gamma} < 0,$$

so $\hat{\mu} = \bar{t}$ maximises the likelihood.

5.3.2 Computing Multivariate MLEs

In many practical problems, the statistical model involves more than a single parameter θ. In this case, we work with a **vector of parameters**

$$\underline{\theta} = (\theta_1, \dots, \theta_p).$$

The definitions of the likelihood function and the MLE extend in a natural way. The main difference is that the optimisation problem now requires solving **simultaneous equations** for each parameter.

Definition 5.5 (Multivariate MLE)**.** Let $L(\underline{\theta}; \underline{x})$ and $\ell(\underline{\theta}; \underline{x})$ denote the **likelihood** and **log-likelihood** functions respectively, based on observations $\underline{x} = (x_1, \dots, x_n)$ from a model with parameters $\underline{\theta} = (\theta_1, \dots, \theta_p)$.

The maximum likelihood estimator $\hat{\underline{\theta}}$ is obtained by solving the system

$$\nabla_{\underline{\theta}}\, \ell(\underline{\theta}; \underline{x}) \;=\; \begin{pmatrix} \dfrac{\partial}{\partial \theta_1}\, \ell(\underline{\theta}; \underline{x}) \\ \dfrac{\partial}{\partial \theta_2}\, \ell(\underline{\theta}; \underline{x}) \\ \vdots \\ \dfrac{\partial}{\partial \theta_p}\, \ell(\underline{\theta}; \underline{x}) \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{pmatrix}.$$

In general, solving this system of equations cannot be done in closed form, so in practice we usually rely on *numerical optimisation methods* to approximate the MLE. We will not cover these methods in this course. Instead, we will focus on a few important cases where the MLE can be worked out explicitly.

Example 5.13 (Normal MLE for μ and γ)**.** Suppose we have n i.i.d. observations $X_i \sim \mathcal{N}(\mu, \gamma)$. Determine the **maximum likelihood estimators** for μ and γ.

Step 1. Compute log-likelihood:

The likelihood function is a relabelling of the data terms in Example 5.3:

$$
\begin{aligned}
L(\mu, \gamma; \underline{x}) &= \prod_{i=1}^{n} \frac{1}{\sqrt{2\pi\gamma}} \exp\left\{-\frac{1}{2\gamma}(x_i - \mu)^2\right\} \\
&= (2\pi\gamma)^{-\frac{n}{2}} \exp\left\{\sum_{i=1}^{n} -\frac{1}{2\gamma}(x_i - \mu)^2\right\}.
\end{aligned}
$$

The log-likelihood is thus

$$\ell(\mu, \sigma^2; \underline{x}) = -\frac{n}{2}\log(2\pi\sigma^2) - \frac{1}{2\sigma^2}\sum_{i=1}^{n}(x_i - \mu)^2.$$

Step 2. Differentiate w.r.t. μ:

$$\frac{\partial \ell}{\partial \mu} = \frac{1}{\sigma^2}\sum_{i=1}^{n}(x_i - \mu).$$

Setting this equal to 0 gives

$$\hat{\mu} = \bar{x}.$$

Step 2. Differentiate w.r.t. σ^2:

$$\frac{\partial \ell}{\partial \sigma^2} = -\frac{n}{2\sigma^2} + \frac{1}{2(\sigma^2)^2}\sum_{i=1}^{n}(x_i - \mu)^2.$$

Setting this equal to 0 and substituting $\mu = \hat{\mu}$ gives

$$\hat{\sigma}^2 = \frac{1}{n}\sum_{i=1}^{n}(x_i - \bar{x})^2.$$

Final Answer: The MLEs are

$$\hat{\mu} = \bar{X}, \qquad \hat{\sigma}^2 = \frac{1}{n}\sum_{i=1}^{n}(X_i - \bar{X})^2.$$

Note: $\hat{\sigma}^2$ is biased (its expectation is $\frac{n-1}{n}\sigma^2$), but it is the maximum likelihood estimate.

5.4 Information and Efficiency of Estimators

In the previous chapter we studied two kinds of *properties of estimators*:

- In Properties of Estimators 4.2, we introduced: bias 4.4, variance 4.5, and MSE 4.8.

- In Asymptotic Properties of Estimators 4.4, we introduced: asymptotic unbiasedness 4.9 and consistency 4.10.

Now that we have the *likelihood function*, we can go further and introduce two powerful ideas:

- **Cramér–Rao Lower Bound (CRLB):** a universal lower bound on the variance of any unbiased estimator.
- **Efficiency:** whether an estimator actually attains this bound (at least asymptotically).

! Important 9: Motivation

Our central measure of estimator quality was the **mean squared error (MSE)**. From the bias-variance decomposition 4.4 of the MSE, we know:

$$\mathrm{MSE}(\hat{\theta}) = \mathrm{Bias}(\hat{\theta})^2 + \mathrm{Var}(\hat{\theta}).$$

For **unbiased estimators** ($\mathrm{Bias}(\hat{\theta}) = 0$), this reduces to

$$\mathrm{MSE}(\hat{\theta}) = \mathrm{Var}(\hat{\theta}).$$

So if we restrict attention to unbiased estimators, the question of "how good" becomes:

> *Which unbiased estimator has the **smallest possible variance**?*

The Cramér–Rao Lower Bound (CRLB) answers this: it gives a universal benchmark for how small the variance can ever be.

Before introducing the CRLB and efficiency, we first need two key definitions derived from the likelihood function.

5.4.1 Score and Fisher Information

Definition 5.6 (Score Statistic)**.** The **score** is the derivative of the log-likelihood with respect to the parameter:

$$U(\theta; \underline{X}) \;=\; \frac{\partial}{\partial\theta}\,\ell(\theta; \underline{X}) = \frac{\partial}{\partial\theta}\,\log f(\underline{X} \mid \theta).$$

It is a **random quantity**.

Definition 5.7 (Fisher Information)**.** The **Fisher Information** (or *expected information*) for data $\underline{X}$ with log-likelihood $\ell(\theta; \underline{X})$ is

$$\mathcal{I}(\theta; \underline{X}) \;=\; -\,\mathrm{E}\!\left[\frac{\partial^2}{\partial\theta^2}\,\ell(\theta; \underline{X})\right].$$

Intuition: Fisher information measures how much the data tell us about θ.

- A sharply peaked log-likelihood $\Rightarrow$ high information (data strongly constrain θ).
- A flat log-likelihood $\Rightarrow$ low information (many θ's are plausible). The curvature (second derivative) captures this "peakedness."

Proposition 5.1 (Fisher Information for i.i.d. Data)**.** *Suppose we have an i.i.d. sample $X_1, \dots, X_n$ drawn from the same distribution with PDF/PMF $f(x \mid \theta)$. Define:*

- *The Fisher information from **one observation**:*

$$\mathcal{I}(\theta) = -\mathrm{E}\left[\tfrac{\partial^2}{\partial\theta^2}\log f(X \mid \theta)\right]$$

- *The Fisher information from the **whole sample**:*

$$\mathcal{I}_n(\theta) = -\mathrm{E}\left[\tfrac{\partial^2}{\partial\theta^2}\ell(\theta; \underline{X}^{(n)})\right]$$

Then

$$\mathcal{I}_n(\theta) = n\,\mathcal{I}(\theta).$$

Proof. For n IID observations, the log-likelihood is

$$\ell(\theta; \underline{X}) \;=\; \sum_{i=1}^{n} \log f(X_i \mid \theta).$$

Differentiating twice,

$$\frac{\partial^2}{\partial\theta^2}\ell(\theta;\underline{X}^{(n)}) \;=\; \sum_{i=1}^{n}\frac{\partial^2}{\partial\theta^2}\log f(X_i \mid \theta).$$

Now take expectations:

$$\mathcal{I}_n(\theta) \;=\; -\,\mathrm{E}\left[\frac{\partial^2}{\partial\theta^2}\ell(\theta;\underline{X}^{(n)})\right] \;=\; -\,\mathrm{E}\left[\sum_{i=1}^{n}\frac{\partial^2}{\partial\theta^2}\log f(X_i \mid \theta)\right].$$

By linearity of expectation,

$$\mathcal{I}_n(\theta) \;=\; \sum_{i=1}^{n}\left(-\,\mathrm{E}\left[\frac{\partial^2}{\partial\theta^2}\log f(X_i \mid \theta)\right]\right).$$

Since the X_i are IID, each term in the sum is the same:

$$\mathcal{I}_n(\theta) = n\,\mathcal{I}(\theta).$$

□

Example 5.14 (Fisher Information for the Poisson Model)**.** Suppose we have n i.i.d. observations $X_i \sim \text{Poisson}(\alpha)$. From Example 5.4, we saw the likelihood took the form

$$L(\alpha;\underline{x}) = \frac{e^{-\alpha n}\alpha^{n\bar{x}}}{\prod_{i=1}^{n} x_i!}$$

Determine the Fisher information 5.1 for:

(a) The **whole sample**
(b) A **single observation**

a. Whole sample

We have

$$\ell(\alpha;\underline{x}) = -n\alpha + \left(\sum_{i=1}^{n} x_i\right)\log\alpha - \sum_{i=1}^{n}\log x_i!, \qquad \alpha > 0.$$

Differentiate:

$$\frac{\partial \ell}{\partial \alpha} = -n + \frac{\sum_{i=1}^{n} x_i}{\alpha}, \qquad \frac{\partial^2 \ell}{\partial \alpha^2} = -\frac{\sum_{i=1}^{n} x_i}{\alpha^2}.$$

Hence the Fisher information is

$$\mathcal{I}_n(\alpha) = -\,\mathrm{E}\left[\frac{\partial^2 \ell}{\partial \alpha^2}\right] = -\,\mathrm{E}\left[-\frac{\sum_{i=1}^n X_i}{\alpha^2}\right] = \frac{n\alpha}{\alpha^2} = \frac{n}{\alpha}.$$

---

b. Single observation

For one $X \sim \text{Poisson}(\alpha)$,

$$\ell_1(\alpha; x) = -\alpha + x \log \alpha - \log x!, \qquad \frac{\partial^2 \ell_1}{\partial \alpha^2} = -\frac{x}{\alpha^2}.$$

Therefore

$$\mathcal{I}_1(\alpha) = -\,\mathrm{E}\left[\frac{\partial^2 \ell_1}{\partial \alpha^2}\right] = -\,\mathrm{E}\left[-\frac{X}{\alpha^2}\right] = \frac{\alpha}{\alpha^2} = \frac{1}{\alpha}.$$

5.4.2 Cramér–Rao Lower Bound and Efficiency

We can now state one of the most important results in statistical inference: the **Cramér–Rao inequality**. It tells us how small the variance of an unbiased estimator can ever be.

Theorem 5.2 (The Cramér–Rao Inequality and the CRLB)**.** *Let $X_1, \dots, X_n$ be i.i.d. random variables with likelihood function $L(\theta; \underline{x})$, and let $\hat{\theta}$ be an unbiased estimator of θ.*

Then, under regularity conditions*,*

$$\mathrm{Var}(\hat{\theta}) \;\geq\; \frac{1}{\mathcal{I}_n(\theta)},$$

*where $\mathcal{I}_n(\theta)$ is the Fisher information 5.1 for the **whole sample**.*

*The quantity $\frac{1}{\mathcal{I}_n(\theta)}$ is called the **Cramér–Rao Lower Bound (CRLB)**. It represents the* smallest possible variance *achievable by any unbiased estimator of θ.*

Proof of **??**. The proof of the Cramér–Rao inequality follows from the **covariance inequality**:

$$\operatorname{Cov}(X, Y)^2 \leq \operatorname{Var}(X) \operatorname{Var}(Y),$$

applied to the score random variable $X = U(\theta; \underline{X})$ and the estimator $Y = \hat{\theta}(\underline{X})$. Substituting these in and rearranging for the variance of $\hat{\theta}$ gives

$$\operatorname{Var}(\hat{\theta}) \ \geq \ \frac{\operatorname{Cov}(\hat{\theta}, U)^2}{\operatorname{Var}(U)}.$$

Then, under *regularity conditions*, one can show that the variance of U is the Fisher information, and the covariance term equals 1.

Thus, the proof can be broken into steps:

1. **Prove the Covariance Inequality**
2. **Show:** $\mathrm{E}[U] = 0$
3. **Show:** $\operatorname{Var}(U) = \mathcal{I}(\theta)$
4. **Show:** $\operatorname{Cov}(\hat{\theta}, U) = 1$

Along the way, we will present the regularity conditions.

Step 1 (Proving Covariance Inequality):

The covariance inequality follows from the Cauchy-Schwarz inequality for random variables:

Cauchy-Schwarz Inequality for Random Variables

For any random variables X and Y of finite variance,

$$\mathrm{E}[XY]^2 \leq \mathrm{E}[X^2]\mathrm{E}[Y^2].$$

Proof of Cauchy-Schwarz Inequality
Consider $\mathrm{E}\left[(aX + Y)^2\right]$, which is clearly non-negative for real a.
Multiplying out gives:

$$a^2\mathrm{E}[X^2] + 2a\mathrm{E}[XY] + \mathrm{E}[Y^2] \geq 0.$$

Consider the left hand side as a quadratic in a. Since it is always non-negative it cannot have two real roots. Using the quadratic formula the roots satisfy:

$$a = \frac{-2\mathrm{E}[XY] \pm \sqrt{4\mathrm{E}[XY]^2 - 4\mathrm{E}[X^2]\mathrm{E}[Y^2]}}{2\mathrm{E}[X^2]}$$

For there to be zero or one real roots, the square root term must be non-positive i.e.

$$\mathrm{E}[XY]^2 - \mathrm{E}[X^2]\mathrm{E}[Y^2] \leq 0.$$

which gives the required result.

Applying Cauchy-Schwarz, to the mean-zero shifted random variables, $X - \mathrm{E}[X]$ and $Y - \mathrm{E}[Y]$, we obtain

$$\begin{aligned}\mathrm{Cov}(X,Y)^2 &= \mathrm{E}\left[(X-\mathrm{E}[X])(Y-\mathrm{E}[Y])\right]^2 \\ &\le \mathrm{E}\left[(X-\mathrm{E}[X])^2\right]\mathrm{E}\left[(Y-\mathrm{E}[Y])^2\right] = \mathrm{Var}(X)\mathrm{Var}(Y).\end{aligned}$$

Aside: You may recall that the *correlation* between two random variables is defined as

$$\mathrm{Cor}(X,Y) = \frac{\mathrm{Cov}(X,Y)}{\mathrm{Var}(X)^{1/2}\mathrm{Var}(Y)^{1/2}}$$

and satisfies $-1 \le \mathrm{Cor}(X,Y) \le 1$. This inequality is derived from the **covariance inequality**.

Step 2 (Proving $\mathrm{E}[U] = 0$):

By the definition of the log-likelihood:

$$\tfrac{\partial}{\partial\theta}\ell(\theta;\underline{x}) = \frac{\partial}{\partial\theta}\log f(\underline{x} \mid \theta).$$

Using the chain rule, we have

$$\frac{\partial}{\partial\theta}\ell(\theta;\underline{x}) = \frac{\partial}{\partial\theta}\log f(\underline{x} \mid \theta) = \frac{\frac{\partial}{\partial\theta}f(\underline{x} \mid \theta)}{f(\underline{x} \mid \theta)}.$$

Since $U(\theta;\underline{x}) = \frac{\partial}{\partial\theta}\ell(\theta;\underline{x})$, we can compute its expectation as follows

$$
\begin{aligned}
\mathrm{E}[U(\theta;\underline{X})] &= \int U(\theta;\underline{x}) f(\underline{x} \mid \theta)\,\mathrm{d}\underline{x} \\
&= \int \frac{\frac{\partial}{\partial\theta} f(\underline{x} \mid \theta)}{f(\underline{x} \mid \theta)} f(\underline{x} \mid \theta)\,\mathrm{d}\underline{x} \\
&= \int \frac{\partial}{\partial\theta} f(\underline{x} \mid \theta)\,\mathrm{d}\underline{x} \\
&= \frac{\partial}{\partial\theta} \int f(\underline{x} \mid \theta)\,\mathrm{d}\underline{x} \\
&= \frac{\partial}{\partial\theta} 1 \\
&= 0.
\end{aligned}
$$

Here, we assumed the first *regularity condition*:

> **Regularity Condition 1:** We assumed we can interchange the integral and derivative in the step
>
> $$\int \frac{\partial}{\partial\theta} f(\underline{x} \mid \theta)\,\mathrm{d}\underline{x} = \frac{\partial}{\partial\theta} \int f(\underline{x} \mid \theta)\,\mathrm{d}\underline{x}$$

Step 3 (Proving $\mathrm{Var}(U) = \mathcal{I}_n(\theta)$**):**

By definition,

$$\mathrm{Var}(U) = \mathrm{E}[U^2] - \left(\mathrm{E}[U]\right)^2.$$

From Step 2 we know $\mathrm{E}[U] = 0$, so

$$\mathrm{Var}(U) = \mathrm{E}[U^2] = \mathrm{E}\left[\left(\frac{\partial}{\partial\theta}\,\ell(\theta;\underline{X})\right)^2\right].$$

Recall the identity derived in Step 2:

$$\frac{\partial}{\partial\theta}\ell(\theta;\underline{x}) = \frac{\frac{\partial}{\partial\theta} f(\underline{x} \mid \theta)}{f(\underline{x} \mid \theta)}.$$

Differentiating again, using the product and chain rules, yields:

$$\frac{\partial^2}{\partial\theta^2}\ell(\theta;\underline{x}) = -\frac{\left(\frac{\partial}{\partial\theta}f(\underline{x}\mid\theta)\right)^2}{f(\underline{x}\mid\theta)^2} + \frac{\frac{\partial^2}{\partial\theta^2}f(\underline{x}\mid\theta)}{f(\underline{x}\mid\theta)}.$$

Therefore,

$$\frac{\partial^2}{\partial\theta^2}\ell(\theta;\underline{x}) = -\left(\frac{\partial}{\partial\theta}\ell(\theta;\underline{x})\right)^2 + \frac{\frac{\partial^2}{\partial\theta^2}f(\underline{x}\mid\theta)}{f(\underline{x}\mid\theta)}.$$

Rearranging for $\left(\frac{\partial}{\partial\theta}\ell(\theta;\underline{x})\right)^2$ yields:

$$\left(\frac{\partial}{\partial\theta}\ell(\theta;\underline{x})\right)^2 = \frac{\frac{\partial^2}{\partial\theta^2}f(\underline{x}\mid\theta)}{f(\underline{x}\mid\theta)} - \frac{\partial^2}{\partial\theta^2}\ell(\theta;\underline{x}).$$

Computing the expectation, we have

$$\begin{aligned}
\operatorname{Var}(U) &= \mathrm{E}\left[\frac{\frac{\partial^2}{\partial\theta^2}f(\underline{X}\mid\theta)}{f(\underline{X}\mid\theta)} - \frac{\partial^2}{\partial\theta^2}\ell(\theta;\underline{X})\right] \\
&= \mathrm{E}\left[\frac{\frac{\partial^2}{\partial\theta^2}f(\underline{X}\mid\theta)}{f(\underline{X}\mid\theta)}\right] + \underbrace{\mathrm{E}\left[-\frac{\partial^2}{\partial\theta^2}\ell(\theta;\underline{X})\right]}_{=\mathcal{I}_n(\theta)}.
\end{aligned}$$

Concentrating on the remaining expectation:

$$\begin{aligned}
\mathrm{E}\left[\frac{\frac{\partial^2}{\partial\theta^2}f(\underline{X}\mid\theta)}{f(\underline{X}\mid\theta)}\right] &= \int \frac{\partial^2}{\partial\theta^2}f(\underline{x}\mid\theta)\,\mathrm{d}\underline{x} \\
&= \frac{\partial^2}{\partial\theta^2}\int f(\underline{x}\mid\theta)\,\mathrm{d}\underline{x} \\
&= \frac{\partial^2}{\partial\theta^2}1 \\
&= 0.
\end{aligned}$$

Here, we assumed the second *regularity condition*:

> **Regularity Condition 2:** We assumed we can interchange the integral and second derivative in the step
>
> $$\int \frac{\partial^2}{\partial\theta^2}f(\underline{x}\mid\theta)\,\mathrm{d}\underline{x} = \frac{\partial^2}{\partial\theta^2}\int f(\underline{x}\mid\theta)\,\mathrm{d}\underline{x}$$

Therefore, we can conclude Step 3 with the final result:

$$\operatorname{Var}(U) = \mathcal{I}_n(\theta).$$

Step 4 (Proving $\operatorname{Cov}(\hat{\theta}, U) = 1$):

By definition,

$$\operatorname{Cov}(\hat{\theta}, U) = \mathrm{E}\big[(\hat{\theta} - \theta)\, U\big].$$

Using the same result used in steps 2 and 3:

$$U(\theta; \underline{x}) = \frac{\partial}{\partial \theta} \log f(\underline{x} \mid \theta) = \frac{\frac{\partial}{\partial \theta} f(\underline{x} \mid \theta)}{f(\underline{x} \mid \theta)},$$

we have

$$\mathrm{E}[\hat{\theta} U] = \int \hat{\theta}(\underline{x})\, \frac{\partial}{\partial \theta} f(\underline{x} \mid \theta)\, \mathrm{d}\underline{x}.$$

Since $\hat{\theta}$ is *constant* in θ and assuming we can interchange the derivative and integral, we arrive at:

$$\mathrm{E}[\hat{\theta} U] = \frac{\partial}{\partial \theta} \int \hat{\theta}(\underline{x}) f(\underline{x} \mid \theta)\, \mathrm{d}\underline{x}.$$

This assumption forms our third regularity condition:

> **Regularity Condition 3:** We assumed we can interchange the integral and derivative in the step
>
> $$\int \frac{\partial}{\partial \theta} \left(\hat{\theta}(\underline{x}) f(\underline{x} \mid \theta)\right)\, \mathrm{d}\underline{x} = \frac{\partial}{\partial \theta} \int \hat{\theta}(\underline{x}) f(\underline{x} \mid \theta)\, \mathrm{d}\underline{x}$$

Note now that $\int \hat{\theta}(\underline{x}) f(\underline{x} \mid \theta)\, \mathrm{d}\underline{x} = \mathrm{E}[\hat{\theta}]$ and, since we assumed $\hat{\theta}$ is **unbiased**, we have

$$\int \hat{\theta}(\underline{x}) f(\underline{x} \mid \theta)\, \mathrm{d}\underline{x} = \mathrm{E}[\hat{\theta}] = \theta.$$

Hence,

$$\mathrm{E}[\hat{\theta} U] = \frac{\partial}{\partial \theta}\, \theta = 1.$$

Also, from Step 2, $\mathrm{E}[U] = 0$. Therefore,

$$\mathrm{Cov}(\hat{\theta}, U) = \mathrm{E}[\hat{\theta}U] - \mathrm{E}[\hat{\theta}]\,\mathrm{E}[U] = 1 - \theta \cdot 0 = 1.$$

Conclusion:

Putting Steps 1–4 together:

$$\mathrm{Var}(\hat{\theta}) \;\geq\; \frac{\mathrm{Cov}(\hat{\theta}, U)^2}{\mathrm{Var}(U)} \;=\; \frac{1^2}{\mathcal{I}_n(\theta)} \;=\; \frac{1}{\mathcal{I}_n(\theta)}.$$

This completes the proof of the **Cramér–Rao inequality**.

□

> ⚠ Non-Examinable Content
>
> The *regularity conditions* required for the CRLB are **not examinable**.
> Informally, these conditions rule out "badly behaved" cases; for example, densities $f(x \mid \theta)$ with infinite variance, infinite expectation, or parameter-dependent support. They also justify steps such as interchanging derivatives and integrals in the proof.
> If you are curious, see the non-examinable proof in Proof 5.4.2.

Definition 5.8 (Efficiency of an Estimator)**.**

- An unbiased estimator is called **efficient** if its variance *achieves* the CRLB exactly:

$$\mathrm{Var}(\hat{\theta}) = \frac{1}{\mathcal{I}_n(\theta)}.$$

- A sequence of estimators $\hat{\theta}_n$ is **asymptotically efficient** if, as the sample size grows, its variance approaches the CRLB[1]:

$$\lim_{n\to\infty} \frac{\mathrm{Var}(\hat{\theta}_n)}{1/\mathcal{I}_n(\theta)} \;=\; 1,$$

  where $\mathcal{I}_n(\theta)$ is the Fisher information 5.1 for the **whole** sample of size n.

Corollary 5.1. *By the CRLB and Fisher Information for IID Data 5.1, the variance of an efficient estimator decreases at rate $1/n$ as the sample size increases.*

[1] **Asymptotic Equivalence.** Equivalently, we could say $\mathrm{Var}(\hat{\theta}_n) \sim \frac{1}{\mathcal{I}_n(\theta)}$, where "$\sim$" here means *asymptotically equivalent.*

5.5 The Asymptotic Distribution of the MLE

Previously, we introduced asymptotic 6 properties of estimators.

In particular, we saw that we can determine the asymptotic sampling distribution 4.3 of an estimator:

> The Central Limit Theorem 4.1 told us that the sampling distribution of the sample mean 4.1 is approximately Normal for large n.

This gave us a first example of the principle: with enough data, many estimators behave like a Normal random variable around the true parameter.

A much more general and powerful result is that maximum likelihood estimators also satisfy this property.

Theorem 5.3 (Asymptotic Distribution of the MLE)**.** *Let $\hat{\theta}_n$ denote the maximum likelihood estimator 5.3 based on n i.i.d. samples $X_1, \dots, X_n$ drawn from the same distribution with PDF/PMF $f(x \mid \theta)$.*

Under regularity conditions*, the MLE has the following asymptotic distribution:*

$$\sqrt{n}\,(\hat{\theta}_n - \theta) \xrightarrow{d} \mathcal{N}(0, \mathcal{I}(\theta)^{-1}),$$

*where $\mathcal{I}(\theta) = -\mathrm{E}\left[\frac{\partial^2}{\partial\theta^2} \log f(X \mid \theta)\right]$ is the Fisher Information 5.1 for **a single observation**.*

Proof of **??**. The full proof uses advanced tools such as Taylor expansions of the score function and the Central Limit Theorem for sums of random variables.

Rough sketch:

□

> ⚠ Non-Examinable Content
>
> The *regularity conditions* required for the Asymptotic Distribution of the MLE are **not examinable**.

In this course, the main use of the Theorem 5.3 will be to approximate the MLE distribution using the normal distribution for sufficiently large n:

Definition 5.9 (Approximate Sampling Distribution of the MLE)**.** From the asymptotic distribution of the MLE 5.3, we obtain the following approximation:

$$\hat{\theta}_n \overset{\text{approx.}}{\sim} \mathcal{N}\big(\theta,\ \tfrac{1}{n}\,\mathcal{I}(\theta)^{-1}\big), \qquad \text{for large } n.$$

In practice, since the true parameter θ is unknown, we replace θ by the observed MLE $\hat{\theta}_{\text{obs}}$. This gives the practical approximation

$$\hat{\theta}_n \overset{\text{approx.}}{\sim} \mathcal{N}\left(\theta,\ \tfrac{1}{n}\,\mathcal{I}(\hat{\theta}_{\text{obs}})^{-1}\right).$$

This Normal distribution serves as an **approximate sampling distribution** of the MLE and underpins methods such as constructing confidence intervals and hypothesis tests.

Example 5.15 (Asymptotic Distribution of the Poisson MLE)**.** Suppose $X_1, X_2, \dots, X_n$ is an i.i.d. random sample from a Poisson(λ) distribution.

We know that the maximum likelihood estimator for λ is

$$\hat{\lambda} = \bar{X}.$$

What is the asymptotic distribution of $\hat{\lambda} = \bar{X}$ as $n \to \infty$?

In Example 5.14, we saw the Fisher information for one Poisson observation is

$$\mathcal{I}(\lambda) = \frac{1}{\lambda}.$$

For n independent observations this scales to

$$\mathcal{I}_n(\lambda) = \frac{n}{\lambda}.$$

Therefore,

$$\frac{1}{\mathcal{I}_n(\lambda)} = \frac{\lambda}{n}.$$

It follows immediately from Theorem 5.3 that

$$\hat{\lambda} \overset{\text{approx.}}{\sim} \mathcal{N}\big(\lambda,\ \tfrac{\lambda}{n}\big), \quad \text{as } n \to \infty.$$

6 Bayesian Inference

6.1 Introduction to the Bayesian Approach

So far we have taken a **frequentist perspective**: parameters such as θ are fixed but unknown, and all randomness comes from the sample.

Bayesian inference takes a different view:

- **Frequentist**: the parameter θ is **fixed but unknown**.
- **Bayesian**: the parameter θ is treated as a **random variable**.

The word *random* here does **not** mean that the parameter is physically fluctuating. Instead, it represents our **uncertainty** about its true value.

6.1.1 Probability as Uncertainty

In the Bayesian framework, probability is interpreted as a way of **quantifying uncertainty** about unknown quantities.

Rather than saying "*the parameter is a fixed number we don't know*", we describe how plausible different values are using a probability distribution.

Example 6.1 (Umbrella Decisions)**.** Imagine the weather forecast says there is a 70% chance of rain tomorrow.

- The sky doesn't "rain in 70% of parallel universes."
- Instead, the number 70% reflects our uncertainty given current information (satellite data, forecasts, etc.).

Example 6.2 (Exam Marks)**.** Suppose θ is the average exam mark of a module. From past years, we believe:

- It's usually around 60 marks,
- but could reasonably be anywhere between 50 and 70.

We could represent this belief as a Normal prior distribution centred at 60 with standard deviation 10.

Example 6.3 (Disease Prevalance)**.** Let θ be the proportion of people with a rare condition.

Before collecting data, experts might believe θ is "probably around 2%, but definitely less than 10%".

We could express this with a **?@sec-beta** prior.

This provides a **language of degrees of belief**: probability distributions describe how plausible different values of θ are, given what we know.

6.1.2 The Bayesian Inference Procedure

The Bayesian approach to inference has three ingredients:

1. **Prior** — a probability distribution that captures what we believe about θ *before* seeing the data.
2. **Likelihood** — the distribution of the data $\underline{x}$ given θ.
3. **Posterior** — the updated distribution for θ that combines prior information with the evidence from the data.

The update is performed using **Bayes' theorem** (introduced in the next section). Intuitively:

$$\text{Posterior} \quad \propto \quad \text{Likelihood} \times \text{Prior}.$$

This equation expresses the central idea: *our beliefs about θ are revised in light of the observed data.*

6.2 Bayes' Theorem for Densities

Bayesian inference is named after Bayes' Theorem 1.2, a general identity in probability theory. For two events A and B, the conditional probability is given by

$$\Pr(A \mid B) = \frac{\Pr(B \mid A)\Pr(A)}{\Pr(B)}.$$

This rule is sometimes described as **reversing conditioning**: it lets us swap the order, turning "B given A" into "A given B," with a correction for their overall probabilities.

To apply this idea in Bayesian inference, we extend it from events to random variables and their densities. Using conditional densities 1.7,

$$f_{X|Y}(x \mid y) = \frac{f_{XY}(x,y)}{f_Y(y)},$$

we arrive at Bayes' theorem for densities.

Theorem 6.1 (Bayes' Theorem for Densities.)**.** *For random variables X and Y with joint density (or PMF) $f(x,y)$:*

$$f(x \mid y) = \frac{f(y \mid x)\, f(x)}{f(y)}.$$

Proof. By the definition of conditional density:

$$f(x \mid y) = \frac{f(x,y)}{f(y)}, \qquad f(y \mid x) = \frac{f(x,y)}{f(x)}.$$

Rearranging the second expression gives $f(x,y) = f(y \mid x)f(x)$. Substituting this into the first gives

$$f(x \mid y) = \frac{f(y \mid x)f(x)}{f(y)}.$$

□

6.3 Bayes' Theorem in Statistics

Once we have specified a parameter θ, the model tells us the **distribution of the data**:

$$f(\underline{x} \mid \theta).$$

This is the likelihood 5 (the joint density/PMF of the observed sample under θ).

Our Bayesian goal is the reverse: to describe the **distribution of the parameter given the data**:

$$\pi(\theta \mid \underline{x}).$$

This is exactly the "reversing conditioning" step that Bayes' theorem 1.2 makes possible. It takes the likelihood $f(\underline{x} \mid \theta)$ and the prior $\pi(\theta)$ and turns them into the posterior $\pi(\theta \mid \underline{x})$:

$$\text{Known: } f(\underline{x} \mid \theta) \quad \xRightarrow[\text{Bayes' theorem}]{} \quad \text{Wanted: } \pi(\theta \mid \underline{x})$$

Remark (Likelihood vs. Likelihood Function). Here we use **likelihood** to mean the joint density/PMF $f(\underline{x} \mid \theta)$ itself.

When viewed as a function of θ (with the data fixed), this is the likelihood function 5.1. In this course, we will not stress this distinction further.

> **! Important 10: Notation in Bayesian Statistics**
>
> Both frequentist and Bayesian inference involve probability distributions for the **data**.
>
> - In the **frequentist setting** we write
>
> $$X_i \sim \text{Distribution}(\theta),$$
>
> where θ is a fixed but unknown parameter.
>
> - In the **Bayesian setting** we write
>
> $$X_i \mid \theta \sim \text{Distribution}(\theta),$$
>
> where the conditioning bar $\mid \theta$ emphasises that θ itself is random, and we are describing the distribution of the data **given** a particular value of θ.
>
> In both cases, $f(x_i \mid \theta)$ denotes the PDF/PMF of the data conditional on θ.
>
> ---
>
> **The Bayesian addition:** we also introduce probability distributions *over the parameter θ itself*. To emphasise the difference in notation:
>
> - **Prior distribution** (before data): $\pi(\theta)$
> - **Posterior distribution** (after data): $\pi(\theta \mid \underline{x})$
>
> In Bayesian inference there are therefore two distinct kinds of distributions, which our notation keeps separate:
>
> - **Data distributions:** $f(x \mid \theta)$ describe how the data would look *if we knew θ*.
> - **Parameter distributions:** $\pi(\theta)$ and $\pi(\theta \mid \underline{x})$ describe our *uncertainty about θ*.

6.3.1 From Bayes' theorem to the Posterior

To apply Bayes' theorem 6.1 in statistics we make the identification:

- $x = \theta$: the **parameter**, treated as a random variable.

- $y = \underline{x}$: the **observed data**.

This leads directly to the **posterior distribution**:

$$\pi(\theta \mid \underline{x}) = \frac{f(\underline{x} \mid \theta)\,\pi(\theta)}{f(\underline{x})}.$$

Definition 6.1 (Prior Distribution.)**.** The **prior distribution** represents our uncertainty about θ *before observing the data.*

We denote its density (PDF or PMF) by

$$\pi(\theta).$$

Definition 6.2 (Posterior Distribution)**.** The **posterior distribution** represents our uncertainty about θ *after observing the data.*

It is defined by

$$\pi(\theta \mid \underline{x}) = \frac{f(\underline{x} \mid \theta)\,\pi(\theta)}{f(\underline{x})}.$$

- The **likelihood** $f(\underline{x} \mid \theta)$ is the term where data and parameter appear together; it is this dependence that updates the prior into the posterior.
- The **prior** $\pi(\theta)$ reflects what we believed about θ before seeing the data.
- The denominator $f(\underline{x})$ is a normalising constant ensuring the posterior integrates to 1.

6.3.2 Posteriors in Practice

We now work through a simple example to see how this update works.

Example 6.4 (Biased Coin)**.** Let $\theta = \Pr(\text{Head})$ for a possibly biased coin.

- **Prior:** Before seeing data, we express no preference over $\theta \in (0, 1)$ by choosing a **uniform prior**:

$$\theta \sim \text{Uniform}(0, 1) \equiv \text{Beta}(1, 1), \qquad \pi(\theta) = 1 \text{ for } 0 < \theta < 1.$$

  This prior has mean $\mathrm{E}[\theta] = 0.5$.

- **Data:** We toss the coin $n = 5$ times and observe $x = 1$ head.

- **Posterior:** Bayes' theorem combines the Binomial likelihood with the prior to give $\pi(\theta \mid x)$.

Let X be the number of heads in 5 tosses. Assuming that each coin throw is *independent and identically distributed*, X is Binomial[1]:

$$X \mid \theta \sim \text{Bin}(5, \theta).$$

Note that there is only a **single observation** here: one experiment is "tossing the coin 5 times and seeing how many come up heads". The probability of observing $X = 1$ is

$$f(x = 1 \mid \theta) = 5\theta(1 - \theta)^4$$

If we plot this:

```
import matplotlib.pyplot as plt
import numpy as np
import math

set_plot_theme("light")

def binom_lik(theta):
    return 5 * theta * (1 - theta) ** 4


theta_grid = np.linspace(0, 1, 200)
binom_lik_vals = binom_lik(theta_grid)

plt.figure(figsize=(5,3))
plt.plot(theta_grid, binom_lik_vals)
plt.xlabel(r"$\theta$")
plt.ylabel("Likelihood")
plt.title("Binomial Likelihood (with $n=5$ and $X=1$)")
plt.tight_layout()
plt.show()

set_plot_theme("dark")

plt.figure(figsize=(5,3))
```

[1]**Binomial distribution.** If we assume each coin throw C_i is an i.i.d. Bernoulli(θ) trial with a 1 representing a heads landing, and a 0 representing a tails, the total number of heads out of 5 is $X = C_1 + C_2 + C_3 + C_4 + C_5$. Therefore:

$$X \mid \theta \sim \text{Bin}(5, \theta).$$

```python
plt.plot(theta_grid, binom_lik_vals)
plt.xlabel(r"$\theta$")
plt.ylabel("Likelihood")
plt.title("Binomial Likelihood (with $n=5$ and $X=1$)")
plt.tight_layout()
plt.show()
```

we see that this favours θ around 0.2. In fact, in Example 5.9, we saw that the MLE is $\hat{\theta} = 0.2$.

Our prior for θ was $\pi(\theta) = 1$, for $0 < \theta < 1$. To update our beliefs, by conditioning on the single observation $x = 2$, we use Bayes' Theorem 6.2:

$$\pi(\theta|x = 1) = \frac{\pi(\theta)f(x = 1|\theta)}{f(x = 1)} = \frac{1 \times 5\theta(1-\theta)^4}{f(x = 1)}, \quad 0 < \theta < 1.$$

To compute the *denominator*, $f(x = 1)$, we use the law of total probability 1.3:

$$\begin{aligned} f(x = 1) &= \int_{\Theta} \pi(\theta) f(x = 1 \,|\, \theta)\, d\theta = \int_0^1 1 \times 5\theta(1-\theta)^4\, d\theta \\ &= \int_0^1 \theta \times 5(1-\theta)^4\, d\theta = \left[-(1-\theta)^5\,\theta\right]_0^1 + \int_0^1 (1-\theta)^5\, d\theta \\ &= 0 + \left[-\frac{(1-\theta)^6}{6}\right]_0^1 = \frac{1}{6}. \end{aligned}$$

Therefore, the posterior density is (for $0 < \theta < 1$):

$$\begin{aligned} \pi(\theta|x = 1) &= \frac{\pi(\theta)f(x = 1|\theta)}{f(x = 1)} = \frac{5\theta(1-\theta)^4}{1/6} \\ &= 30\,\theta(1-\theta)^4 = \frac{\theta(1-\theta)^4}{\mathrm{B}(2,5)}, \qquad 0 < \theta < 1, \end{aligned}$$

and so the posterior distribution is $\theta|x = 1 \sim \mathrm{Beta}(2, 5)$. This distribution has its mode at $\theta = 0.2$, and mean at $\mathrm{E}[\theta|x = 1] = 2/7 = 0.286$.

The main difficulty in calculating the posterior distribution was in obtaining the $f(x)$ term. However, in many cases we can recognise the posterior distribution without the need to calculate this constant term (constant with respect to θ). In this example, we can calculate the posterior distribution as

$$\begin{aligned}\pi(\theta|\underline{x}) &\propto \pi(\theta) f(x=1|\theta) \\ &\propto 1 \times 5\theta(1-\theta)^4, \qquad 0<\theta<1 \\ &= k\theta(1-\theta)^4, \qquad 0<\theta<1.\end{aligned}$$

As θ is a continuous quantity, what we would like to know is what continuous distribution defined on $(0,1)$ has a probability density function which takes the form $k\theta^{g-1}(1-\theta)^{h-1}$. The answer is the Beta$(g,h)$ distribution. Therefore, choosing g and h appropriately, we can see that the posterior distribution is $\theta|x=1 \sim \text{Beta}(2,5)$.

Interpretation:

- **Most likely value (mode):** $\hat{\theta} \approx 0.2$, the same as the sample proportion $1/5$.
- **Posterior mean:** $\text{E}[\theta \mid x] \approx 0.286$, slightly larger than 0.2 because the prior was centred at 0.5.
- **Uncertainty reduced:** the posterior standard deviation shrinks from 0.289 (prior) to 0.160 (posterior).

So, after seeing the data, our beliefs about θ have been updated: we now think small values of θ are more plausible, but not as extremely as the raw data alone would suggest.

Example 6.5. Consider an experiment to determine how good a music expert is at distinguishing between pages from Haydn and Mozart scores. Let $\theta = \Pr(\text{correct choice})$. Suppose that, before conducting the experiment, we have been told that the expert is very competent. In fact, it is suggested that we should have a prior distribution which has a mode around $\theta = 0.95$ and for which $\Pr(\theta < 0.8)$ is very small. We choose $\theta \sim \text{Beta}(77,5)$, with probability density function

$$\pi(\theta) = 128107980\,\theta^{76}(1-\theta)^4, \qquad 0<\theta<1.$$

A graph of this prior density is given as follows:

6.3.3 Python

```python
import numpy as np
import math
import matplotlib.pyplot as plt

set_plot_theme("light")

# Beta( , ) with =77, =5 → ( ) ^(76) (1- )^(4)
alpha, beta = 77, 5

# B( , ) = Γ( )Γ( )/Γ( + )     1/B = exp(lgamma( + ) - lgamma( ) - lgamma( ))
# Note the lgamma function computes the logarithm of the Gamma function.
beta_const_inv = math.exp(math.lgamma(alpha + beta) - math.lgamma(alpha) - math.lgamma(beta))

def beta_pdf(theta):
    return beta_const_inv * np.power(theta, alpha - 1) * np.power(1 - theta, beta - 1)

theta_grid = np.linspace(0.0, 1.0, 500)
pdf_values = beta_pdf(theta_grid)

plt.figure(figsize=(7,4))
plt.plot(theta_grid, pdf_values)
plt.xlabel(r"$\theta$")
plt.ylabel("Density")
plt.title(rf"Beta PDF: $\alpha={alpha}$, $\beta={beta}$")
plt.tight_layout()
plt.show()

set_plot_theme("dark")

plt.figure(figsize=(7,4))
plt.plot(theta_grid, pdf_values)
plt.xlabel(r"$\theta$")
plt.ylabel("Density")
plt.title(rf"Beta PDF: $\alpha={alpha}$, $\beta={beta}$")
plt.tight_layout()
plt.show()
```

6.3.4 R

```r
#| echo: true
#| code-fold: true
```

```r
#| code-summary: "Show code"
#| fig-align: center

alpha <- 77
beta  <- 5

# Normalising constant via log-gamma (works like Python's lgamma)
beta_const_inv <- exp(lgamma(alpha + beta) - lgamma(alpha) - lgamma(beta))

beta_pdf <- function(theta) {
  beta_const_inv * theta^(alpha - 1) * (1 - theta)^(beta - 1)
}

theta_grid <- seq(0, 1, length.out = 500)
pdf_values <- beta_pdf(theta_grid)

plot(theta_grid, pdf_values, type = "l",
     xlab = expression(theta), ylab = "Density",
     main = bquote("Beta PDF: " ~ alpha == .(alpha) ~ "," ~ beta == .(beta)))
```

Example 6.6. Software engineers at **Instagram** are interested in the click-through rate θ of their advertisement recommender.

The **click-through rate** is the proportion of clicks out of the total number of impressions of an online advertisement.

The engineers assess the click-through rate by testing their recommender on users of Instagram. They obtain a total of n impressions.

(a) Using a **?@sec-beta**, quantify your prior beliefs about θ.
(b) After running the test, the engineers obtained a total of $X = 47$ successful clicks out of a total of $n = 10,000$ impressions. Using a **Binomial** model and your prior specified in part (a), obtain the posterior distribution for θ.

a.

I personally never click on adverts, apart from on accident. I don't think many people do. So, I believe that the click-through rate θ should be close to 0.

I am not very confident, so I will pick:

$$\theta \sim \text{Beta}(5,\, 100).$$

This prior has mean

$$\frac{5}{5+100} \approx 0.047,$$

and places the vast majority of its probability mass on values $\theta < 0.1$.

---

b.

The data model is

$$X \mid \theta \sim \text{Binomial}(n = 10000,\, \theta), \qquad X = 47,$$

which has PMF

$$f(x = 47 \mid \theta) = \binom{10000}{47} \theta^{47}(1-\theta)^{9953}.$$

Using Bayes' theorem:

$$\pi(\theta \mid x) \;\propto\; f(x \mid \theta)\, \pi(\theta).$$

From part (a) the prior is

$$\pi(\theta) \;\propto\; \theta^{\alpha-1}(1-\theta)^{\beta-1}, \qquad \alpha = 5,\; \beta = 100.$$

Therefore,

$$\pi(\theta \mid x) \;\propto\; \theta^{47}(1-\theta)^{9953} \;\cdot\; \theta^{\alpha-1}(1-\theta)^{\beta-1}.$$

Simplifying exponents:

$$\pi(\theta \mid x) \;\propto\; \theta^{(\alpha+47)-1}(1-\theta)^{(\beta+9953)-1}.$$

Thus, the posterior distribution is

$$\theta \mid x \sim \text{Beta}(\alpha + 47,\; \beta + 9953).$$

Substituting $\alpha = 5$, $\beta = 100$:

$$\theta \mid X \sim \text{Beta}(5 + 47,\ 100 + 9953) = \text{Beta}(52,\ 10053).$$

Interpretation.

- The posterior mean is

$$\text{E}[\theta \mid X] = \frac{52}{52 + 10053} \approx 0.0052,$$

which is about **0.52% click-through rate**.

- The posterior variance is very small because $n = 10{,}000$ is large, so the engineers can be quite confident that θ is very close to 0.5.

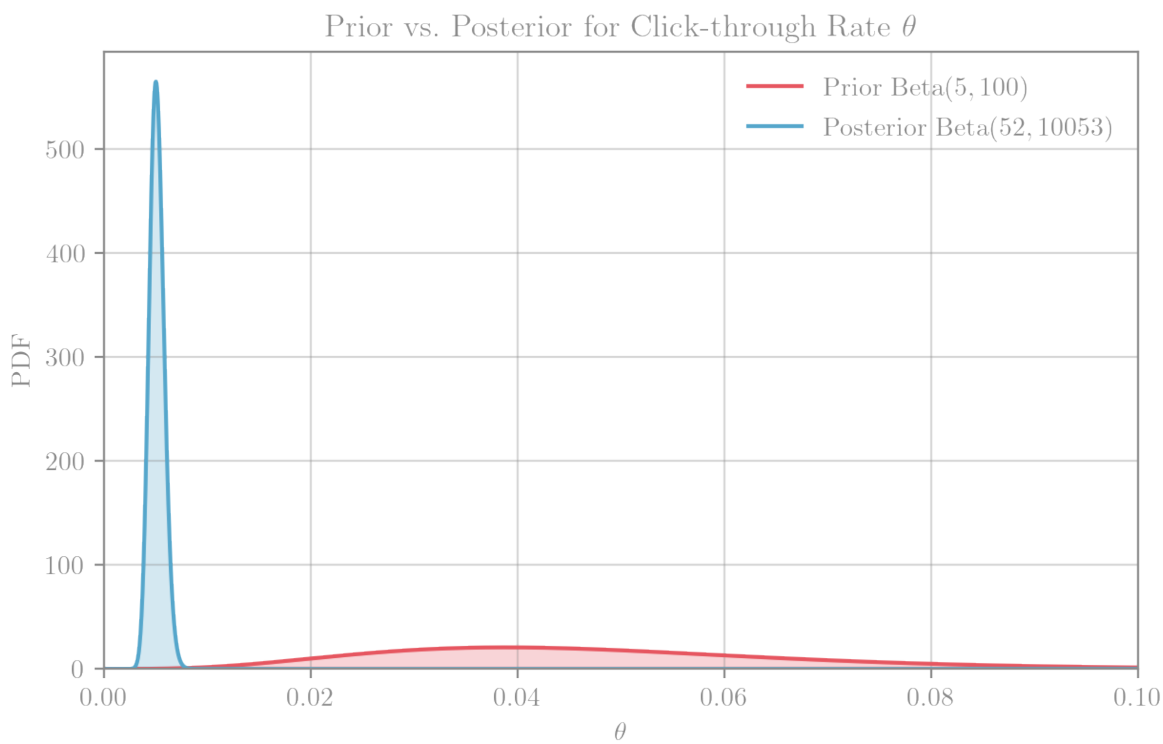

Figure 6.1: Beta prior and posterior for the Instagram click-through rate parameter θ after observing $X = 47$ clicks out of $n = 10{,}000$ impressions. The posterior is sharply concentrated near ~0.5%

Example 6.7 (Measuring a particle's speed)**.** You are attempting to measure the **speed of a particle** θ and quantify the uncertainty in the measurement. The experiment is set up such that the particle **travels** 1 km and the measurements are the speeds $X_i \mid \theta$ for $i = 1, \dots, n$ in kilometres per second (km/s).

There are a **multitude of factors** that additively result in **measurement error** for the speed of the particle.

(a) Using a **normal prior**, quantify **your prior beliefs** about θ.

Since speed is physically constrained, a normal prior is imperfect because it places mass on the whole real line. For speed we should have $0 < \theta < c$, where $c = 299{,}792$km/s (speed of light).

Nonetheless, if we are asked to use a normal prior and have little prior knowledge about the particle type, a weakly informative prior is reasonable, e.g.

$$\theta \sim \mathcal{N}(1.5 \times 10^5,\ (4 \times 10^4)^2).$$

This concentrates most mass in the physically plausible range while remaining diffuse.

(b) Why is a **normal distribution** appropriate for the likelihood? Write down the likelihood.

Because many small, additive, independent sources of error contribute to each measurement, the Central Limit Theorem 4.1 motivates

$$X_i \mid \theta \overset{\text{iid}}{\sim} \mathcal{N}(\theta, \sigma^2).$$

Hence the likelihood is

$$\begin{aligned} f(\underline{x} \mid \theta, \sigma) &= (2\pi)^{-n/2} \sigma^{-n} \exp\left\{ -\frac{1}{2\sigma^2} \sum_{i=1}^{n} (x_i - \theta)^2 \right\} \\ &= (2\pi)^{-n/2} \sigma^{-n} \exp\left\{ -\frac{1}{2\sigma^2} \left(\sum_{i=1}^{n} x_i^2 - 2\theta \sum_{i=1}^{n} x_i + n\theta^2 \right) \right\}. \end{aligned}$$

(c) Assuming the **standard deviation** σ of the measurements is known, derive the posterior under a prior $\theta \sim \mathcal{N}(\mu_0, \sigma_0^2)$.

Up to proportionality,

$$f(\underline{x} \mid \theta, \sigma) \;\propto\; \exp\left\{-\frac{1}{2\sigma^2}(n\theta^2 - 2n\bar{x}\,\theta)\right\}, \qquad \pi(\theta) \;\propto\; \exp\left\{-\frac{1}{2\sigma_0^2}(\theta^2 - 2\mu_0\theta)\right\}.$$

Multiplying and completing the square, the posterior is normal with precision (inverse variance)

$$\frac{1}{V} = \frac{n}{\sigma^2} + \frac{1}{\sigma_0^2}, \qquad M = V\left(\frac{n\bar{x}}{\sigma^2} + \frac{\mu_0}{\sigma_0^2}\right).$$

Therefore,

$$\boxed{\theta \mid \underline{x} \;\sim\; \mathcal{N}(M, V)}.$$

(d) Suppose you measured the speed 4 times and obtained (km/s): 306135, 293227, 307985, 301298. Using the prior from (a), set $\sigma = 5000$ and derive your **posterior distribution**.

Compute $\bar{x} = (306135 + 293227 + 307985 + 301298)/4 = 302{,}161.25$. With $\mu_0 = 150{,}000$, $\sigma_0^2 = (40{,}000)^2$, $\sigma^2 = (5{,}000)^2$, $n = 4$:

$$\frac{1}{V} = \frac{n}{\sigma^2} + \frac{1}{\sigma_0^2} = \frac{4}{25{,}000{,}000} + \frac{1}{1{,}600{,}000{,}000} = 1.60625 \times 10^{-7}, \quad V = \frac{1}{1.60625 \times 10^{-7}} \approx 6{,}225{,}681.$$

$$M = V\left(\frac{n\bar{x}}{\sigma^2} + \frac{\mu_0}{\sigma_0^2}\right) \approx 6{,}225{,}681 \left(\frac{4 \cdot 302{,}161.25}{25{,}000{,}000} + \frac{150{,}000}{1{,}600{,}000{,}000}\right) \approx 301{,}569.18.$$

Thus,

$$\boxed{\theta \mid \underline{x} \;\sim\; \mathcal{N}(301{,}569.2,\; 6{,}225{,}681)} \quad \text{(i.e., SD } \approx 2{,}495.1 \text{ km/s).}$$

6.3.5 Conjugacy

In all the examples in Posteriors in Practice 6.3.2, a notable phenomenon occurred:

1. The **prior** distribution $\pi(\theta)$
2. The resulting **posterior** distribution $\pi(\theta \mid \underline{x})$

belonged to the same *family of distributions*.

In Bayesian inference, this property is called **conjugacy**, and it allows us to derive posterior distributions in closed form without approximation.

Definition 6.3 (Conjugate Prior)**.** Suppose $f(\underline{x} \mid \theta)$ is the joint PDF/PMF of the observations.

A prior distribution for θ is said to be **conjugate** if, after updating with the data, the posterior distribution $\pi(\theta \mid \underline{x})$ remains in the same family as the prior $\pi(\theta)$.

Remark 6.1 (Why is conjugacy useful?).

- Conjugacy makes Bayesian updating *algebraically simple*: posterior parameters can often be obtained by just "updating counts" or "adding observations".
- This provides intuition for how prior information and data combine, and gives closed-form formulas that are easy to compute.

6.4 Asymptotic Posterior Distribution

In **frequentist statistics**, we saw two important results about the asymptotic 6 sampling distribution of estimators:

1. The Central Limit Theorem 4.1 showed that the sampling distribution of the sample mean 4.1 is approximately Normal for large n.
2. The Asymptotic Distribution of the MLE 5.3 showed that the sampling distribution of the maximum likelihood estimator is approximately Normal for large n.

A similar phenomenon occurs in **Bayesian inference**: for large samples, the **posterior distribution** itself becomes approximately Normal, regardless of the prior (as long as it is not too extreme).

Theorem 6.2 (Bernstein–von Mises)**.** *Suppose we have an i.i.d. sample $X_1, \dots, X_n$ drawn from the same distribution with PDF/PMF $f(x \mid \theta)$ and let $\pi(\theta \mid \underline{X}^{(n)})$ denote the posterior distribution for θ.*

Under regularity conditions*, as $n \to \infty$ the posterior distribution satisfies*

$$\theta \mid \underline{X}^{(n)} \overset{approx.}{\sim} \mathcal{N}\Big(\widehat{\theta}_n,\, \mathcal{I}_n(\theta)^{-1}\Big),$$

where

- $\widehat{\theta}_n$ *is the MLE,*
- $\mathcal{I}_n(\theta)$ *is the Fisher information 5.1 for the* ***whole sample****.*

Part III

Part III - Interval Estimation

7 Frequentist Interval Estimation

Recalling the main inference tasks 3.2, we now move from *point estimation* to *interval estimation*.

The key idea is that instead of giving a single "best guess"[1] of a parameter, we want to provide a **range of plausible values**. In other words, an **interval**.

7.1 The Frequentist Perspective

Recall that, for frequentists, the parameter θ is **fixed but unknown**.

The randomness comes from the data: if we repeated the experiment many times, we would get different samples, different estimates, and hence different intervals.

- **Point Estimate:** a single statistic $\hat{\theta}(\underline{X})$.
- **Interval estimate:** two statistics $L(\underline{X})$ and $U(\underline{X})$ that form a *random interval*.

Definition 7.1 (Confidence Interval)**.** A $100(1-\alpha)\%$ **confidence interval** for θ is a *random interval*

$$[L(\underline{X}),\ U(\underline{X})]$$

satisfying

$$\Pr\big(L(\underline{X}) < \theta < U(\underline{X})\big) = 1 - \alpha.$$

This property is called **coverage**: in the long run, a proportion $1-\alpha$ of such intervals (constructed from repeated experiments) will contain the true θ.

[1] Recall from Chapter 6 that Bayesians do not return a single "best guess," but a full **posterior distribution** over parameter values. From this distribution we can compute **summaries** such as plausible ranges of θ (see Chapter 8), which play a role similar to confidence intervals but with a different interpretation.

Important 11: Terminology: Exact vs. Approximate Confidence Intervals

A **confidence interval (CI)** is a random interval, constructed from the data, that aims to contain the true parameter value with a specified probability (the *coverage probability*).

- **Exact CI.** The coverage probability is *exactly* equal to the nominal level (e.g. 95%). Such intervals are sometimes called *well-calibrated*. These are rare, but they can be derived in some classical settings (e.g. certain normal models).
- **Approximate CI.** The coverage probability is only guaranteed *asymptotically* (typically as $n \to \infty$). In practice, most commonly used intervals are of this type.

7.2 Interpreting Confidence Intervals

In practice, we only ever **observe one dataset** $\underline{x}$. So we calculate the observed interval:

$$[L_{\text{obs}}, U_{\text{obs}}] = [L(\underline{x}), U(\underline{x})].$$

Once the data are observed, this interval is **fixed, not random**.

Therefore, a frequentist would **not** say

$$\Pr(L_{\text{obs}} < \theta < U_{\text{obs}}) = 1 - \alpha.$$

That statement is meaningless, because θ is not random in the frequentist view.

Instead, the correct interpretation is: **our method has long-run coverage $1 - \alpha$.**

7.3 Exact Confidence Intervals

Before turning to specific examples, it is useful to outline the *general strategy* for constructing an exact confidence intervals 11.

7.3.1 General Recipe

1. **Choose an estimator.** Start with an estimator $\hat{\theta} = \hat{\theta}(X_1, \dots, X_n)$ for the parameter of interest θ.

2. **Identify its sampling distribution.** For exact intervals, the *exact* distribution of $\hat{\theta}$ under the model must be known.
3. **Form a pivotal quantity.** Construct a function $Z = Z(\underline{X}, \theta)$ whose distribution does not depend on any unknown parameters. Such a Z is called a **pivot**.
4. **Use quantiles of the pivot.** From the known distribution of Z, find constants a, b such that

$$\Pr\big(a \leq Z \leq b\big) = 1 - \alpha.$$

5. **Invert the inequalities.** Solve $a \leq Z(\underline{X}, \theta) \leq b$ for θ. The result is an exact $(1 - \alpha)$ confidence interval for θ.

> ⚠ Non-Examinable Content
>
> This *general strategy* for constructing exact confidence intervals is **not examinable**. In particular, you do **not** need to remember the term *pivot*.

7.3.2 The Normal Case (Known Variance)

Let's apply the general strategy 7.3.1 to construct an exact confidence interval for the mean of a normal distribution with known variance.

Suppose we observe a normal random sample:

$$X_i \sim \mathcal{N}(\mu, \sigma^2), \quad i = 1, \ldots, n.$$

1. **Choose an estimator.** Use the sample mean:

$$\bar{X} = \frac{1}{n} \sum_{i=1}^{n} X_i.$$

2. **Identify its sampling distribution.** Since the X_i are normal,

$$\bar{X} \sim \mathcal{N}\left(\mu, \tfrac{\sigma^2}{n}\right).$$

3. **Form a pivotal quantity.** Standardise to remove dependence on μ:

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim \mathcal{N}(0, 1).$$

4. **Use quantiles of the pivot.** For the standard normal,

$$\Pr\Big(-z_{1-\alpha/2} \le Z \le z_{1-\alpha/2}\Big) = 1-\alpha,$$

   where $z_{1-\alpha/2}$ is the $(1-\alpha/2)$ quantile.

5. **Invert the inequalities.** Substituting Z and rearranging gives

$$\Pr\Big(\bar{X} - z_{1-\alpha/2}\,\tfrac{\sigma}{\sqrt{n}} \;\le\; \mu \;\le\; \bar{X} + z_{1-\alpha/2}\,\tfrac{\sigma}{\sqrt{n}}\Big) = 1-\alpha.$$

Therefore, an *exact* $(1-\alpha)$ confidence interval for μ is

$$\mu \in \Big[\bar{X} - z_{1-\alpha/2}\,\tfrac{\sigma}{\sqrt{n}},\;\; \bar{X} + z_{1-\alpha/2}\,\tfrac{\sigma}{\sqrt{n}}\Big],$$

or equivalently,

$$\bar{X} \;\pm\; z_{1-\alpha/2}\,\frac{\sigma}{\sqrt{n}}.$$

Definition 7.2 (Confidence Interval for the Normal Mean (Known Variance))**.** Suppose $X_1, \dots, X_n \overset{\text{iid}}{\sim} \mathcal{N}(\mu, \sigma^2)$ with **known variance** σ^2. Then an exact $(1-\alpha)$ confidence interval for μ is

$$\mu \;\in\; \left[\bar{X} - z_{1-\alpha/2}\,\frac{\sigma}{\sqrt{n}},\;\; \bar{X} + z_{1-\alpha/2}\,\frac{\sigma}{\sqrt{n}}\right],$$

or equivalently

$$\bar{X} \;\pm\; z_{1-\alpha/2}\,\frac{\sigma}{\sqrt{n}},$$

where

- $\bar{X}$ is the sample mean 4.1, and
- $z_{1-\alpha/2}$ is the $(1-\alpha/2)$ quantile of the standard Normal distribution $\mathcal{N}(0,1)$.

Remark 7.1 (Quantiles of the pivot)*.* When constructing a confidence interval from a pivot, the choice of quantiles is not unique.

- In the **symmetric case**, we use

$$\Pr\Big(-z_{1-\alpha/2} \le Z \le z_{1-\alpha/2}\Big) = 1 - \alpha,$$

which gives the familiar two-sided interval.

- But we could instead use **any** pair of quantiles $z_{\alpha_1}, z_{1-\alpha_2}$ with $\alpha_1 + \alpha_2 = \alpha$:

$$\Pr\Big(z_{\alpha_1} \le Z \le z_{1-\alpha_2}\Big) = 1 - \alpha.$$

- As a special case, a **one-sided interval** arises from

$$\Pr(Z \le z_{1-\alpha}) = 1 - \alpha.$$

The symmetric interval is usually preferred because it places equal probability mass in both tails, making it "balanced" around the estimator. This choice minimises the interval length for symmetric distributions (like the normal), which is why it is *standard*.

Example 7.1 (Archaeology and Skeleton Lengths: CI with Known Variance)**.** An archaeologist found preserved in a peat bog the skeletons of 30 adult males from an unknown human population. She is interested in skeleton lengths.

The sample mean is $\bar{x} = 161.72$ cm. Assume that the skeleton lengths come from a $\mathcal{N}(\mu, \sigma^2)$ distribution.
Further, measurements from other human populations at the time suggest that it is reasonable to assume the population variance is $\sigma^2 = 100$ cm^2.

1. Construct 95% and 99% confidence intervals for μ.

2. How many skeletons would be needed to ensure that the width of a 95% confidence interval for μ is no greater than 4 cm?

Step 1. Confidence intervals for μ.

We have $\bar{x} = 161.72$, $n = 30$, $\sigma = 10$.

For a confidence interval, the formula is

$$\bar{x} \;\pm\; z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}}.$$

- For 95% confidence, $z_{0.05/2} = 1.96$:

$$161.72 \pm 1.96 \cdot \frac{10}{\sqrt{30}} = 161.72 \pm 3.578 = (158.14,\; 165.30).$$

- For 99% confidence, $z_{0.01/2} = 2.576$:

$$161.72 \pm 2.576 \cdot \frac{10}{\sqrt{30}} = 161.72 \pm 4.703 = (157.02,\ 166.42).$$

Step 2. Required sample size for desired width.

The width of a 95% CI is

$$\text{Width} = 2 \cdot z_{0.05/2} \cdot \frac{\sigma}{\sqrt{n}} = 2 \cdot 1.96 \cdot \frac{10}{\sqrt{n}}.$$

We want this ≤ 4:

$$2 \cdot 1.96 \cdot \frac{10}{\sqrt{n}} \leq 4 \quad \Rightarrow \quad \sqrt{n} \geq \frac{2 \cdot 1.96 \cdot 10}{4} = 9.8.$$

So

$$n \geq (9.8)^2 = 96.04.$$

Therefore, at least **97 skeletons** are needed.

7.3.3 The Normal Case (Unknown Variance)

Let's apply the general strategy 7.3.1 to construct an exact confidence interval for the mean of a normal distribution when the variance is **unknown**.

Suppose we observe a normal random sample with unknown mean *and* variance:

$$X_i \sim \mathcal{N}(\mu, \sigma^2), \quad i = 1, \dots, n.$$

Since σ^2 is unknown, we estimate it using the sample variance 4.2:

$$S^2 = \frac{1}{n-1} \sum_{i=1}^{n} (X_i - \bar{X})^2.$$

Aside: The Student's t Distribution

- The **?@sec-t** distribution was discovered in 1908 by William Sealy Gosset, who published under the pseudonym *"Student"*.
- It arises when we replace the (unknown) variance σ^2 in a Normal problem with its estimate from the data. Formally,

$$T = \frac{Z}{\sqrt{U/\nu}}, \quad Z \sim \mathcal{N}(0,1), \;\; U \sim \chi^2_\nu,$$

  where Z and U are independent.
- The t distribution looks very similar to the standard Normal, but has **heavier tails** - this accounts for the extra uncertainty caused by estimating σ^2.
- As the degrees of freedom ($\nu \to \infty$), the t distribution becomes almost indistinguishable from $\mathcal{N}(0,1)$.

---

1. **Choose an estimator.** Use the sample mean:

$$\bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i.$$

2. **Identify its sampling distribution.** As before,

$$\bar{X} \sim \mathcal{N}\left(\mu, \tfrac{\sigma^2}{n}\right).$$

   But now σ^2 is unknown and must be estimated by S^2.

3. **Form a pivotal quantity.** Standardise using S:

$$T = \frac{\bar{X} - \mu}{S/\sqrt{n}}.$$

   It can be shown (using χ^2 theory and independence between $\bar{X}$ and S^2) that

$$T \sim t_{n-1}.$$

4. **Use quantiles of the pivot.** For the t distribution with $n-1$ degrees of freedom,

$$\Pr\left(-t_{n-1,\,1-\alpha/2} \le T \le t_{n-1,\,1-\alpha/2}\right) = 1-\alpha,$$

   where $t_{n-1,\,1-\alpha/2}$ is the $(1-\alpha/2)$ quantile.

5. **Invert the inequalities.** Substituting T and rearranging gives

$$\Pr\left(\bar{X} - t_{n-1,\,1-\alpha/2}\,\frac{S}{\sqrt{n}} \ \le\ \mu\ \le\ \bar{X} + t_{n-1,\,1-\alpha/2}\,\frac{S}{\sqrt{n}}\right) = 1-\alpha.$$

Therefore, an *exact* $(1-\alpha)$ confidence interval for μ is

$$\mu \in \left[\bar{X} - t_{n-1,\,1-\alpha/2}\,\tfrac{S}{\sqrt{n}},\ \ \bar{X} + t_{n-1,\,1-\alpha/2}\,\tfrac{S}{\sqrt{n}}\right],$$

or equivalently,

$$\bar{X}\ \pm\ t_{n-1,\,1-\alpha/2}\,\frac{S}{\sqrt{n}}.$$

Definition 7.3 (Confidence Interval for the Normal Mean (Unknown Variance))**.** Suppose $X_1, \dots, X_n \overset{\text{iid}}{\sim} \mathcal{N}(\mu, \sigma^2)$ with **unknown variance** σ^2.

Then an exact $(1-\alpha)$ confidence interval for μ is

$$\mu\ \in\ \left[\bar{X} - t_{n-1,\,1-\alpha/2}\,\frac{S}{\sqrt{n}},\ \ \bar{X} + t_{n-1,\,1-\alpha/2}\,\frac{S}{\sqrt{n}}\right],$$

or equivalently

$$\bar{X}\ \pm\ t_{n-1,\,1-\alpha/2}\,\frac{S}{\sqrt{n}},$$

where

- $\bar{X}$ is the sample mean 4.1,
- S^2 is the sample variance 4.2, and
- $t_{n-1,1-\alpha/2}$ is the $(1-\alpha/2)$ quantile of the Student's t distribution with $n-1$ degrees of freedom.

Example 7.2 (Archaeology and Skeleton Lengths: CI with Unknown Variance)**.** Continuing with Example 7.1, suppose now that we don't assume we know the population variance.

In addition to the sample mean $\bar{x} = 161.72$, we also compute the sample variance $s^2 = 86.63$.

Provide the archaeologist with 95% and 99% confidence intervals for μ.

We need the critical values $t_{29,\,0.975}$ and $t_{29,\,0.995}$.

From using R, Python or the above visualisation with degrees of freedon $\nu = n - 1 = 29$, we have

$$t_{29,\,0.975} = 2.045 \quad \text{and} \quad t_{29,\,0.995} = 2.756.$$

Using the one-sample t CI from Definition 7.3:

$$\bar{X} \;\pm\; t_{n-1,\,1-\alpha/2}\,\frac{S}{\sqrt{n}}, \qquad n = 30,\ \bar{x} = 161.72,\ S^2 = 86.63 \;\Rightarrow\; \frac{S}{\sqrt{n}} = \sqrt{\frac{86.63}{30}} \approx 1.6993.$$

Therefore:

1. **95% CI**:
$$161.72 \;\pm\; 2.045 \times \sqrt{\frac{86.63}{30}} \;=\; (158.24,\ 165.20).$$
2. **99% CI**
$$161.72 \;\pm\; 2.756 \times \sqrt{\frac{86.63}{30}} \;=\; (157.04,\ 166.40).$$

7.4 Approximate Confidence Intervals

For more general distributions (not necessarily normal), exact CIs are not available. In such cases we rely on asymptotics 6 to derive **approximate** CIs. For example:

1. The Central Limit Theorem 4.1.
2. The Asymptotic Distribution of the MLE 5.3.

7.4.1 CLT: Approximate CI for the Mean

Suppose

$$X_1, \dots, X_n \overset{\text{iid}}{\sim} \text{Distribution}(\underline{\theta})$$

with mean μ and variance σ^2.

The sample mean 4.1 $\bar{X}$ is an estimator of μ. By the Central Limit Theorem 4.1, for large n:

$$\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \overset{\text{approx.}}{\sim} \mathcal{N}(0, 1).$$

Since σ^2 is usually unknown, we replace it with the sample variance 4.2 S^2, noting that S^2 is a consistent estimator of σ^2.

Definition 7.4 (CLT: Approximate CI for the Mean)**.** Suppose $X_1, \dots, X_n \overset{\text{iid}}{\sim} \text{Distribution}(\underline{\theta})$ with mean μ. Then an **approximate** $(1 - \alpha)$ confidence interval for μ is

$$\mu \in \left[\bar{X} - z_{1-\alpha/2} \frac{S}{\sqrt{n}}, \ \bar{X} + z_{1-\alpha/2} \frac{S}{\sqrt{n}}\right],$$

or equivalently

$$\bar{X} \pm z_{1-\alpha/2} \frac{S}{\sqrt{n}},$$

where

- $\bar{X}$ is the sample mean 4.1,
- $S = \sqrt{S^2}$ and S^2 is the sample variance 4.2,
- $z_{1-\alpha/2}$ is the $(1 - \alpha/2)$ quantile of the standard Normal distribution $\mathcal{N}(0, 1)$.

Example 7.3 (Poisson Goals: CLT Confidence Interval)**.** Suppose the **total** number of goals in a Premier league football match is assumed to be Poisson(λ) distributed.

Using the available data in Table 2.4, and assuming i.i.d. observations, and using the Central Limit Theorem 4.1 approximation, construct an approximate 95% confidence interval for the parameter λ.

Step 1: Define the model and estimator

Let $X_i \sim \text{Poisson}(\lambda)$ denote the **total** number of goals per football match. Note that the **?@sec-pois** satisfies $\mathrm{E}[X_i] = \lambda$.

Therefore, we use the sample mean 4.1

$$\hat{\lambda} = \frac{1}{n}\sum_{i=1}^{n} X_i$$

as our estimator of λ.

Step 2: Apply the CLT

From the CLT 4.1, for sufficiently large n and *known* variance $\sigma^2 = \mathrm{Var}(X_i)$, we have

$$\hat{\lambda} \overset{\text{approx.}}{\sim} \mathcal{N}\left(\lambda, \frac{\sigma^2}{n}\right).$$

We need an estimate of $\sigma^2 = \mathrm{Var}(X_i)$:

- **Option 1 (empirical variance):** $\widehat{\sigma^2} = S^2$ (sample variance).
- **Option 2 (Poisson variance):** since $\mathrm{Var}(X_i) = \lambda$, use $\widehat{\sigma^2} = \hat{\lambda}$.

Step 3: Construct the CIs

With $z_{0.975} = 1.96$:

- **Option 1 (using S^2):**

$$\lambda \in \left[\ \hat{\lambda} - 1.96 \cdot \frac{S}{\sqrt{n}}\ ,\ \hat{\lambda} + 1.96 \cdot \frac{S}{\sqrt{n}}\ \right].$$

- **Option 2 (using $\hat{\lambda}$ for Poisson variance):**

$$\lambda \in \left[\ \hat{\lambda} - 1.96 \cdot \sqrt{\frac{\hat{\lambda}}{n}}\ ,\ \hat{\lambda} + 1.96 \cdot \sqrt{\frac{\hat{\lambda}}{n}}\ \right].$$

Step 4: Apply to the data (first 10 matches shown)

From Table 2.4, the **total number** of goals per match gives the data:

Match	Home goals	Away goals	Total goals
Newcastle United vs Arsenal	7	0	$x_1 = 7$
Manchester United vs Liverpool F.C.	0	1	$x_2 = 1$
Tottenham Hotspur vs Chelsea F.C.	1	3	$x_3 = 4$
Aston Villa vs Leeds United	3	2	$x_4 = 5$
West Ham United vs Manchester City	0	4	$x_5 = 4$

Match	Home goals	Away goals	Total goals
Brighton & Hove Albion vs Everton	2	2	$x_6 = 4$
Leicester City vs Southampton	1	0	$x_7 = 1$
Wolverhampton Wanderers vs Brentford	2	1	$x_8 = 3$
Crystal Palace vs Fulham	0	1	$x_9 = 1$
Nottingham Forest vs Bournemouth	1	1	$x_{10} = 2$
...	...	...	...

Then

$$\hat{\lambda} = \bar{x} = \frac{32}{10} = 3.2, \qquad S^2 \approx 3.956.$$

- **Option 1:** $\mathrm{SE} = S/\sqrt{n} \approx 0.629 \Rightarrow$

$$\lambda \in [\,3.20 \pm 1.96 \times 0.629\,] \;=\; [\,1.97,\; 4.43\,].$$

- **Option 2:** $\mathrm{SE} = \sqrt{\hat{\lambda}/n} \approx 0.566 \Rightarrow$

$$\lambda \in [\,3.20 \pm 1.96 \times 0.566\,] \;=\; [\,2.09,\; 4.31\,].$$

Result. With these 10 matches, the approximate 95% CIs are

$$\boxed{[1.97,\; 4.43] \text{ (using } S^2) \quad \text{and} \quad [2.09,\; 4.31] \text{ (using } \hat{\lambda})}.$$

Notes:

- **Option 2** uses the Poisson property $\mathrm{Var}(X_i) = \lambda$, whereas **Option 1** is more agnostic (i.e. can be used more broadly) and can be preferable if there's overdispersion (variance > mean) or mild model misspecification.
- If the model is well-specified (i.e. the total number of goals **is** Poisson distributed), as we observe more matches (larger n) the two intervals will get closer.

7.4.2 Asymptotic MLE: Approximate CI for the MLE

Suppose $\hat{\theta}_n$ is the maximum likelihood estimator of θ based on n IID observations.

From the Asymptotic Distribution of the MLE 5.3,

$$\hat{\theta}_n \overset{\text{approx.}}{\sim} \mathcal{N}\Big(\theta,\ \tfrac{1}{n}\,\mathcal{I}(\theta)^{-1}\Big),$$

where $\mathcal{I}(\theta)$ is the Fisher information 5.1 for a **single observation**.

But in practice, θ is unknown - that's the very reason we're estimating it! So we cannot evaluate $\mathcal{I}(\theta)$ directly.

Just as in the CLT confidence interval 7.4 we replaced the unknown σ with its estimator S, here we replace the unknown θ inside $\mathcal{I}(\theta)$ with the estimator $\hat{\theta}_n$.

- This gives a **computable** estimate of the Fisher information:

$$\mathcal{I}(\theta) \ \approx\ \mathcal{I}(\hat{\theta}_n).$$

- Since $\hat{\theta}_n \to \theta$ as $n \to \infty$ (consistency of the MLE), this substitution is asymptotically valid.

In other words, this *plug-in idea* ensures the confidence interval uses only observable quantities, while still being asymptotically correct.

Definition 7.5 (Asymptotic MLE: Approximate CI)**.** Suppose $\hat{\theta}_n$ is the maximum likelihood estimator of θ based on n IID observations. Then an **approximate** $(1-\alpha)$ confidence interval for θ is

$$\theta \ \in\ \left[\hat{\theta}_n - z_{1-\alpha/2}\sqrt{\tfrac{1}{n}\,\mathcal{I}(\hat{\theta}_n)^{-1}},\ \ \hat{\theta}_n + z_{1-\alpha/2}\sqrt{\tfrac{1}{n}\,\mathcal{I}(\hat{\theta}_n)^{-1}}\right],$$

or equivalently

$$\hat{\theta}_n \ \pm\ z_{1-\alpha/2}\sqrt{\tfrac{1}{n}\,\mathcal{I}(\hat{\theta}_n)^{-1}},$$

where

- $\hat{\theta}_n$ is the maximum likelihood estimator,
- $\mathcal{I}(\theta)$ is the Fisher information 5.1 for a single observation, evaluated at $\hat{\theta}_n$, and
- $z_{1-\alpha/2}$ is the $(1-\alpha/2)$ quantile of the standard Normal distribution $\mathcal{N}(0,1)$.

Example 7.4 (Poisson Goals: MLE Confidence Interval)**.** Following from Example 7.3, but now using the Asymptotic Distribution of the MLE 5.3, construct an approximate 95% confidence interval for λ.

Step 1: Relevant quantities

From Poisson MLE 5.11, the MLE is

$$\hat{\lambda} = \bar{X}.$$

From Poisson Fisher Information 5.14, the Fisher information for **one observation** is

$$\mathcal{I}(\lambda) = \frac{1}{\lambda}.$$

Step 2: Asymptotic MLE distribution and CI

By Asymptotic MLE 5.3,

$$\hat{\lambda} \overset{\text{approx.}}{\sim} \mathcal{N}\Big(\lambda,\ \tfrac{1}{n}\,\mathcal{I}(\lambda)^{-1}\Big) = \mathcal{N}\Big(\lambda,\ \tfrac{\lambda}{n}\Big).$$

Since the variance involves the unknown λ, we follow Definition 7.5 by plugging-in the MLE $\hat{\lambda}$ into λ:

$$\tfrac{\lambda}{n} \approx \sqrt{\tfrac{\hat{\lambda}}{n}}.$$

Hence an approximate $(1-\alpha)$ CI is

$$\lambda \ \in \ \Big[\hat{\lambda} - z_{1-\alpha/2}\sqrt{\tfrac{\hat{\lambda}}{n}},\ \ \hat{\lambda} + z_{1-\alpha/2}\sqrt{\tfrac{\hat{\lambda}}{n}}\Big].$$

Step 3: Apply to data

From Table 2.4 (first 10 rows),

$$\underline{x} = (7, 1, 4, 5, 4, 4, 1, 3, 1, 2), \quad n = 10,$$

so

$$\hat{\lambda} = \bar{X} = \frac{32}{10} = 3.2, \qquad \widehat{\text{SE}} = \sqrt{\frac{3.2}{10}} = \sqrt{0.32} \approx 0.5657.$$

With $z_{0.975} = 1.96$,

$$\lambda \in [\,3.2 \pm 1.96 \times 0.5657\,] = [\,3.2 \pm 1.108\,] = \boxed{[\,2.09,\ 4.31\,]} \quad \text{(to 2 d.p.)}.$$

Notes.

This matches the CLT option that used $\mathrm{Var}(X) = \lambda$ (since the Poisson model implies variance equals mean). If there were overdispersion, the sample-variance based CLT interval would be wider.

8 Bayesian Intervals

```
gap = 20
containerW = width
plotW = Math.round(Math.max(240, Math.min(600, (containerW - gap) / 2 - 15)))
plotH = Math.round(plotW * 0.8)

marginTop = 20
marginRight = 35
marginBottom = 35
marginLeft = 35
innerW = plotW - marginLeft - marginRight
innerH = plotH - marginTop - marginBottom

bodyFont = getComputedStyle(document.body).fontFamily
```

We now consider the **Bayesian analogue** of frequentist confidence intervals.

8.1 The Bayesian perspective

Recall from Chapter 6 that Bayesian inference returns a *full distribution* over the parameter - the posterior 6.2:

$$\pi(\theta \mid \underline{x}) \;\propto\; \pi(\theta)\, f_{\underline{X}}(\underline{x} \mid \theta).$$

From this posterior we can construct a **Bayesian credible interval**, which represents a range of parameter values that are *plausible* given the data and prior assumptions.

Definition 8.1 (Credible interval)**.** A $100(1-\alpha)\%$ **credible interval** for θ is any set C_α such that

$$\Pr(\theta \in C_\alpha \mid \underline{x}) \;=\; \int_{C_\alpha} \pi(\theta \mid \underline{x})\, \mathrm{d}\theta \;=\; 1-\alpha.$$

For discrete θ, replace the integral by a sum.

For any posterior distribution there are infinitely many valid $(1-\alpha)$ credible intervals.

This mirrors the frequentist setting, where infinitely many $(1-\alpha)$ confidence intervals also exist, though in practice one often chooses a canonical version (for example, the symmetric interval around the estimator).

8.2 Interpreting Credible Intervals

A 95% **credible interval** means that, *given the observed data and the prior*, there is a 95% posterior probability that θ lies inside the chosen interval. The probability statement is about the parameter itself, conditional on the data at hand.

In contrast, a 95% **confidence interval** is a procedure: if we were to repeatedly draw new samples and reconstruct a confidence interval each time, then 95% of those intervals would cover the true parameter. It does **not** mean there is a 95% chance that θ lies inside the one interval we have.

Thus, credible intervals are often more natural to interpret, but they depend explicitly on the prior as well as the data.

8.3 Highest Density Intervals

Definition 8.2 (Highest Density Interval (HDI))**.** A $100(1-\alpha)\%$ **highest density interval (HDI)** for θ is the region

$$C_\alpha = \{\theta : \ \pi(\theta \mid \underline{x}) \geq \gamma\},$$

where γ is chosen so that $\Pr(\theta \in C_\alpha \mid \underline{x}) = 1-\alpha$.

Among all $(1-\alpha)$ credible intervals 8.1, the HDI has **shortest length**.

If the posterior distribution has many modes, then it is possible that the HDI will be the union of several disjoint regions. For example,

$$C_\alpha = (a,b) \cup (c,d) \cup (e,f), \quad a < b < c < d < e < f.$$

Example 8.1 (Beta HDI)**.** Suppose that the posterior distribution for θ is a $\text{Beta}(1,24)$ distribution, with probability density function

$$\pi(\theta \mid \underline{x}) = 24(1-\theta)^{23}, \quad 0 < \theta < 1.$$

Task. Determine the $100(1-\alpha)\%$ HDI for~θ.

The HDI must include those values of θ with highest posterior density and so must take the form $C_\alpha = (0, b)$. The end-point b must satisfy

$$\int_0^b 24(1-\theta)^{23}\, d\theta = 1-\alpha.$$

Now,

$$\int_0^b 24(1-\theta)^{23}\, d\theta = \left[-(1-\theta)^{24}\right]_0^b = 1-(1-b)^{24}.$$

Hence,

$$1-(1-b)^{24} = 1-\alpha \quad \Longrightarrow \quad 1-b = \alpha^{1/24} \quad \Longrightarrow \quad b = 1-\alpha^{1/24}.$$

Therefore, a $100(1-\alpha)\%$ HDI for θ is

$$(0,\, 1-\alpha^{1/24}).$$

Example 8.2 (Normal HDI)**.** Suppose we have a random sample $\underset{\sim}{x}e$ from a $\mathcal{N}(\mu, 1/\tau)$ distribution (where τ is known).

We have seen that, assuming vague prior knowledge, the posterior distribution is

$$\mu \mid \underline{x} \sim \mathcal{N}\big(\bar{x},\ \tfrac{1}{n\tau}\big)\,.$$

Task. Determine the $100(1-\alpha)\%$ HDI for~μ.

Solution 8.1. This distribution has a symmetric bell shape and so the HDI takes the form $C_\alpha = (a, b)$ with end-points

$$a = \bar{x} - \frac{z_{\alpha/2}}{\sqrt{n\tau}}, \qquad b = \bar{x} + \frac{z_{\alpha/2}}{\sqrt{n\tau}},$$

where z_α is the upper α-quantile of the $\mathcal{N}(0,1)$ distribution.

Therefore, the 95% HDI for~μ is

$$\left(\bar{x} - \frac{1.96}{\sqrt{n\tau}},\ \bar{x} + \frac{1.96}{\sqrt{n\tau}}\right).$$

Note that this interval is numerically identical to the 95% frequentist confidence interval for the (population) mean of a normal random sample with known variance.

However, the interpretation is very different.

Part IV

Part IV — Hypothesis Testing

9 Frequentist Hypothesis Testing

Recalling the main inference tasks 3.2, we now move from *interval estimation* to *hypothesis testing*.

The name **hypothesis testing** comes from the **scientific method**:

1. Propose a hypothesis about the world.
2. Collect data through experiment or observation.
3. Assess whether the data are consistent with the hypothesis.

If the data are consistent, this provides **support** for the hypothesis (though never proof). If the data look very unlikely under the hypothesis, this is evidence *against* it.

Unlike **interval estimation** (which gives a range of plausible values), hypothesis testing asks a **yes/no question** about the parameter θ.

To make this precise, we specify a **null hypothesis** H_0, a concrete mathematical statement about θ:

$$H_0 : \theta \in \Theta_0,$$

where $\Theta_0 \subseteq \Theta$ is part of the parameter space.

The **alternative hypothesis** is simply the complement set:

$$H_1 : \theta \in \Theta \setminus \Theta_0,$$

i.e. the parameter values in Θ which are not in Θ_0.

So every hypothesis test is really just a partition 1.3 of the parameter space into two regions: one where the null holds, and one where it does not.

We then use the observed data $\underline{x}$ to evaluate whether H_0 is plausible. The frequentist reasoning is:

1. **Assume H_0 is true.**
2. Use the statistical model to work out what kinds of data would be typical under H_0.
3. Compare the actual data with this prediction. If the data look very unusual under H_0, this counts against the hypothesis.

Terminology: Reject vs. Accept

A hypothesis test produces a **decision rule**, not a logical proof:

- **Reject** H_0: the data look very unlikely under H_0, so we treat this as evidence against H_0.

- **Fail to reject** H_0 (sometimes phrased as "accept H_0"): the data did not provide strong enough evidence against H_0.

Neither outcome proves anything about the *truth* of H_0. These terms simply describe which side of the rejection rule the observed data fall on.
In this course we will use **"fail to reject"**, since it emphasises that not rejecting is weaker than proving H_0 true. But you should be aware that "accept" is often used informally in practice to mean the same decision.

9.1 The Testing Procedure

To make this comparison practical, we do not usually work with the full data distribution. Instead, we reduce the sample to a **test statistic** $T(\underline{X})$ that captures the evidence relevant to H_0.

9.1.1 General Recipe

1. **Specify the hypotheses.**
   Write down the null hypothesis H_0 and (optionally) an alternative H_1.
   H_1 is not always needed at this stage, but it helps to define what counts as "extreme."

2. **Choose a test statistic.**
   Select a statistic $T(\underline{X})$ that would tend to take unusually large or small values if H_0 were false.

3. **Find its distribution under H_0.**
   Work out (exactly or approximately) the sampling distribution of $T(\underline{X})$ assuming H_0 is true.

4. **Define a rejection region R_α.**
   For a chosen significance level α, determine a set R_α of "extreme" values of $T(\underline{X})$ such that

$$\max_{\theta \in H_0} \Pr(T \in R_\alpha \mid \theta) \;\leq\; \alpha.$$

- If H_0 is a **single value** (simple null) and T is a **continuous random variable**, this maximum occurs exactly at that value, so we can choose R_α so that the probability equals α.
- If H_0 is **composite** or T is **discrete**, the test may be **conservative**: the probability is strictly less than α for some $\theta \in H_0$.

5. **Compute the observed statistic.**
From the data, calculate $T_{\text{obs}} = T(\underline{x})$.

6. **Decision.**

   - If $T_{\text{obs}} \in R_\alpha$, reject H_0.

   - If $T_{\text{obs}} \notin R_\alpha$, fail to reject[1] H_0.

! Important 12: Ingredients of a Hypothesis Test

To define a hypothesis test we need **two choices**:

1. A **test statistic** $T(\underline{X})$ that captures evidence against H_0.

2. A method of constructing the **rejection region** R_α for T at level α.

Together, (T, R_α) specify the test completely.
However:

- There are usually several possible choices of test statistic $T(\underline{X})$.
We prefer ones that are **powerful** - i.e. they give a high chance of detecting departures from H_0.

- Even once T is chosen, there can be different possible **rejection regions** R.
By convention we use regions that capture the most extreme values of T in the direction(s) specified by H_1.

In addition, the sampling distribution 4.3 the test statistic $T(\underline{X})$ under H_0 must be computed. However, it may be obtained *exactly*, or *approximately* by using asymptotic distributions 6.

Definition 9.1 (Rejection Region)**.** Given a test statistic $T(\underline{X})$ and significance level α, a **rejection region** is a subset R_α of the sample space of T such that

[1] It is common to say "reject H_0" when the p-value is small, and "fail to reject H_0" otherwise. Both phrases should be interpreted with care: rejecting H_0 does **not** prove it is false, and failing to reject H_0 does **not** prove it is true. Instead, hypothesis testing provides a decision rule about whether the data are sufficiently inconsistent with H_0 at a chosen significance level.

$$\max_{\theta \in H_0} \Pr(T \in R_\alpha \mid \theta) \;\le\; \alpha.$$

This ensures that, no matter which parameter value in H_0 is true, the probability of a false rejection (Type I error) never exceeds α.

- For a **simple null** with a continuous test statistic, the maximum occurs at the null value itself, and we can usually choose R_α so that the probability equals α.

- For a **composite null** or a **discrete statistic**, the maximum may be strictly less than α, making the test *conservative*.

In practice:
- For a **two-sided test**, R_α is typically chosen as both tails of the distribution of T under H_0.
- For a **one-sided test**, R_α is typically one tail (upper or lower), depending on the alternative H_1.

Other choices of R_α are mathematically possible, but by convention we use the **most extreme** values of T, since this usually maximises power.

As our first example of a hypothesis test, we will pick a **bad** choice of test statistic T and a couple of choices of rejection region R.

Example 9.1 (A Terrible Hypothesis Test)**.** Suppose $X_1, \dots, X_n \overset{\text{iid}}{\sim} \mathcal{N}(\mu, 1)$ and we wish to test

1. $H_0 : \mu = 0$.
2. $H_1 : \mu \neq 0$.

We choose $T = X_1$ as our **test statistic**.

(a) For a chosen α, determine *two different* rejection regions.
(b) With the observed sample

$$x_1 = 0.4,\; x_2 = 3.2,\; x_3 = 2.1,\; x_4 = 2.3,\; x_5 = 1.8.$$

For which α would you **reject** H_0?

a. Two possible rejection regions

Under H_0, $T \sim \mathcal{N}(0, 1)$.

- **Conventional two-sided test:** Reject if T is very large in magnitude:

$$R_\alpha^{(1)} = \{\, |T| > z_{1-\alpha/2} \,\}.$$

- **Unconventional one-sided test:** Reject if T is very large and positive only:

$$R_\alpha^{(2)} = \{\, T > z_{1-\alpha} \,\}.$$

  This region also has probability α under H_0, but it only detects positive departures from 0.

Other choices are mathematically possible (e.g. a left-tail test, or even a symmetric band around 0 with total mass α), but these would be poor at detecting the alternatives we care about.

---

b. Observed value

Here $T_{\text{obs}} = x_1 = 0.4$.

- For the two-sided region $R_\alpha^{(1)}$: Reject if $|0.4| > z_{1-\alpha/2}$. This happens only if $\alpha \gtrsim 0.689$.
- For the one-sided region $R_\alpha^{(2)}$: Reject if $0.4 > z_{1-\alpha}$. This requires $\alpha \gtrsim 0.345$.

So in both cases you would only reject H_0 at extremely large (non-standard) significance levels. At any usual choice like $\alpha = 0.05$ or 0.01, you fail to reject.

Why this test is terrible (and instructive):

- It **ignores $n-1$ observations**, so the *power* is extremely poor (see Example 9.2).
- Different R_α give different answers, showing that both T *and* R matter.
- The statistic X_1 is a very unstable indicator of μ.

In Example 9.1 we effectively solved for the smallest significance level α at which the observed test statistic would fall in the rejection region.

This motivates the formal definition of the ***p*-value**:

Definition 9.2 (p-value)**.** Given an observed test statistic T_{obs}, the ***p*-value** is the smallest significance level α for which $T_{\text{obs}} \in R_\alpha$.

Equivalently, it is the probability, under H_0, of observing a test statistic at least as extreme as T_{obs}:

$$p = \Pr\left(T \text{ is at least as extreme as } T_{\text{obs}} \mid H_0\right).$$

- For a **two-sided test**, "extreme" means either very large or very small values of T.
- For a **one-sided test**, "extreme" means only one direction, determined by H_1.

Almost all tests in practice are either **one-sided** or **two-sided**. Restricting to these two cases fixes the form of the rejection region:

Definition 9.3 (One-Sided and Two-Sided Tests)**.**

- In a **one-sided test**, the alternative hypothesis H_1 specifies a direction.
  - Example: $H_0 : \mu \leq \mu_0$ vs. $H_1 : \mu > \mu_0$.
  - The rejection region is of the form

    $$R_\alpha = \{T \mid T \geq c_\alpha\},$$

    i.e. we only reject for *large* values of T.
  - Similarly, if H_1 is in the other direction, $H_1 : \mu < \mu_0$, then

    $$R_\alpha = \{T \mid T \leq c_\alpha\}.$$

- In a **two-sided test**, the alternative hypothesis H_1 does not specify a direction (e.g. $H_1 : \mu \neq \mu_0$).
  - The rejection region is then in both tails:

    $$R_\alpha = \{T \mid |T| \geq c_\alpha\}.$$

  - Here, the α level is usually split between the two tails, so each tail has probability $\alpha/2$ under H_0.

9.1.2 Types of Errors

Because decisions are based on data (which vary by chance), mistakes are inevitable.

A hypothesis test produces a yes/no decision about H_0. Comparing this decision with the true state of H_0 gives four logical possibilities.

Decision	Reality: H_0 true	Reality: H_0 false
Reject H_0	**Type I Error** (wrongly rejecting a true H_0)	Correct decision
Fail to reject H_0	Correct decision	**Type II Error** (missing a false H_0)

Definition 9.4 (Errors in Hypothesis Tests)**.**

- **Type I Error (False Positive):** Rejecting H_0 when it is actually true.
- **Type II Error (False Negative):** Failing to reject H_0 when the alternative is true.

By the definition of the rejection region 9.1, the probability of a Type I Error is exactly the **significance level** α.

9.1.3 Power of a Test

To describe both types of error within a single framework, we define the **power function**:

Definition 9.5 (Power Function)**.** The **power function** of a test is

$$\beta(\theta) = \Pr(T \in R_\alpha \mid \theta), \qquad \theta \in \Theta.$$

- If $\theta \in H_0$, this is the **type I error probability**, which by design is at most α.
- If $\theta \in H_1$, this is the probability of **correctly rejecting** H_0 at that parameter value.

Thus the power function unifies the error probabilities: it is small on H_0, and we hope it is large on H_1.

- **High power** means the test is good at detecting departures from H_0.
- **High specificity** means the test is good at not raising false alarms.

In practice, there is a **trade-off**: lowering α reduces false positives but usually reduces power too.

Computing the **power** of a test depends on the true value of the parameter θ:

Example 9.2 (The Power of a Terrible Hypothesis Test)**.** Continuing from Example 9.1, for each of the *rejection regions*

1. $R_\alpha^{(1)} = \{\, |T| > z_{1-\alpha/2} \,\}$, and

2. $R_\alpha^{(2)} = \{\, T > z_{1-\alpha} \,\}$,

derive the **power functions** of the test as a function of μ.

The *alternative* hypothesis is $H_1 : \mu \neq 0$. If H_1 were true, then our test statistic would have distribution

$$T = X_1 \sim \mathcal{N}(\mu, 1), \quad \mu \neq 0.$$

1. $R_\alpha^{(1)}$

For $R_\alpha^{(1)}$, we reject if the test statistic T satisfies $|T| > z_{1-\alpha/2}$, we can compute this as:

$$\begin{aligned}
\Pr(|T| > z_{1-\alpha/2}) &= \Pr(T > z_{1-\alpha/2}) + \Pr(T < -z_{1-\alpha/2}) \\
&= \Pr\left(\frac{T-\mu}{1} > z_{1-\alpha/2}\right) + \Pr\left(\frac{T-\mu}{1} < -z_{1-\alpha/2}\right) \\
&= \Pr\left(Z > \mu + z_{1-\alpha/2}\right) + \Pr\left(Z < \mu - z_{1-\alpha/2}\right) \\
&= 1 - \Phi(\mu + z_{1-\alpha/2}) + \Phi(\mu - z_{1-\alpha/2}),
\end{aligned}$$

where $Z \sim \mathcal{N}(0, 1)$ and Φ is its CDF.

2. $R_\alpha^{(2)}$

For $R_\alpha^{(2)}$, we reject if the test statistic T satisfies $T > z_{1-\alpha}$, we can compute this as:

$$\begin{aligned}
\Pr(T > z_{1-\alpha}) &= \Pr\left(\frac{Z-\mu}{1} > z_{1-\alpha}\right) \\
&= \Pr(Z > z_{1-\alpha} + \mu) \\
&= 1 - \Phi(z_{1-\alpha} + \mu),
\end{aligned}$$

where $Z \sim \mathcal{N}(0, 1)$ and Φ is its CDF.

Summary

- For the **two-sided test**

$$\beta(\mu) = \Pr(|T| > z_{1-\alpha/2} \mid \mu) = 1 - \Big[\Phi(z_{1-\alpha/2} - \mu) - \Phi(-z_{1-\alpha/2} - \mu)\Big].$$

  This is one minus the probability that T falls inside the acceptance region.

- For the **one-sided test**

$$\beta(\mu) = \Pr(T > z_{1-\alpha} \mid \mu) = 1 - \Phi(z_{1-\alpha} - \mu).$$

Both expressions reduce to $\beta(0) = \alpha$ when $\mu = 0$, as required.

9.2 Exact Hypothesis Tests

In this section, we derive commonly used hypothesis tests and consider their power.

9.2.1 One-Sample Normal (known variance)

In Example 9.1, we considered i.i.d. normally distributed data with known variance. However, we considered a **bad** test statistic. For general i.i.d. $X_i \sim \mathcal{N}(\mu, \sigma^2)$, it is much more natural to use the sample mean 4.1,

$$\bar{X} \sim \mathcal{N}\left(\mu, \frac{\sigma^2}{n}\right),$$

to test $H_0 : \mu = \mu_0$ vs. either a one-sided or two-sided alternative. Let's do this:

Example 9.3 (One-Sample Normal Test (known variance))**.** Suppose we have i.i.d. data $X_i \sim \mathcal{N}(\mu, \sigma^2)$. Suppose we wish to test:

1. $H_0 : \mu = \mu_0$, $H_1 : \mu \neq \mu_0$.
2. $H_0 : \mu \leq \mu_0$, $H_1 : \mu > \mu_0$.
3. $H_0 : \mu \geq \mu_0$, $H_1 : \mu < \mu_0$.

Using the test statistic:

$$T = \bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i,$$

derive the **one-sided** and **two-sided** rejection regions at significance level α.

1.

Under the null hypothesis, the test statistic T has distribution:

$$T \mid \mu = \mu_0 \sim \mathcal{N}\left(\mu_0, \frac{\sigma^2}{n}\right).$$

Since the alternative is two-sided, we want power in both directions and so pick a two-sided rejection region. We have

$$\frac{T - \mu_0}{\sigma/\sqrt{n}} \sim \mathcal{N}(0, 1).$$

Therefore, since

$$\begin{aligned}
\Pr\left(\frac{T-\mu_0}{\sigma/\sqrt{n}} \geq z_{1-\alpha/2} \,\middle|\, \mu=\mu_0\right) + \Pr\left(\frac{T-\mu_0}{\sigma/\sqrt{n}} \leq -z_{1-\alpha/2} \,\middle|\, \mu=\mu_0\right) &= 1-\alpha \\
\Pr\left(|T-\mu_0| \geq \frac{z_{1-\alpha/2}\sigma}{\sqrt{n}} \,\middle|\, \mu=\mu_0\right) &= 1-\alpha.
\end{aligned}$$

Therefore, a valid two-sided $1-\alpha$ rejection region takes the form:

$$R_\alpha = \left\{T \mid \ |T-\mu_0| > \frac{\sigma\, z_{1-\alpha/2}}{\sqrt{n}}\right\}.$$

2.

Under the null hypothesis, the test statistic T has distribution:

$$T \mid \mu \sim \mathcal{N}\left(\mu, \frac{\sigma^2}{n}\right), \quad \text{for } \mu \leq \mu_0.$$

The null hypothesis does not fully specify the distribution of the null.

9.2.2 One-Sample Normal (unknown variance)

9.2.3 Two-Sample Normal (unknown variances)

So far, we have only ever considered a **single sample**

9.3 Approximate Hypothesis Tests

9.3.1 CLT: Approximate Hypothesis Tests

Example 9.4 (Test of Proportions)**.**

9.3.2 Asymptotic MLE: Approximate Hypothesis Tests

9.4 Likelihood Ratio Test

10 Bayesian Hypothesis Testing

Recalling the main inference tasks 3.2, we now move from *interval estimation* to *hypothesis testing* in the **Bayesian** framework, and consider the Bayesian analogue of frequentist hypothesis testing..

The Bayesian perspective is slightly different from the frequentist one: instead of asking whether data look *unlikely* under a hypothesis, we ask:

> **How probable is each hypothesis given the observed data?**

10.1 The Bayesian Setup

Just like in frequentist hypothesis testing 9, the Bayesian approach compares two hypotheses:

$$H_0 : \theta \in \Theta_0, \qquad H_1 : \theta \in \Theta_1,$$

where $\Theta_1 = \Theta \setminus \Theta_0$.

The **Bayesian perspective** is to assign probabilities to hypotheses and to parameter values. This involves two ingredients:

1. **Prior probabilities of the hypotheses:**

$$\Pr(H_0) \quad \text{and} \quad \Pr(H_1),$$

   such that $\Pr(H_0) + \Pr(H_1) = 1$.

2. **Conditional priors for the parameter under each hypothesis:**

$$\pi(\theta \mid H_0) \quad \text{and} \quad \pi(\theta \mid H_1).$$

10.1.1 Posterior Probabilities of Hypotheses

After observing data $\underline{x}$, Bayes' theorem gives

$$\Pr(H_0 \mid \underline{x}) = \frac{f(\underline{x} \mid H_0)\ \Pr(H_0)}{f(\underline{x})}.$$

If H_0 is not simple, the likelihood $f(\underline{x} \mid H_0)$ is obtained by averaging over $\theta \in \Theta_0$ with the conditional prior:

$$f(\underline{x} \mid H_0) = \int_{\Theta_0} f(\underline{x} \mid \theta)\, \pi(\theta \mid H_0)\, \mathrm{d}\theta.$$

So we can write

$$\Pr(H_0 \mid \underline{x}) = \frac{m_0(\underline{x})\ \Pr(H_0)}{m(\underline{x})},$$

where

- $m_0(\underline{x}) = \int_{\Theta_0} f(\underline{x} \mid \theta)\, \pi(\theta \mid H_0)\, \mathrm{d}\theta$ is the **marginal likelihood under H_0,**
- $m_1(\underline{x}) = \int_{\Theta_1} f(\underline{x} \mid \theta)\, \pi(\theta \mid H_1)\, \mathrm{d}\theta$ is the **marginal likelihood under H_1,**
- $m(\underline{x}) = m_0(\underline{x}) \Pr(H_0) + m_1(\underline{x}) \Pr(H_1)$ is the **overall marginal likelihood.**

Thus the Bayesian procedure directly yields the posterior probabilities $\Pr(H_0 \mid \underline{x})$ and $\Pr(H_1 \mid \underline{x})$ for the competing hypotheses.

10.1.2 Bayes Factor

The **Bayes factor** summarises how strongly the data update our beliefs:

$$B_{10} = \frac{m_1(\underline{x})}{m_0(\underline{x})}.$$

It measures how much more (or less) the data support H_1 compared to H_0.

Posterior odds are obtained from prior odds via:

$$\frac{\Pr(H_1 \mid \underline{x})}{\Pr(H_0 \mid \underline{x})} = B_{10} \times \frac{\Pr(H_1)}{\Pr(H_0)}.$$

- If $B_{10} > 1$: the data increase belief in H_1.
- If $B_{10} < 1$: the data increase belief in H_0.

Part V

Additional Material

Formula Sheet

Discrete distributions

Bernoulli

Model. $X \sim \text{Bernoulli}(p)$ with $0 < p < 1$.
Support. $x \in \{0, 1\}$.
PMF.

$$\Pr(X = x) = p^x (1 - p)^{1-x}, \quad x \in \{0, 1\}.$$

Mean/Var. $\mathrm{E}[X] = p, \quad \mathrm{Var}(X) = p(1 - p)$.

Binomial

Model. $X \sim \text{Bin}(n, p)$, $n \in \mathbb{N}$, $0 < p < 1$.
Support. $x = 0, 1, \ldots, n$.
PMF.

$$\Pr(X = x) = \binom{n}{x} p^x (1 - p)^{n-x}.$$

Mean/Var. $\mathrm{E}[X] = np, \quad \mathrm{Var}(X) = np(1 - p)$.

Multinomial

Model. $\mathbf{X} = (X_1, \dots, X_k) \sim \text{Multinomial}(n, \mathbf{p})$, with $n \in \mathbb{N}$, $k \geq 2$, $\mathbf{p} = (p_1, \dots, p_k)$, $p_i > 0$, and $\sum_{i=1}^{k} p_i = 1$.
Support. $\mathbf{x} = (x_1, \dots, x_k)$ where $x_i \in \{0, 1, \dots, n\}$ and $\sum_{i=1}^{k} x_i = n$.
PMF.

$$\Pr(\mathbf{X} = \mathbf{x}) = \frac{n!}{x_1! \cdots x_k!} \prod_{i=1}^{k} p_i^{x_i}.$$

Mean/Cov.

$$\mathrm{E}[X_i] = np_i, \qquad \mathrm{Var}(X_i) = np_i(1 - p_i), \qquad \mathrm{Cov}(X_i, X_j) = -np_i p_j \quad (i \neq j).$$

Vector form.

$$\mathrm{E}[\mathbf{X}] = n\,\mathbf{p}, \qquad \mathrm{Cov}(\mathbf{X}) = n\left(\mathrm{diag}(\mathbf{p}) - \mathbf{p}\mathbf{p}^\top\right).$$

Note. For $k = 2$, the first component $X_1 \sim \mathrm{Bin}(n, p_1)$ (Binomial).

Poisson

Model. $X \sim \mathrm{Po}(\lambda)$, $\lambda > 0$.
Support. $x = 0, 1, 2, \dots$.
PMF.

$$\Pr(X = x) = \frac{\lambda^x e^{-\lambda}}{x!}.$$

Mean/Var. $\mathrm{E}[X] = \lambda, \quad \mathrm{Var}(X) = \lambda.$

Geometric

Model. $X \sim \mathrm{Geom}(p)$ (trials **until first success**, counting the success), $0 < p < 1$.
Support. $x = 1, 2, \dots$.
PMF.

$$\Pr(X = x) = p(1 - p)^{x-1}.$$

Mean/Var. $\mathrm{E}[X] = \frac{1}{p}, \quad \mathrm{Var}(X) = \frac{1-p}{p^2}.$

⚠ Parameterisation alert

Some texts start at $x = 0$ (failures before the first success).

Negative Binomial

Model. $X \sim \mathrm{NegBin}(r, p)$: number of trials to achieve the r-th success (counts the successes), $r > 0$ (often integer), $0 < p < 1$.
Support. $x = r, r+1, \ldots$.
PMF.

$$\Pr(X = x) = \binom{x-1}{r-1} p^r (1-p)^{x-r}.$$

Mean/Var. $\mathrm{E}[X] = \frac{r}{p}, \quad \mathrm{Var}(X) = \frac{r(1-p)}{p^2}$.

⚠ Parameterisation alert

Another common version counts **failures** before the r-th success ($x = 0, 1, \ldots$) with different mean/variance. Always check which one is used.

Continuous distributions

Uniform (continuous)

Model. $X \sim \mathrm{Uniform}(a, b)$, $a < b$.
Support. $a < x < b$.
PDF.

$$f_X(x; a, b) = \frac{1}{b-a}.$$

Mean/Var. $\mathrm{E}[X] = \frac{a+b}{2}, \quad \mathrm{Var}(X) = \frac{(b-a)^2}{12}$.

Exponential

Model. $X \sim \text{Exponential}(\lambda)$ with **rate** $\lambda > 0$.
Support. $x > 0$.
PDF.

$$f_X(x; \lambda) = \lambda e^{-\lambda x}.$$

Mean/Var. $\mathrm{E}[X] = \frac{1}{\lambda}, \quad \mathrm{Var}(X) = \frac{1}{\lambda^2}.$

Gamma

Model. $X \sim \text{Gamma}(a, b)$ with **shape** $a > 0$ and **rate** $b > 0$.
Support. $x > 0$.
PDF.

$$f_X(x; a, b) = \frac{b^a}{\Gamma(a)} x^{a-1} e^{-bx}.$$

Mean/Var. $\mathrm{E}[X] = \frac{a}{b}, \quad \mathrm{Var}(X) = \frac{a}{b^2}.$

> ⚠ Parameterisation alert
>
> Many sources use **scale** $\beta = 1/b$: then $f(x) = \frac{1}{\Gamma(a)\beta^a} x^{a-1} e^{-x/\beta}$ and $\mathrm{E}[X] = a\beta$.

Normal

Model. $X \sim \mathcal{N}(\mu, \sigma^2)$ with $\sigma > 0$.
Support. $x \in \mathbb{R}$.
PDF.

$$f_X(x; \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right).$$

Mean/Var. $\mathrm{E}[X] = \mu, \quad \mathrm{Var}(X) = \sigma^2.$

Properties

Suppose $X_1, \dots, X_n$ are independent normal random variables, then:

Lognormal

Model. $X \sim \text{LogNormal}(\mu, \sigma^2)$ meaning $\ln X \sim \mathcal{N}(\mu, \sigma^2)$.
Support. $x > 0$.
PDF.

$$f_X(x; \mu, \sigma^2) = \frac{1}{x\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(\ln x - \mu)^2}{2\sigma^2}\right).$$

Mean/Var. $\mathrm{E}[X] = e^{\mu+\sigma^2/2}, \quad \mathrm{Var}(X) = (e^{\sigma^2} - 1)e^{2\mu+\sigma^2}$.

Chi-squared

Model. $X \sim \chi^2_\nu$ with $\nu > 0$ degrees of freedom.
Support. $x > 0$.
PDF.

$$f_X(x; \nu) = \frac{1}{2^{\nu/2}\Gamma(\nu/2)} x^{\nu/2-1} e^{-x/2}.$$

Mean/Var. $\mathrm{E}[X] = \nu, \quad \mathrm{Var}(X) = 2\nu$.

> **Note**
>
> χ^2_ν is a special case of Gamma with $a = \nu/2$, $b = 1/2$.

Student-t

Model. $X \sim t_\nu$ with $\nu > 0$ degrees of freedom.
Support. $x \in \mathbb{R}$.
PDF.

$$f_X(x; \nu) = \frac{\Gamma\left(\frac{\nu+1}{2}\right)}{\sqrt{\nu\pi}\,\Gamma\left(\frac{\nu}{2}\right)} \left(1 + \frac{x^2}{\nu}\right)^{-(\nu+1)/2}.$$

Mean/Var. $\mathrm{E}[X] = 0$ for $\nu > 1$; undefined for $\nu \leq 1$.
$\mathrm{Var}(X) = \frac{\nu}{\nu-2}$ for $\nu > 2$; infinite for $1 < \nu \leq 2$; undefined for $\nu \leq 1$.

F distribution

Model. $X \sim F_{d_1,d_2}$ with $d_1, d_2 > 0$ degrees of freedom.
Support. $x > 0$.
PDF.

$$f_X(x; d_1, d_2) = \frac{\Gamma\left(\frac{d_1+d_2}{2}\right)}{\Gamma\left(\frac{d_1}{2}\right)\Gamma\left(\frac{d_2}{2}\right)} \left(\frac{d_1}{d_2}\right)^{d_1/2} \frac{x^{d_1/2-1}}{\left(1 + \frac{d_1}{d_2}x\right)^{(d_1+d_2)/2}}.$$

Mean/Var. $\mathrm{E}[X] = \frac{d_2}{d_2-2}$ for $d_2 > 2$.
$\mathrm{Var}(X) = \frac{2d_2^2(d_1+d_2-2)}{d_1(d_2-2)^2(d_2-4)}$ for $d_2 > 4$.

Beta

Model. $X \sim \mathrm{Beta}(a, b)$, $a > 0, b > 0$.
Support. $0 < x < 1$.
PDF.

$$f_X(x; a, b) = \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)} x^{a-1}(1-x)^{b-1}.$$

Mean/Var. $\mathrm{E}[X] = \frac{a}{a+b}, \quad \mathrm{Var}(X) = \frac{ab}{(a+b)^2(a+b+1)}$.

Multivariate Distributions

Multivariate Normal

Model. $\mathbf{X} \sim \mathcal{N}_p(\mu, \Sigma)$ with Σ positive definite.
Support. $\mathbf{x} \in \mathbb{R}^p$.
PDF.

$$f_{\mathbf{X}}(\mathbf{x}; \mu, \Sigma) = \frac{1}{(2\pi)^{p/2}(\det\Sigma)^{1/2}} \exp\left(-\tfrac{1}{2}(\mathbf{x}-\mu)^\top \Sigma^{-1}(\mathbf{x}-\mu)\right).$$

Mean/Cov. $\mathrm{E}[\mathbf{X}] = \mu, \quad \mathrm{Var}(\mathbf{X}) = \Sigma$.

Revision Sheet

Under Construction...

Distribution Explorer

Question Bank

Probability of heads

A fair coin is tossed 10 times. What is the probability of exactly 6 heads?
Solution
Work it via the Binomial:

$$\Pr(X=6) = \binom{10}{6}(1/2)^{10}.$$

Newspaper subscriptions

In a certain city, 50% of families subscribe to the morning newspaper, 65% subscribe to the afternoon newspaper, and 30% subscribe to both.
What proportion of families subscribe to **at least one** newspaper?
Solution
Let M = "morning" and A = "afternoon".
By the addition rule of probability:

$$\Pr(M \cup A) \;=\; \Pr(M) + \Pr(A) - \Pr(M \cap A).$$

Substituting:

$$0.5 + 0.65 - 0.3 \;=\; 0.85.$$

So, 85% of families subscribe to at least one newspaper.

Normal Data MLE

Let $X_1, \ldots, X_n \overset{iid}{\sim} \mathcal{N}(\mu, 1)$. Find the MLE of μ.
Solution
The likelihood is

$$L(\mu; \underline{x}) = \mathrm{Prod}_{i=1}^{n} \frac{1}{\sqrt{2\pi}} \exp\big(-\tfrac{1}{2}(x_i-\mu)^2\big) = (2\pi)^{-n/2} \exp\left(-\frac{1}{2}\sum_{i=1}^{n}(x_i-\mu)^2\right).$$

Taking logs:

$$\ell(\mu) = -\tfrac{n}{2}\log(2\pi) - \tfrac{1}{2}\sum_{i=1}^{n}(x_i-\mu)^2.$$

Differentiate with respect to μ:

$$\frac{\partial \ell}{\partial \mu} = \sum_{i=1}^{n}(x_i-\mu).$$

Setting equal to 0:

$$\sum_{i=1}^{n}(x_i-\mu) = 0 \quad \Longrightarrow \quad \hat{\mu} = \frac{1}{n}\sum_{i=1}^{n} x_i = \bar{X}.$$

So the MLE of μ is the **sample mean** $\bar{X}$.

Converting mean and standard deviation

October temperatures in Newcastle (daily maximum) have population mean 13°C and population standard deviation 3°C.
What would the population mean and standard deviation be if temperature was measured in °F?
Note: If X is in °C then $Y = 1.8X + 32$ is in °F.
Solution
Using linearity of expectation:

$$\mathrm{E}[Y] \;=\; 1.8\,\mathrm{E}[X] + 32 \;=\; 1.8 \times 13 + 32 \;=\; 55.4^{\circ}\mathrm{F}.$$

For the standard deviation, the shift does not matter and the scale multiplies:

$$\mathrm{SD}(Y) \;=\; 1.8\,\mathrm{SD}(X) \;=\; 1.8 \times 3 \;=\; 5.4^{\circ}\mathrm{F}.$$

PDF, normalisation constant, and CDF

The random variable X has PDF

$$f(x) = \begin{cases} kx(1-x), & 0 \le x \le 1, \\ 0, & \text{otherwise,} \end{cases}$$

where k is a constant.

1. Find the value of k.

2. Roughly sketch $f(x)$. State, without calculation, the value of $\mathrm{E}[X]$.

3. Find and roughly sketch the CDF $F(x) = \Pr(X \le x)$.

Solution

a. Since the area under a PDF is 1:

$$1 = \int_{-\infty}^{\infty} f(x)\,dx = \int_0^1 k(x - x^2)\,dx = k\left[\tfrac{x^2}{2} - \tfrac{x^3}{3}\right]_0^1 = \tfrac{k}{6}.$$

So $k = 6$.

---

b. The PDF is symmetric about $x = \frac{1}{2}$, so

$$\mathrm{E}[X] = \tfrac{1}{2}.$$

c. For $0 \le x \le 1$:

$$F(x) = \int_0^x 6(t - t^2)\,dt = 6\left[\tfrac{t^2}{2} - \tfrac{t^3}{3}\right]_0^x = x^2(3 - 2x).$$

So the full CDF is:

$$F(x) = \begin{cases} 0, & x < 0, \\ x^2(3 - 2x), & 0 \le x \le 1, \\ 1, & x > 1. \end{cases}$$

```
//| label: fig-PDF-CDF-example
//| fig-cap: Probability density function $f(x)$ and cumulative distribution function $F(x)
//| fig-align: center

// domain
xs = d3.range(-0.1, 1.1, 0.01)

// PDF
PDF = xs.map(x => ({
  x: x,
  y: (x >= 0 && x <= 1) ? 6*x*(1-x) : 0,
  which: "PDF f(x)"
}))

// CDF
CDF = xs.map(x => {
  let y;
  if (x < 0) y = 0;
  else if (x > 1) y = 1;
  else y = x**2 * (3 - 2*x);
  return {x: x, y: y, which: "CDF F(x)"};
})

viewof chart = Plot.plot({
  height: 300,
  width: 500,
  style: { background: "var(--plot-panel-bg)",
           fontSize: 14,
           fontFamily: bodyFont,  },
  y: {domain: [0,1.8]},
  marks: [
    Plot.line(PDF, {x: "x", y: "y", stroke: "steelblue"}),
    Plot.line(CDF, {x: "x", y: "y", stroke: "orange"}),
    Plot.ruleY([0]),
  ],
  color: {legend: true}
})
```

Linear combinations of independent random variables

Suppose that X_1 and X_2 are **independent** random variables with

$$\mathrm{E}[X_1] = 4, \mathrm{Var}(X_1) = 5, \mathrm{E}[X_2] = 3, \mathrm{Var}(X_2) = 1.$$

Let

$$Y_1 = 2X_1 + X_2, \qquad Y_2 = X_1 - 2X_2.$$

Find:

(a) $\mathrm{E}[Y_1],\ \mathrm{Var}(Y_1)$

(b) $\mathrm{E}[Y_2],\ \mathrm{Var}(Y_2)$

(c) $\mathrm{Cov}(Y_1, Y_2)$ and the correlation between Y_1 and Y_2.

Solution
a.

$$\mathrm{E}[Y_1] = 2\mathrm{E}[X_1] + \mathrm{E}[X_2] = 2 \times 4 + 3 = 11.$$
$$\mathrm{Var}(Y_1) = 4\mathrm{Var}(X_1) + \mathrm{Var}(X_2) = 4 \times 5 + 1 = 21.$$

---

b.

$$\mathrm{E}[Y_2] = \mathrm{E}[X_1] - 2\mathrm{E}[X_2] = 4 - 6 = -2.$$
$$\mathrm{Var}(Y_2) = \mathrm{Var}(X_1) + 4\mathrm{Var}(X_2) = 5 + 4 \times 1 = 9.$$

---

c.

$$\begin{aligned}\mathrm{Cov}(Y_1, Y_2) &= \mathrm{Cov}(2X_1 + X_2,\ X_1 - 2X_2) \\ &= 2\mathrm{Var}(X_1) - 2\mathrm{Var}(X_2) + (-4 + 1)\mathrm{Cov}(X_1, X_2).\end{aligned}$$

Since X_1 and X_2 are independent, $\mathrm{Cov}(X_1, X_2) = 0$. So

$$\mathrm{Cov}(Y_1, Y_2) = 2 \times 5 - 2 \times 1 = 8.$$

Finally, the correlation is

$$\mathrm{Corr}(Y_1, Y_2) = \frac{\mathrm{Cov}(Y_1, Y_2)}{\sqrt{\mathrm{Var}(Y_1)\mathrm{Var}(Y_2)}} = \frac{8}{\sqrt{21 \times 9}} \approx 0.582.$$

Linear combinations of dependent random variables

Suppose that X_1 and X_2 are **dependent** random variables with

$$\mathrm{E}[X_1] = 4, \mathrm{Var}(X_1) = 5, \mathrm{E}[X_2] = 3, \mathrm{Var}(X_2) = 1$$

and

$$\mathrm{Cov}(X_1, X_2) = 2.$$

Let

$$Y_1 = 2X_1 + X_2, \qquad Y_2 = X_1 - 2X_2.$$

Find:

(a) $\mathrm{E}[Y_1],\ \mathrm{Var}(Y_1)$

(b) $\mathrm{E}[Y_2],\ \mathrm{Var}(Y_2)$

(c) $\mathrm{Cov}(Y_1, Y_2)$ and the correlation between Y_1 and Y_2.

Solution

a.

$$\mathrm{E}[Y_1] = 2\mathrm{E}[X_1] + \mathrm{E}[X_2] = 11.$$

$$\begin{aligned}
\mathrm{Var}(Y_1) &= \mathrm{Var}(2X_1) + \mathrm{Var}(X_2) + 2\mathrm{Cov}(2X_1, X_2) \\
&= 4\mathrm{Var}(X_1) + \mathrm{Var}(X_2) + 4\mathrm{Cov}(X_1, X_2) \\
&= 4 \times 5 + 1 + 4 \times 2 = 29.
\end{aligned}$$

---

b.

$$\mathrm{E}[Y_2] = \mathrm{E}[X_1] - 2\mathrm{E}[X_2] = -2.$$

$$\begin{aligned}
\mathrm{Var}(Y_2) &= \mathrm{Var}(X_1) + \mathrm{Var}(2X_2) - 2\mathrm{Cov}(X_1, 2X_2) \\
&= \mathrm{Var}(X_1) + 4\mathrm{Var}(X_2) - 4\mathrm{Cov}(X_1, X_2) \\
&= 5 + 4 - 8 = 1.
\end{aligned}$$

---

c.

$$\begin{aligned}\mathrm{Cov}(Y_1, Y_2) &= \mathrm{Cov}(2X_1 + X_2,\, X_1 - 2X_2)\\ &= 2\mathrm{Var}(X_1) - 2\mathrm{Var}(X_2) + (-4+1)\mathrm{Cov}(X_1, X_2)\\ &= 10 - 2 - 6 = 2.\end{aligned}$$

So the correlation is

$$\mathrm{Corr}(Y_1, Y_2) = \frac{\mathrm{Cov}(Y_1, Y_2)}{\sqrt{\mathrm{Var}(Y_1)\mathrm{Var}(Y_2)}} = \frac{2}{\sqrt{29 \times 1}} \approx 0.371.$$

Minimiser of $\mathrm{E}[(X-c)^2]$

Suppose that X is a continuous random variable. Show that $\mathrm{E}[(X-c)^2]$ is minimised when c is the population **mean**. State any assumptions.
Solution
As X is continuous,

$$\mathrm{E}[(X-c)^2] = \int_{-\infty}^{\infty} (x-c)^2 f(x)\, dx.$$

We assume we can interchange differentiation with respect to c and integration with respect to x. Then

$$\frac{\partial}{\partial c}\mathrm{E}[(X-c)^2] = \mathrm{E}\left[\frac{\partial}{\partial c}(X-c)^2\right] = \int_{-\infty}^{\infty} \frac{\partial}{\partial c}(x-c)^2 f(x)\, dx.$$

Differentiating:

$$\frac{\partial}{\partial c}\mathrm{E}[(X-c)^2] = -2\int_{-\infty}^{\infty} (x-c) f(x)\, dx = -2\{\mathrm{E}[X] - c\}.$$

This derivative is zero when $c = \mathrm{E}[X]$.
The second derivative is positive, so this is indeed a minimum.

Minimiser of $\mathrm{E}[|X-c|]$

Suppose that X is a continuous random variable. Show that $\mathrm{E}[|X-c|]$ is minimised when c is the population **median**. State any assumptions.
Hint: The derivative of $|X-c|$ with respect to c is -1 when $x > c$ and $+1$ when $x < c$.
Solution
We assume the technical conditions allowing us to interchange differentiation with respect to c and integration with respect to x.

Using the hint:

$$\frac{\partial}{\partial c}\,\mathrm{E}[|X-c|] = \mathrm{E}\left[\frac{\partial}{\partial c}|X-c|\right] = \mathrm{E}[I(X \le c) - I(X > c)],$$

where $I(\cdot)$ is the indicator function.
So,

$$\frac{\partial}{\partial c}\,\mathrm{E}[|X-c|] = \Pr(X \le c) - \Pr(X > c).$$

Since $\Pr(X > c) = 1 - \Pr(X \le c)$, this simplifies to

$$\frac{\partial}{\partial c}\,\mathrm{E}[|X-c|] = 2\Pr(X \le c) - 1.$$

This derivative is zero when $\Pr(X \le c) = \frac{1}{2}$, i.e. when c is the **median** of the distribution of X.
(Aside: this argument ignores the non-differentiability of $|X-c|$ at $x = c$ — a more careful treatment requires subgradients.)

Beta Conjugacy

With a Beta prior $p \sim \mathrm{Beta}(\alpha, \beta)$ and $X \mid p \sim \mathrm{Bin}(n, p)$, show the posterior is $p \mid x \sim \mathrm{Beta}(\alpha + x, \beta + n - x)$.
Solution
Intuition.
The Beta prior has the same algebraic form as the Binomial likelihood in p: powers of p and $(1-p)$. Multiplying them keeps the form, so the posterior is also Beta (this is *conjugacy*).
Formal derivation.
Prior:

$$\pi(p) = \mathrm{Beta}(p \mid \alpha, \beta) = \frac{1}{B(\alpha,\beta)}\,p^{\alpha-1}(1-p)^{\beta-1}, \qquad 0 < p < 1.$$

Likelihood (for x successes in n trials):

$$L(p; x)\mathrm{Pr}opto p^{x}(1-p)^{\,n-x}.$$

Posterior (up to proportionality):

$$\pi(p \mid x)\ \mathrm{Pr}opto\ \pi(p)\,L(p \mid x)\ \mathrm{Pr}opto\ p^{\alpha-1+x}\,(1-p)^{\beta-1+(n-x)}.$$

This is in the form of the Beta distribution with updated parameters:

$$p \mid x \sim \mathrm{Beta}(\alpha + x,\ \beta + n - x).$$

Binomial Probabilities

Suppose $X \sim \text{Bin}(n, p)$.

a. When $n = 6$ and $p = 0.3$, find, either by hand or using R or Python, each of:

   i. $\Pr(X < 2)$

   ii. $\Pr(1 < X \leq 2)$

   iii. $\Pr(X > 1)$

b. When $n = 10$ and $p = 0.75$, find the probability that X is within one of its population mean.

Solution

a.

As X is binomial, it can only take the values $0, 1, 2, \ldots, 6$.

i.

$$\Pr(X < 2) = \Pr(X = 0) + \Pr(X = 1).$$

$$\Pr(X = 0) = \binom{6}{0}(0.3)^0(0.7)^6 = 0.7^6.$$

$$\Pr(X = 1) = \binom{6}{1}(0.3)^1(0.7)^5 = 6 \times 0.3 \times 0.7^5.$$

So,

$$\Pr(X < 2) = 0.7^6 + 6 \times 0.3 \times 0.7^5 \approx 0.4202.$$

ii.

$$\Pr(1 < X \leq 2) = \Pr(X = 2) = \binom{6}{2}(0.3)^2(0.7)^4 = 15 \times 0.09 \times 0.2401 \approx 0.3241.$$

iii.

$$\Pr(X > 1) = 1 - \Pr(X \leq 1) = 1 - \Pr(X < 2).$$

So,

$$\Pr(X > 1) = 1 - 0.4202 = 0.5798.$$

---

b.

For $X \sim \text{Bin}(10, 0.75)$:

- The mean is $\mathrm{E}[X] = np = 10 \times 0.75 = 7.5$.
- "Within one of the mean" means $6.5 \leq X \leq 8.5$.
  Since X is integer-valued, this corresponds to $X \in \{7, 8\}$.

$$\Pr(6.5 \leq X \leq 8.5) = \Pr(X = 7) + \Pr(X = 8).$$

$$\Pr(X = 7) = \binom{10}{7}(0.75)^7(0.25)^3,$$

$$\Pr(X = 8) = \binom{10}{8}(0.75)^8(0.25)^2.$$

Numerically,

$$\Pr(X = 7) \approx 0.2503, \quad \Pr(X = 8) \approx 0.2816.$$

So,

$$\Pr(6.5 \leq X \leq 8.5) \approx 0.2503 + 0.2816 = 0.5319.$$

In R or Python:

10.1.1 Python

```python
from math import comb
from scipy.stats import binom

# Part (a): n=6, p=0.3
n, p = 6, 0.3

pr_less_2 = binom.cdf(1, n, p)              # P(X < 2)
pr_eq_2   = binom.pmf(2, n, p)              # P(X = 2)
pr_greater_1 = 1 - pr_less_2                # P(X > 1)

# Part (b): n=10, p=0.75
n, p = 10, 0.75

pr_within_mean = binom.pmf(7, n, p) + binom.pmf(8, n, p)

print(
    f"Pr(X < 2) = {pr_less_2:.4f}\n"
    f"Pr(1 < X <= 2) = {pr_eq_2:.4f}\n"
    f"Pr(X > 1) = {pr_greater_1:.4f}\n"
    f"Pr(X within 1 of mean) = {pr_within_mean:.4f}"
)
```

10.1.2 R

```r
#| echo: true
#| code-fold: true
#| code-summary: "Show code"

# Part (a): n=6, p=0.3
n <- 6; p <- 0.3

pr_less_2 <- pbinom(1, n, p)     # P(X < 2)
pr_eq_2   <- dbinom(2, n, p)     # P(X = 2)
pr_greater_1 <- 1 - pr_less_2    # P(X > 1)

# Part (b): n=10, p=0.75
n <- 10; p <- 0.75

pr_within_mean <- dbinom(7, n, p) + dbinom(8, n, p)

cat(
  "Pr(X < 2) =", round(pr_less_2,4), "\n",
  "Pr(1 < X <= 2) =", round(pr_eq_2,4), "\n",
  "Pr(X > 1) =", round(pr_greater_1,4), "\n",
  "Pr(X within 1 of mean) =", round(pr_within_mean,4), "\n"
)
```

Mean and Standard Deviation of Linear Combinations

Suppose $X \sim \mathcal{N}(3, 2)$ and, independently, $Y \sim \mathcal{N}(-2, 7)$.
Find the population mean and standard deviation of:

(a) $3X + 2$

(b) $5 - Y$

(c) $3X + 7 - Y$

Solution
a.

$$\mathrm{E}[3X + 2] = 3\mathrm{E}[X] + 2 = 3 \times 3 + 2 = 11.$$

$$\mathrm{Var}(3X + 2) = 3^2\mathrm{Var}(X) = 9 \times 2 = 18.$$

So the standard deviation is

$$\sqrt{18} \approx 4.2426.$$

b.

$$E[5 - Y] = 5 - E[Y] = 5 - (-2) = 7.$$

$$\text{Var}(5 - Y) = (-1)^2\text{Var}(Y) = 1 \times 7 = 7.$$

So the standard deviation is

$$\sqrt{7} \approx 2.6458.$$

(c)

$$E[3X + 7 - Y] = 3E[X] + 7 - E[Y] = 3 \times 3 + 7 - (-2) = 18.$$

Since X and Y are independent,

$$\text{Var}(3X + 7 - Y) = 3^2\text{Var}(X) + (-1)^2\text{Var}(Y) = 9 \times 2 + 1 \times 7 = 25.$$

So the standard deviation is

$$\sqrt{25} = 5.$$

Expectation with a Poisson Distribution

The number of breakdowns per week in a computer cluster is a random variable Y having a **Poisson distribution** with population mean μ.
The weekly cost of repairing breakdowns is

$$C = 3Y + Y^2.$$

Find $E[C]$.
Solution
We have

$$E[C] = E[3Y + Y^2] = 3E[Y] + E[Y^2].$$

Since $Y \sim \text{Poisson}(\mu)$:

- $E[Y] = \mu$
- $\text{Var}(Y) = \mu$
- Recall that $\text{Var}(Y) = E[Y^2] - (E[Y])^2$, so

$$E[Y^2] = \mu + \mu^2.$$

Therefore,

$$\mathrm{E}[C] = 3\mu + (\mu + \mu^2) = 4\mu + \mu^2.$$

Distribution of a Squared Normal Variable

Suppose $Z \sim \mathcal{N}(0, 1)$.
Find the PDF of $Y = Z^2$.
Hint: start by finding a formula for the CDF $F_Y(y) = \Pr(Y \leq y)$ for any $y > 0$.
Solution
For $y > 0$:

$$F_Y(y) = \Pr(Y \leq y) = \Pr(Z^2 \leq y).$$

This is equivalent to

$$F_Y(y) = \Pr(-\sqrt{y} \leq Z \leq \sqrt{y}) = \Pr(Z \leq \sqrt{y}) - \Pr(Z \leq -\sqrt{y}).$$

Using the standard normal CDF Φ,

$$F_Y(y) = \Phi(\sqrt{y}) - \Phi(-\sqrt{y}).$$

Differentiate to find the PDF:

$$f_Y(y) = \frac{\partial}{\partial y}\Big[\Phi(\sqrt{y}) - \Phi(-\sqrt{y})\Big].$$

By the chain rule:

$$f_Y(y) = \phi(\sqrt{y}) \cdot \frac{1}{2\sqrt{y}} + \phi(-\sqrt{y}) \cdot \frac{1}{2\sqrt{y}}.$$

Since ϕ (the standard normal PDF) is symmetric,

$$f_Y(y) = \frac{1}{\sqrt{y}}\, \phi(\sqrt{y}).$$

Substitute $\phi(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}$:

$$f_Y(y) = \frac{1}{\sqrt{2\pi y}} e^{-y/2}, \quad y > 0.$$

For $y \leq 0$, $f_Y(y) = 0$.

Aside.
This is exactly the PDF of a χ^2_1 distribution.

Customer Waiting Times

Customer waiting times at a call centre can be assumed to be **?@sec-gamma** random variables with mean 3 minutes and standard deviation 2 minutes.
What proportion of customers are expected to wait more than 5 minutes?
Hint: you will need R or Python for this.
Solution
Let the waiting time be $X \sim \text{Gamma}(\alpha, \lambda)$, with shape α and rate λ.
From the formulas:

$$\text{E}[X] = \frac{\alpha}{\lambda}, \qquad \text{Var}(X) = \frac{\alpha}{\lambda^2}.$$

We are told:

$$\text{E}[X] = 3, \qquad \text{SD}[X] = 2 \;\Rightarrow\; \text{Var}(X) = 4.$$

So:

$$\frac{\alpha}{\lambda} = 3, \qquad \frac{\alpha}{\lambda^2} = 4.$$

Divide the second equation by the first:

$$\frac{\alpha/\lambda^2}{\alpha/\lambda} = \frac{4}{3} \quad \Rightarrow \quad \frac{1}{\lambda} = \frac{4}{3}.$$

Hence:

$$\lambda = 0.75, \quad \alpha = 3\lambda = 2.25.$$

We want:

$$\Pr(X > 5) = 1 - F_X(5).$$

Exact Binomial Confidence Interval (Clopper–Pearson)

Suppose we have a single **?@sec-binom** observation

$$X \mid p \sim \text{Binomial}(n, p).$$

Following the general strategy 7.3.1, construct an *exact* two-sided $(1-\alpha)$ confidence interval for p.
Note on exact (discrete) coverage. In the discrete case, an "exact" interval $[L(X), U(X)]$ means, for every $0 \le p \le 1$,

$$\Pr\big(L(X) \le p \le U(X)\big) \ge 1 - \alpha, \quad \text{where } X \sim \text{Binomial}(n, p).$$

Because of discreteness, equality for all p is impossible; the interval is conservative.
Solution

1. Choose a statistic. Take $T = X$. Its sampling distribution is

$$X \mid p \sim \text{Binomial}(n, p), \qquad f(x \mid p) = \binom{n}{x} p^x (1-p)^{n-x}.$$

2–3. Form a pivot via tail probabilities and fix central mass. Denote $F_p(y) = \Pr(Y \le y)$ where $Y \sim \text{Binomial}(n, p)$ and denote $\Pr_p(X \le x) = F_p(x)$. For a given observed x, define endpoints by equating the two tails to $\alpha/2$:

$$\Pr_{p_L}(X \le x - 1) = \frac{\alpha}{2}, \qquad \Pr_{p_U}(X \ge x + 1) = \frac{\alpha}{2}.$$

(Edge cases: if $x = 0$ set $p_L = 0$; if $x = n$ set $p_U = 1$.)
4. Invert (Clopper–Pearson). These tail equalities are exactly equivalent to **Beta quantiles**, giving closed-form endpoints:

$$\boxed{p_L = B^{-1}\left(\tfrac{\alpha}{2};\, x,\ n - x + 1\right), \qquad p_U = B^{-1}\left(1 - \tfrac{\alpha}{2};\, x + 1,\ n - x\right)}$$

where $B^{-1}(q; a, b)$ is the q-quantile of $\text{Beta}(a, b)$. With the edge-case conventions above:

$$p_L = \begin{cases} 0, & x = 0, \\ B^{-1}\left(\tfrac{\alpha}{2};\, x,\ n - x + 1\right), & x \ge 1, \end{cases} \quad p_U = \begin{cases} B^{-1}\left(1 - \tfrac{\alpha}{2};\, x + 1,\ n - x\right), & x \le n - 1, \\ 1, & x = n. \end{cases}$$

We obtain the **Clopper–Pearson interval** $[p_L, p_U]$, which is an exact (conservative) two-sided $(1 - \alpha)$ CI for p.

Probability Integral Transform

Suppose X is a continuous random variable with CDF $F(x)$.
Define the random variable

$$U = F(X),$$

that is, the CDF evaluated at a random argument.
Show that $U \sim U(0, 1)$.
Aside. This result is fundamental in simulation: if we can generate uniform random variables $U \sim U(0, 1)$, then we can simulate X by setting $X = F^{-1}(U)$.
Solution
Because $F(\cdot)$ is continuous and strictly increasing, its inverse $F^{-1}(\cdot)$ exists and is also strictly increasing.
Define the CDF of U as

$$G(u) = \Pr(U \le u).$$

Since $U = F(X)$ always lies in $[0, 1]$, we have:

$$G(u) = 0 \quad \text{for } u \le 0, \qquad G(u) = 1 \quad \text{for } u \ge 1.$$

Now consider $u \in (0,1)$:

$$G(u) = \Pr(U \le u) = \Pr(F(X) \le u).$$

Because F is increasing,

$$\Pr(F(X) \le u) = \Pr(X \le F^{-1}(u)).$$

By definition of the CDF F,

$$\Pr(X \le F^{-1}(u)) = F(F^{-1}(u)) = u.$$

Thus the CDF of U is

$$G(u) = \begin{cases} 0, & u \le 0, \\ u, & 0 < u \le 1, \\ 1, & u > 1. \end{cases}$$

This is exactly the CDF of the Uniform$(0,1)$ distribution.
Therefore,

$$U \sim U(0,1).$$

Moment Generating Function of the Exponential

We have seen that properties of a random variable X can be described by the CDF, PMF, or PDF.
Another way is through the **moment generating function (MGF)** $M(t)$, defined by

$$M(t) = \mathrm{E}[e^{tX}],$$

provided the expectation is finite (which is not always the case).
If the MGF exists, it is **unique**: two random variables have the same MGF if and only if they have the same distribution.

Suppose $X \sim \mathrm{Exp}(\lambda)$, with PDF

$$f(x) = \lambda e^{-\lambda x}, \quad x \ge 0.$$

Show that the MGF of X is

$$M(t) = \frac{\lambda}{\lambda - t}, \qquad 0 \le t < \lambda.$$

Solution

Choose t with $0 \le t < \lambda$ to ensure convergence. Then

$$M(t) = \mathrm{E}[e^{tX}] = \int_0^\infty e^{tx} \lambda e^{-\lambda x}\, dx.$$

Combine exponents:

$$M(t) = \int_0^\infty \lambda e^{-(\lambda - t)x}\, dx.$$

Factor out $(\lambda - t)$:

$$M(t) = \frac{\lambda}{\lambda - t} \int_0^\infty (\lambda - t) e^{-(\lambda - t)x}\, dx.$$

But the integrand is the PDF of $\mathrm{Exp}(\lambda - t)$, so the integral equals 1.
Therefore:

$$M(t) = \frac{\lambda}{\lambda - t}, \qquad 0 \le t < \lambda.$$

Estimating λ from Independent Poisson Samples

Suppose $X \sim \mathrm{Poisson}(\lambda)$ and, independently, $Y \sim \mathrm{Poisson}(2\lambda)$.

a. Classify the following as *parameters*, *statistics*, or *neither*:

   i. $X + 2Y$

   ii. $\hat{\lambda} = (X + Y)/3$

   iii. $\lambda / \hat{\lambda}$
   iv. $\Pr(\lambda > \hat{\lambda})$

b. Find the **bias** and **MSE** of each of the following estimators of λ. Which would you recommend?

   i. X

   ii. $Y - X$
   iii. $(X + Y)/3$

c. Consider a general estimator $w_1 X + w_2 Y$. What values of w_1 and w_2 would you recommend?

Solution
a. Classification.

- A *statistic* is a function of data (random variables).

- A *parameter* depends only on the population.
- *Neither* means it involves both.

(i) $X + 2Y$: **Statistic**

(ii) $(X + Y)/3$: **Statistic**

(iii) $\lambda/\hat{\lambda}$: **Neither**

(iv) $\Pr(\lambda > \hat{\lambda})$: **Parameter** (depends on the distribution, not directly observable)

---

b. Bias and MSE.
Recall: for Poisson(μ), $\mathrm{E}[X] = \mathrm{Var}(X) = \mu$.

- $X \sim \text{Poisson}(\lambda)$, so $\mathrm{E}[X] = \lambda$, $\mathrm{Var}(X) = \lambda$.
- $Y \sim \text{Poisson}(2\lambda)$, so $\mathrm{E}[Y] = 2\lambda$, $\mathrm{Var}(Y) = 2\lambda$.
- Independence: $\mathrm{Var}(aX + bY) = a^2\mathrm{Var}(X) + b^2\mathrm{Var}(Y)$.

$$\mathrm{Bias}(\hat{\theta}) = \mathrm{E}[\hat{\theta}] - \lambda, \qquad \mathrm{MSE}(\hat{\theta}) = \mathrm{Var}(\hat{\theta}) + \mathrm{Bias}(\hat{\theta})^2.$$

(i) X:
$\mathrm{E}[X] = \lambda \;\Rightarrow\; \mathrm{Bias} = 0$.
$\mathrm{MSE} = \mathrm{Var}(X) = \lambda$.

(ii) $Y - X$:
$\mathrm{E}[Y - X] = 2\lambda - \lambda = \lambda \;\Rightarrow\; \mathrm{Bias} = 0$.
$\mathrm{Var}(Y - X) = \mathrm{Var}(Y) + \mathrm{Var}(X) = 2\lambda + \lambda = 3\lambda$.
$\mathrm{MSE} = 3\lambda$.

(iii) $(X + Y)/3$:
$\mathrm{E}[(X + Y)/3] = (\lambda + 2\lambda)/3 = \lambda \;\Rightarrow\; \mathrm{Bias} = 0$.
$\mathrm{Var}[(X + Y)/3] = \frac{1}{9}(\lambda + 2\lambda) = \frac{\lambda}{3}$.
$\mathrm{MSE} = \frac{\lambda}{3}$.

Recommendation: $(X + Y)/3$ has the smallest MSE.

---

c. General estimator $w_1 X + w_2 Y$.
Expectation:

$$\mathrm{E}[w_1 X + w_2 Y] = (w_1 + 2w_2)\lambda.$$

For unbiasedness, require $w_1 + 2w_2 = 1$, i.e. $w_1 = 1 - 2w_2$.
Variance:

$$\begin{aligned}\mathrm{Var}((1-2w_2)X + w_2 Y) &= (1-2w_2)^2\lambda + 2w_2^2\lambda \\ &= (6w_2^2 - 4w_2 + 1)\lambda.\end{aligned}$$

This quadratic in w_2 is minimised at $w_2 = \frac{1}{3}$, giving $w_1 = \frac{1}{3}$.
Thus the optimal unbiased estimator is:

$$\hat{\lambda} = \tfrac{1}{3}(X + Y),$$

matching part (b).

Bias, Variance, and MSE of Estimators

A random variable X has expected value $\mathrm{E}[X] = 2\theta + 6$ and variance $\mathrm{Var}(X) = 2\theta^2$.
A random variable Y has expected value $\mathrm{E}[Y] = 3\theta - 12$ and variance $\mathrm{Var}(Y) = 6\theta^2 + 1$.
Assume X and Y are **independent**.

a. Compute the **bias**, **variance**, and **MSE** of the following estimators of θ:

   (i) $\hat{\theta}_1 = 5X - 3Y - 65$

   (ii) $\hat{\theta}_2 = (X + Y)/5$

   (iii) $\hat{\theta}_3 = Y/6 + X/4 + 1/2$

b. Let $\hat{\theta} = aX + bY + c$ be an estimator of θ. Derive a system of linear equations involving a, b, c that ensures $\hat{\theta}$ is **unbiased** for any value of θ.

c. Suppose $\hat{\theta} = aX + bY + c$ is an **unbiased** estimator of θ. Find the values of a, b, c that minimise the **mean squared error** of $\hat{\theta}$ when the true value of $\theta = 1$.

Solution
a. Bias, variance, and MSE.

$$\mathrm{Bias}(\hat{\theta}) = \mathrm{E}[\hat{\theta}] - \theta, \qquad \mathrm{MSE}(\hat{\theta}) = \mathrm{Var}(\hat{\theta}) + \mathrm{Bias}(\hat{\theta})^2.$$

Using independence of X and Y:

$$\mathrm{Var}(aX + bY + c) = a^2\mathrm{Var}(X) + b^2\mathrm{Var}(Y).$$

Estimator $\hat{\theta}$	$\mathrm{E}[\hat{\theta}]$	$\mathrm{Var}[\hat{\theta}]$	Bias	MSE
$5X - 3Y - 65$	$\theta + 1$	$104\theta^2 + 9$	1	$104\theta^2 + 10$
$(X + Y)/5$	$\theta - \frac{6}{5}$	$\frac{8\theta^2+1}{25}$	$-\frac{6}{5}$	$\frac{8\theta^2+37}{25}$
$Y/6 + X/4 + 1/2$	θ	$\frac{7}{24}\theta^2 + \frac{1}{36}$	0	$\frac{7}{24}\theta^2 + \frac{1}{36}$

---

b. Unbiasedness condition.

$$\mathrm{E}[\hat{\theta}] = a\,\mathrm{E}[X] + b\,\mathrm{E}[Y] + c = a(2\theta + 6) + b(3\theta - 12) + c.$$

For unbiasedness: $\mathrm{E}[\hat{\theta}] = \theta$ for all θ.
So

$$(2a + 3b - 1)\theta + (6a - 12b + c) = 0.$$

This gives the system:

$$2a + 3b = 1, \qquad 6a - 12b + c = 0.$$

---

c. Minimising MSE when $\theta = 1$.
For an unbiased estimator, MSE = variance:

$$\mathrm{Var}[\hat{\theta}] = a^2\mathrm{Var}(X) + b^2\mathrm{Var}(Y) = 2a^2\theta^2 + b^2(6\theta^2 + 1).$$

At $\theta = 1$:

$$\mathrm{Var}[\hat{\theta}] = 2a^2 + 7b^2.$$

From part (b): $a = \frac{1-3b}{2}$, $c = 12b - 6a$.
Substitute $a = (1 - 3b)/2$:

$$\mathrm{Var}[\hat{\theta}] = 2\left(\frac{1-3b}{2}\right)^2 + 7b^2 = \frac{(1-3b)^2}{2} + 7b^2 = \frac{23}{2}b^2 - 3b + \frac{1}{2}.$$

Minimise by setting derivative = 0:

$$23b - 3 = 0 \quad \Rightarrow \quad b = \frac{3}{23}.$$

Then

$$a = \frac{1-3b}{2} = \frac{7}{23}, \qquad c = 12b - 6a = -\frac{6}{23}.$$

Final recommended estimator:

$$\hat{\theta} = \frac{7X + 3Y - 6}{23}.$$

Properties of Poisson Estimators

Suppose that $X_1, X_2, \ldots, X_n$, for $n \geq 1$, are independent Poisson(λ) random variables. Two possible estimators of λ are:

$$\hat{\lambda}_1 = \frac{1}{n}\sum_{i=1}^{n} X_i, \qquad \hat{\lambda}_2 = \frac{1}{n^2}\sum_{i=1}^{n} X_i.$$

a. Show that $\hat{\lambda}_1$ is unbiased but $\hat{\lambda}_2$ is not.

b. Show that
$$\mathrm{Var}(\hat{\lambda}_1) = \frac{\lambda}{n}, \qquad \mathrm{Var}(\hat{\lambda}_2) = \frac{\lambda}{n^3}.$$

c. For each estimator, determine whether it can be shown to be **consistent** using its mean and variance.

Solution

a. Unbiasedness.

$$\mathrm{E}[\hat{\lambda}_1] = \frac{1}{n}\sum_{i=1}^{n} \mathrm{E}[X_i] = \frac{1}{n} \cdot n\lambda = \lambda.$$

So $\hat{\lambda}_1$ is unbiased.

$$\mathrm{E}[\hat{\lambda}_2] = \frac{1}{n^2}\sum_{i=1}^{n} \mathrm{E}[X_i] = \frac{1}{n^2} \cdot n\lambda = \frac{\lambda}{n}.$$

So $\hat{\lambda}_2$ is biased.

---

b. Variances.
Since $X_i \sim \text{Poisson}(\lambda)$, $\mathrm{Var}(X_i) = \lambda$.

$$\mathrm{Var}(\hat{\lambda}_1) = \frac{1}{n^2}\sum_{i=1}^{n} \mathrm{Var}(X_i) = \frac{1}{n^2} \cdot n\lambda = \frac{\lambda}{n}.$$
$$\mathrm{Var}(\hat{\lambda}_2) = \frac{1}{n^4}\sum_{i=1}^{n} \mathrm{Var}(X_i) = \frac{1}{n^4} \cdot n\lambda = \frac{\lambda}{n^3}.$$

---

c. Consistency.

- $\hat{\lambda}_1$: unbiased and $\mathrm{Var}(\hat{\lambda}_1) \to 0$ as $n \to \infty$, so $\hat{\lambda}_1$ is consistent.

- $\hat{\lambda}_2$: $\mathrm{E}[\hat{\lambda}_2] = \lambda/n \to 0$ as $n \to \infty$, so it does not converge to λ. Therefore it is not consistent.

MLE for the Gamma Distribution

Suppose that $X_1, \dots, X_n$ is random sample from a Gamma$(1, b)$ distribution.

a. Show that the log-likelihood takes the form

$$\ell(b; \underline{x}) = n \log(b) - b \sum_{i=1}^{n} x_i.$$

b. Using R or Python, plot the likelihood function of b given the data

$$x_1 = 0.043, \quad x_2 = 0.060, \quad x_3 = 0.011, \quad x_4 = 0.142, \quad x_5 = 1.027.$$

c. Derive the maximum likelihood estimate $\hat{b}$ of b. You do not need to verify it is a maximum.

Solution

a. Likelihood and log-likelihood.

The PDF of the Gamma$(1, b)$ distribution is

$$f(x \mid b) = be^{-bx}, \quad x > 0.$$

Thus

$$L(b; \underline{x}) = \prod_{i=1}^{n} f(x_i; b) = \prod_{i=1}^{n} be^{-bx_i} = b^n \exp\left(-b \sum_{i=1}^{n} x_i\right).$$

Therefore

$$\ell(b; \underline{x}) = \log L(b; \underline{x}) = n \log(b) - b \sum_{i=1}^{n} x_i.$$

---

b. Plotting.

10.1.3 Python

```
import numpy as np
import matplotlib.pyplot as plt

x = np.array([0.043, 0.060, 0.011, 0.142, 1.027])
n = len(x)

def likelihood(b):
    return b**n * np.exp(-b * np.sum(x))

b_mesh = np.linspace(0, 12, 200)
lik_values = [likelihood(b) for b in b_mesh]

plt.plot(b_mesh, lik_values)
plt.xlabel("b")
plt.ylabel("Likelihood")
plt.title("Likelihood function for b")
plt.show()
```

10.1.4 R

```
#| echo: true
#| code-fold: true
#| code-summary: "Show code"
#| fig-align: center

x = c(0.043, 0.060, 0.011, 0.142, 1.027)

likelihood = function(b) {
  n = length(x)
  return(b^n * exp(-b * sum(x)))
}

b_mesh = seq(0, 12, length.out = 100)
plot(b_mesh, likelihood(b_mesh), type = "l",
     xlab = "b", ylab = "Likelihood")
```

---

c. Maximum likelihood estimate.

Differentiate the log-likelihood:

$$\frac{\partial}{\partial b}\ell(b;\underline{x}) = \frac{n}{b} - \sum_{i=1}^{n} x_i.$$

Set equal to 0:

$$\frac{n}{b} = \sum_{i=1}^{n} x_i \quad \Rightarrow \quad \hat{b} = \frac{1}{\bar{x}}.$$

So the MLE is $\hat{b} = 1/\bar{x}$.

Deriving the Student-t distribution

Let $X_1, \dots, X_n \overset{\text{iid}}{\sim} \mathcal{N}(\mu, \sigma^2)$. From Proposition 4.2 and Proposition 4.3, we saw that

1. $\bar{X} \sim \mathcal{N}(\mu, \sigma^2/n)$,
2. $\dfrac{(n-1)S^2}{\sigma^2} \sim \chi^2_{n-1}$, and
3. The sample variance S^2 was independent of the sample mean $\bar{X}$.

Derive the PDF of the random variable

$$T = \frac{\bar{X} - \mu}{S/\sqrt{n}}.$$

Solution
Set

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim \mathcal{N}(0,1), \qquad U = \frac{(n-1)S^2}{\sigma^2} \sim \chi^2_{n-1},$$

with Z independent of U. Then

$$T = \frac{\sigma Z}{\sqrt{(\sigma^2 U)/(n-1)}} = \frac{Z}{\sqrt{U/\nu}}, \qquad \nu := n-1.$$

Conditional distribution. Conditioning on $U = u$,

$$T \mid U = u = \frac{Z}{\sqrt{u/\nu}} \sim \mathcal{N}\Big(0, \frac{\nu}{u}\Big),$$

so its conditional density is

$$f_{T|U}(t \mid u) = \sqrt{\frac{u}{2\pi\nu}} \exp\Big(-\frac{u}{2\nu}t^2\Big), \qquad t \in \mathbb{R}.$$

Mixture over U. The marginal PDF of T is

$$f_T(t) = \int_0^\infty f_{T|U}(t \mid u)\, f_U(u)\, \mathrm{d}u,$$

where $f_U(u) = \dfrac{1}{2^{\nu/2}\Gamma(\nu/2)}\, u^{\nu/2-1} e^{-u/2}$ for $u > 0$. Thus

$$\begin{aligned} f_T(t) &= \int_0^\infty \left[\sqrt{\frac{u}{2\pi\nu}}\, e^{-\frac{u}{2\nu}t^2}\right] \left[\frac{1}{2^{\nu/2}\Gamma(\nu/2)}\, u^{\nu/2-1} e^{-u/2}\right] \mathrm{d}u \\ &= \frac{1}{\sqrt{2\pi\nu}\, 2^{\nu/2}\Gamma(\nu/2)} \int_0^\infty u^{\frac{\nu+1}{2}-1} \exp\left\{-\frac{1}{2}\left(1+\frac{t^2}{\nu}\right) u\right\} \mathrm{d}u. \end{aligned}$$

Use the Gamma integral $\displaystyle\int_0^\infty u^{a-1} e^{-bu}\, \mathrm{d}u = \Gamma(a)\, b^{-a}$ with $a = \frac{\nu+1}{2}$ and $b = \frac{1}{2}\left(1+\frac{t^2}{\nu}\right)$:

$$\begin{aligned} f_T(t) &= \frac{1}{\sqrt{2\pi\nu}\, 2^{\nu/2}\Gamma(\nu/2)} \cdot \Gamma\left(\frac{\nu+1}{2}\right) \left[\frac{1}{2}\left(1+\frac{t^2}{\nu}\right)\right]^{-\frac{\nu+1}{2}} \\ &= \frac{\Gamma\left(\frac{\nu+1}{2}\right)}{\sqrt{\nu\pi}\,\Gamma\left(\frac{\nu}{2}\right)} \left(1+\frac{t^2}{\nu}\right)^{-\frac{\nu+1}{2}}, \qquad t \in \mathbb{R}. \end{aligned}$$

This is the **Student-t** density with $\nu = n-1$ degrees of freedom:

$$T \sim t_\nu \quad \text{with} \quad f_T(t) = \frac{\Gamma\left(\frac{\nu+1}{2}\right)}{\sqrt{\nu\pi}\,\Gamma\left(\frac{\nu}{2}\right)} \left(1+\frac{t^2}{\nu}\right)^{-\frac{\nu+1}{2}}.$$

Sample Mean and Standard Deviation

Emergency admissions to 10 hospitals were recorded on a particular day.
The summary statistics were:

- $n = 10$
- $\sum_{i=1}^n x_i = 207$
- $\sum_{i=1}^n x_i^2 = 4501$

Find the **sample mean** $\bar{x}$ and the **sample standard deviation** s.
Solution
The sample mean is

$$\bar{x} = \frac{1}{n}\sum_{i=1}^n x_i = \frac{207}{10} = 20.7.$$

To find the sample variance,

$$s^2 = \frac{1}{n-1}\left(\sum_{i=1}^{n} x_i^2 - n\bar{x}^2\right).$$

Substituting the values:

$$s^2 = \frac{1}{9}\left(4501 - 10 \times (20.7)^2\right) = \frac{1}{9}(4501 - 4284.9) = 24.011.$$

Taking the square root gives the sample standard deviation:

$$s = \sqrt{24.011} \approx 4.900.$$

Central Limit Theorem Approximation

The scores that responders give on a questionnaire can be assumed to be independent random variables, each with the following probability distribution:

Score	1	2	3	4	5
Probability	0.3	0.1	0.2	0.1	0.3

Use the Central Limit Theorem 4.1 to approximate the probability that the average score from $n = 100$ responders lies in the range (2.8, 3.2).
Solution
Let X be the score for a randomly selected responder.
We first compute the moments.
Expected value:

$$\mathrm{E}[X] = \sum_{i=1}^{5} i \Pr(X = i) = 1(0.3) + 2(0.1) + 3(0.2) + 4(0.1) + 5(0.3) = 3.$$

Second moment:

$$\mathrm{E}[X^2] = \sum_{i=1}^{5} i^2 \Pr(X = i) = 1^2(0.3) + 2^2(0.1) + 3^2(0.2) + 4^2(0.1) + 5^2(0.3) = 11.6.$$

Variance:

$$\mathrm{Var}(X) = \mathrm{E}[X^2] - (\mathrm{E}[X])^2 = 11.6 - 9 = 2.6.$$

Let $\bar{X}$ be the sample mean. By the Central Limit Theorem,

$$\bar{X} \overset{\text{approx.}}{\sim} \mathcal{N}\left(3,\ \frac{2.6}{100}\right).$$

We want:

$$\Pr(2.8 < \bar{X} < 3.2) = \Pr\left(\frac{2.8 - 3}{\sqrt{2.6/100}} < Z < \frac{3.2 - 3}{\sqrt{2.6/100}}\right),$$

where $Z \sim N(0, 1)$.
Compute the bounds:

$$\frac{2.8 - 3}{\sqrt{0.026}} = -1.24, \qquad \frac{3.2 - 3}{\sqrt{0.026}} = 1.24.$$

Thus,

$$\Pr(-1.24 < Z < 1.24) = \Pr(Z < 1.24) - \Pr(Z < -1.24).$$

Using R, Python or referring to the standard normal tables:

$$\Pr(Z < 1.24) = 0.8925, \qquad \Pr(Z < -1.24) = 0.1075.$$

So,

$$\Pr(2.8 < \bar{X} < 3.2) \approx 0.8925 - 0.1075 = 0.785.$$

Normal Approximation to the Binomial

Use the Central Limit Theorem 4.1 to justify the Normal approximation to the **?@sec-binom**.
That is, if $X \sim \text{Bin}(n, p)$, show that for large n (with $p \neq 0, 1$),

$$X \overset{\text{approx.}}{\sim} \mathcal{N}(np,\ np(1-p)).$$

Solution
The Binomial distribution counts the number of successes in n independent Bernoulli trials.
For trial i, define the indicator

$$I_i = \begin{cases} 1, & \text{if success,} \\ 0, & \text{if failure.} \end{cases}$$

Each I_i has

$$\text{E}[I_i] = p, \qquad \text{Var}(I_i) = p(1-p).$$

The total number of successes is

$$X = \sum_{i=1}^{n} I_i.$$

By the Central Limit Theorem, the sum of independent, identically distributed random variables is approximately Normal when n is large. Therefore,

$$X \overset{\text{approx.}}{\sim} \mathcal{N}(np,\ np(1-p)).$$

Properties of an Estimator I

Suppose X_1, X_2, X_3 are independent random variables, each with expectation $\text{E}[X] = 2\theta + 1$ and variance $\text{Var}(X) = \theta^2/12$, where θ is an unknown parameter.
Consider the estimator $\hat{\theta}$ of θ, where

$$\hat{\theta} = 2X_1 - X_2 + X_3.$$

1. Find the expectation of $\hat{\theta}$.

2. Find the variance of $\hat{\theta}$.

3. What is the bias of $\hat{\theta}$?

Solution
(a) Expectation

$$\text{E}[\hat{\theta}] = \text{E}[2X_1 - X_2 + X_3] = 2\text{E}[X_1] - \text{E}[X_2] + \text{E}[X_3].$$

Since each X_i has the same mean $\mathrm{E}[X] = 2\theta + 1$,

$$\mathrm{E}[\hat{\theta}] = 2(2\theta + 1) - (2\theta + 1) + (2\theta + 1) = 4\theta + 2.$$

(b) Variance

$$\mathrm{Var}(\hat{\theta}) = \mathrm{Var}(2X_1 - X_2 + X_3).$$

Since the X_i are independent,

$$\mathrm{Var}(\hat{\theta}) = 4\,\mathrm{Var}(X_1) + \mathrm{Var}(X_2) + \mathrm{Var}(X_3).$$

Each has variance $\theta^2/12$, so

$$\mathrm{Var}(\hat{\theta}) = (4 + 1 + 1)\frac{\theta^2}{12} = \frac{\theta^2}{2}.$$

(c) Bias

$$\mathrm{Bias}(\hat{\theta}) = \mathrm{E}[\hat{\theta}] - \theta = (4\theta + 2) - \theta = 3\theta + 2.$$

Properties of an Estimator II

Suppose X_1, X_2, X_3, X_4 are independent random variables, each with $\mathrm{E}[X] = 3\theta$ and $\mathrm{Var}(X) = \theta^2/9$, where θ is an unknown parameter. Consider the estimator

$$\hat{\theta} = X_1 + 2X_2 - X_3 + X_4.$$

1. Find the expectation of $\hat{\theta}$.

2. Find the variance of $\hat{\theta}$.

3. What is the bias of $\hat{\theta}$?

Solution

(a) Expectation

$$\mathrm{E}[\hat{\theta}] = \mathrm{E}[X_1] + 2\mathrm{E}[X_2] - \mathrm{E}[X_3] + \mathrm{E}[X_4].$$

Each has mean 3θ, so

$$\mathrm{E}[\hat{\theta}] = 3\theta + 2(3\theta) - 3\theta + 3\theta = 9\theta.$$

(b) Variance

$$\operatorname{Var}(\hat{\theta}) = \operatorname{Var}(X_1) + 4\operatorname{Var}(X_2) + \operatorname{Var}(X_3) + \operatorname{Var}(X_4).$$

Each has variance $\theta^2/9$, so

$$\operatorname{Var}(\hat{\theta}) = (1+4+1+1)\frac{\theta^2}{9} = \frac{7\theta^2}{9}.$$

(c) Bias

$$\operatorname{Bias}(\hat{\theta}) = \operatorname{E}[\hat{\theta}] - \theta = 9\theta - \theta = 8\theta.$$

Python IDE